

End to End Project ADF

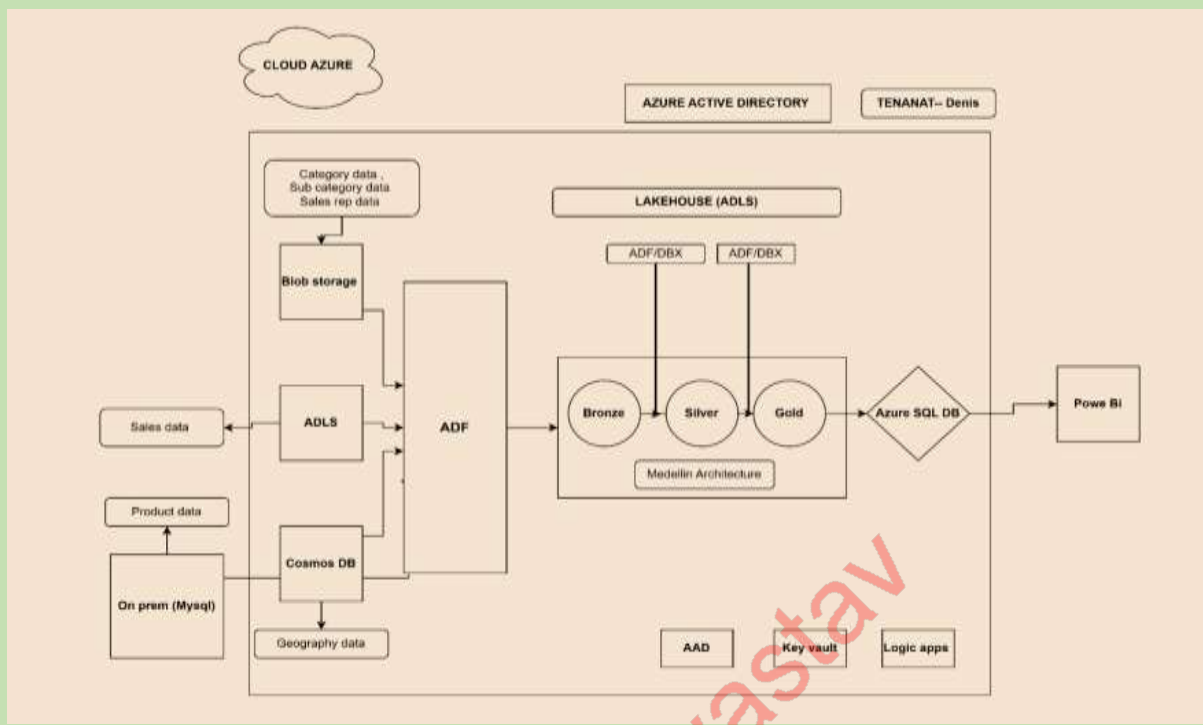
Project process:

This document outlines the implementation of an end-to-end data pipeline using Azure Data Factory (ADF) following the Medellin architecture framework. The project is designed to integrate data from multiple sources, transform it into actionable insights, and deliver it to the reporting layer for visualization.

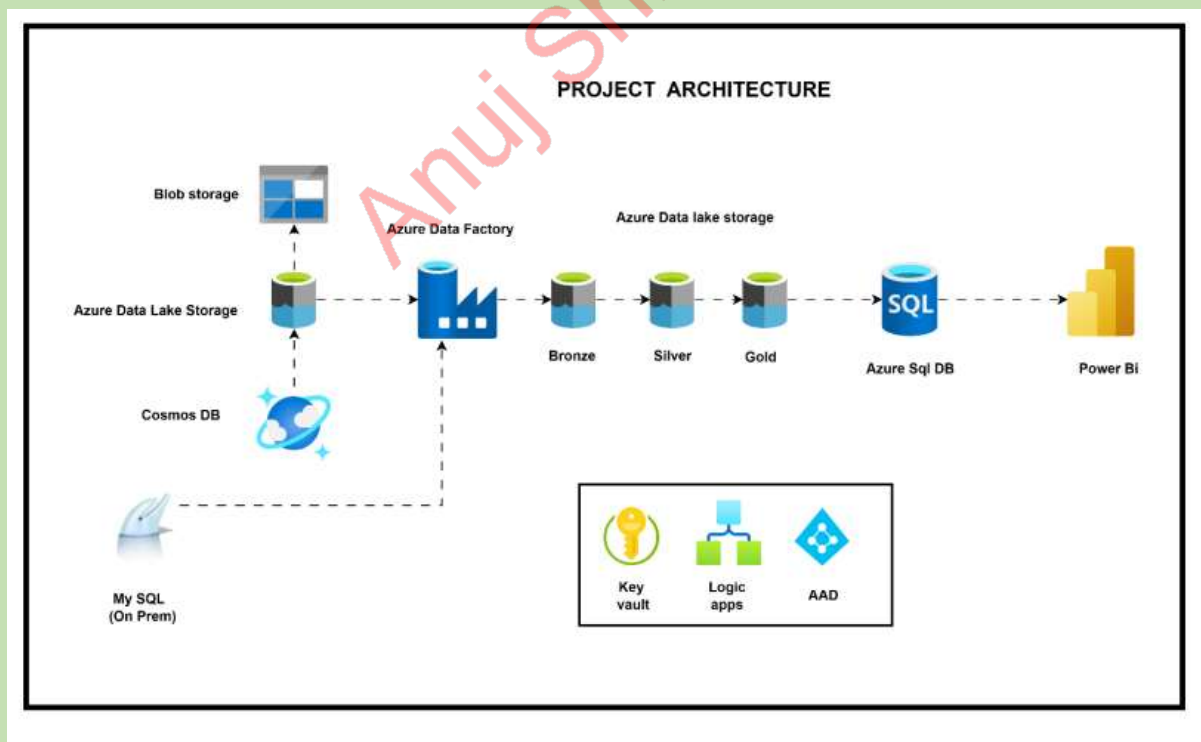
Key Objectives:

1. **Efficient Data Integration:** Automate the ingestion of data from on-premise and cloud-based sources, such as MySQL, Cosmos DB, and Blob Storage, into Azure Data Lake Storage (ADLS).
2. **Data Transformation:** Implement the Bronze-Silver-Gold architecture to incrementally process, clean, and optimize data for analytics.
3. **Seamless Data Delivery:** Load transformed data into Azure SQL Database for Power BI reporting.
4. **Security & Scalability:** Use Azure Active Directory (AAD), Key Vault, and Logic Apps for secure and scalable workflows.

Architecture Diagram:



Project Architecture:



Medallion Architecture: The Medallion Architecture in Azure Data Factory (ADF) structures data into three layers for better data management and analytics:

Bronze Layer (Raw Data - Landing Zone)

- Stores raw data from MySQL, APIs, CSV, etc.
- Stored in: Azure Data Lake Storage (ADLS) Gen2.
- ADF Activity: Copy Activity (loads raw data as Parquet/CSV).

Silver Layer (Cleansed Data - Curated Zone)

- Data is cleaned & transformed (deduplication, type conversion, joins).
- Processed using: ADF Mapping Data Flows / Azure Databricks.
- Stored in: ADLS Gen2 (Parquet/Delta) or Synapse.

Gold Layer (Aggregated Data - Consumption Zone)

- Final business-ready data for reporting & analytics.
- Stored in: Azure Synapse / Power BI datasets.
- Used in: Dashboards, ML models, analytics reports.

ADF Implementation

- Bronze: Copy Activity loads raw data.
- Silver: Data Flows or Databricks cleanses data.
- Gold: Synapse/Power BI for analytics.

Project Overview:

- Based on the architecture, the first step is to migrate data from **on-premises systems, Blob Storage, and Cosmos DB** to **Azure Data Lake Storage**.
- Within **Azure Data Lake Storage**, the data is first stored in the **bronze layer**, which retains a raw copy of the source data.
- The data is then moved to the **silver layer**, where initial **data transformations** are applied to enhance quality and structure.
- Next, the data is transferred to the **gold layer**, where final **data transformations** and optimizations are performed to prepare it for reporting and analytics.
- The processed data is then loaded into **Azure SQL Database** for structured querying and storage.
- Finally, data from **Azure SQL Database** is connected to **Power BI**, where further **data analysis and visualization** are performed to generate business insights.

Required Azure Services: To implement the project, the following **Azure services** need to be created:

- **Azure Blob Storage** – For storing unstructured data.
- **Azure Data Lake Storage** – For scalable and secure data storage.
- **Azure Data Factory** – For orchestrating data movement and transformation.
- **Azure SQL Database** – For structured data storage and querying.
- **Azure Cosmos DB** – For globally distributed NoSQL data management.
- **Azure Key Vault** – For securely managing secrets, keys, and credentials.
- **Azure Logic Apps** – For automating workflows and integrations.

Business / Functional Requirement Document:

Objective: To create one centralized Dataware housing system

1. Data Gathering / Requirement: The objective is to assemble a **data warehousing system** with various tables to provide **Sales Insights** in a single fact table. The data sources include:

1. Sales (folder by year)
2. Categories (Excel)
3. Geography (Excel)
4. Product (CSV /Database)
5. SalesRep (Excel)
6. SubCategories (Excel)

Task 1.1: Sales Fact Table Mechanism

Create a mechanism to load all the files from the sales folder in a single Sales fact table. The mechanism needs to be resilient as:

- Removing a file from the **Sales** folder does **not** cause errors due to missing files.
- Adding a new yearly **Sales** file is automatically included in the fact query upon refresh.

2. Data Modelling:

Task 2.1:

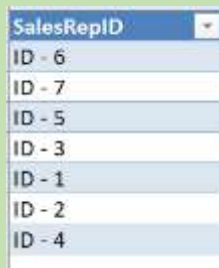
- Do the respective transformations to the Sales fact table in order to split the Country from the city in field "Location". Make sure you set up the correct Data Type to allow Geo maps.
- Do the necessary updates in the Date field to make sure you can setup the Date format.

Task 2.2:

- Create unique key (GeoKey) in Sales and Geography table

Task 2.3:

- The Dimensional queries SalesRep and Sub Category need additional treatment. Some ID columns have the following format:



SalesRepID
ID - 6
ID - 7
ID - 5
ID - 3
ID - 1
ID - 2
ID - 4

Create a small function that removes the “ID - ” part of these columns that you can invoke and reuse for these two queries to clean the IDs.

Task 2.4:

- Create the Data Model connecting all tables and using the Calendar table already set up in the pbix.

3. Additional column calculations:

Task 3.1:

- Calculate **Total Revenue** in Sales table, using the Product’s Retail Price, and multiplying it by the Units.

Task 3.2:

- Calculate **Total Cost** in Sales table, using the Product’s Standard Cost, and multiplying it by the Units.

Task 3.3:

- Calculate **Gross Profit** in Sales: Total Revenue – Total Cost

Task 3.4:

- Calculate a measure for **AVG sales per day** – this is the average sum of **Total Revenue** per day based on the Dates of actual Sales.

Task 3.5:

- Breakdown Analysis by **Product (drop or increase)**

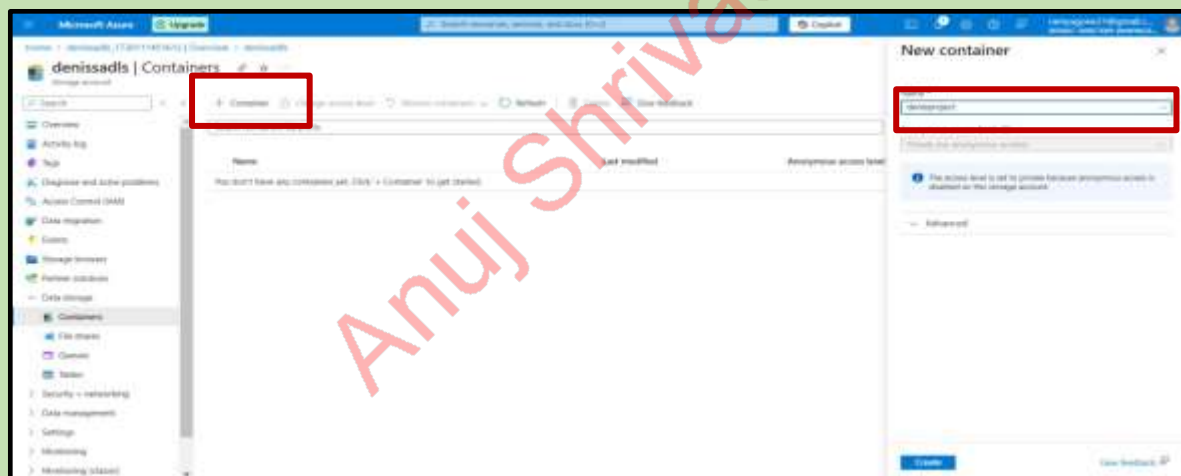
4. Load the target tables into Azure SQL base. The number of analytical tables is developer choice. Feel free to add own ideas and come up with end-to-end process using azure services.

Project Execution: The project execution process is documented with **step-by-step screenshots**, illustrating each stage of implementation in detail.

- Firstly, we will open Azure data lake Storage and go to containers.



- We need to create new container as denisproject.



- Under denisproject container ---create directories---(bronze,silver,gold)

Overview > denisproject | Containers >

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Break down Give feedback

Authentication method: Access key [Switch to Microsoft Entra user account](#)

Location: denisproject

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status	Blob type	Size
No results					

Add Directory

Name

Save Give feedback

Overview > denisproject | Containers >

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Break down Give feedback

Authentication method: Access key [Switch to Microsoft Entra user account](#)

Location: denisproject

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status	Blob type	Size
☐ Bronze					

Add Directory

Name

Save Give feedback

Overview > denisproject | Containers >

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Break down Give feedback

Authentication method: Access key [Switch to Microsoft Entra user account](#)

Location: denisproject

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status	Blob type	Size
☐ Bronze					
☐ Silver					

Add Directory

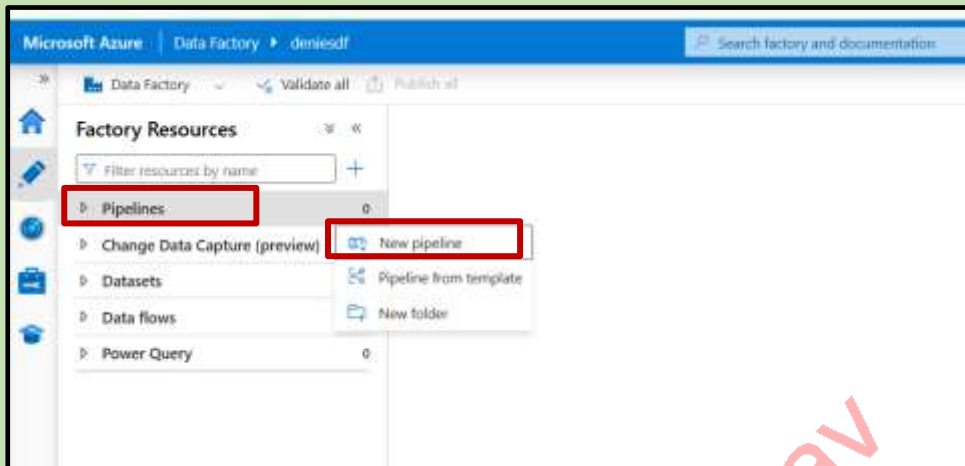
Name

Save Give feedback

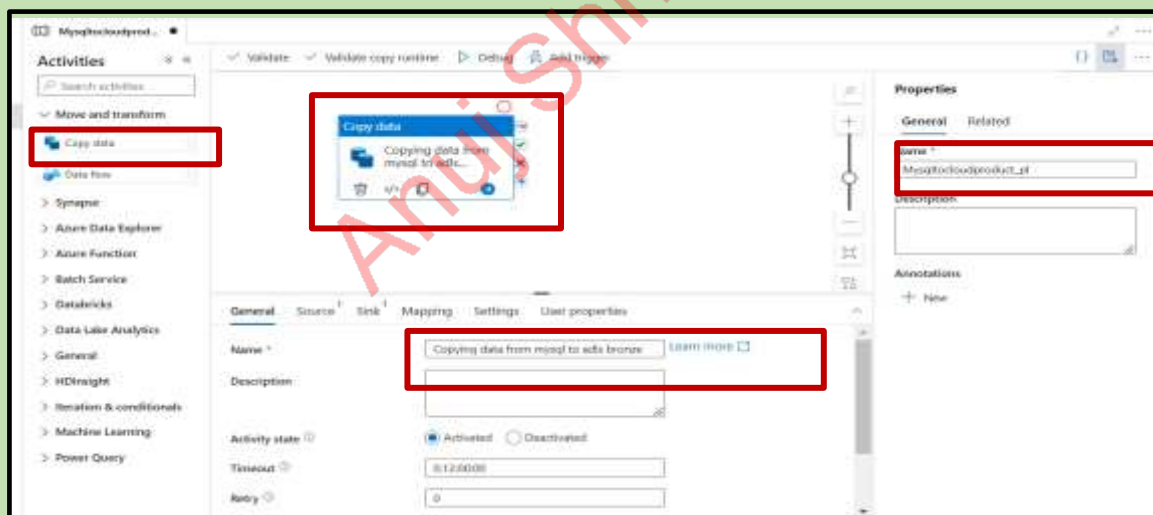
Pipeline 1:

Mysql : Azure data lake storage

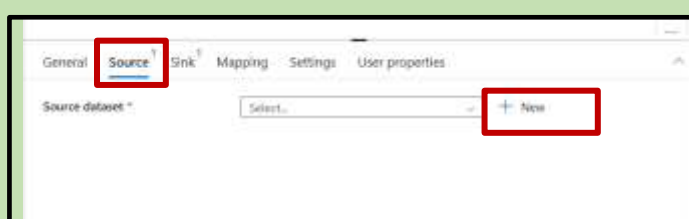
Now open Azure data factory and click on pipeline click on create new



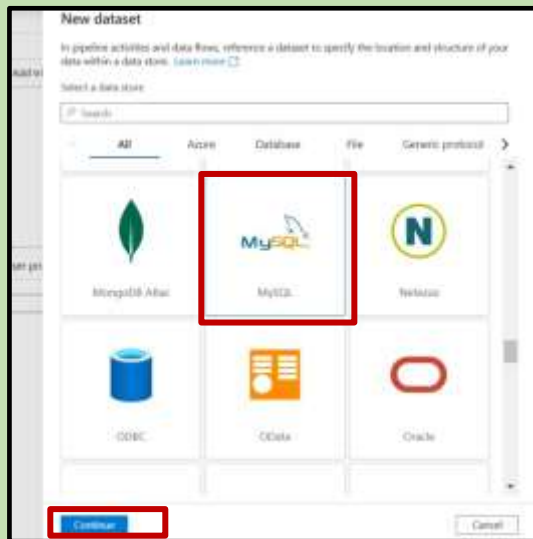
Activities → Copy → give name(mysqltocloudproduct_pl) → (give a name to copy activity)



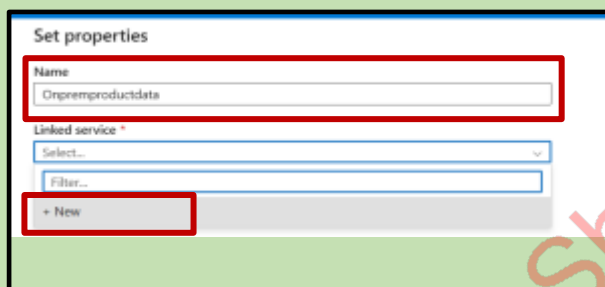
Click on source click on new



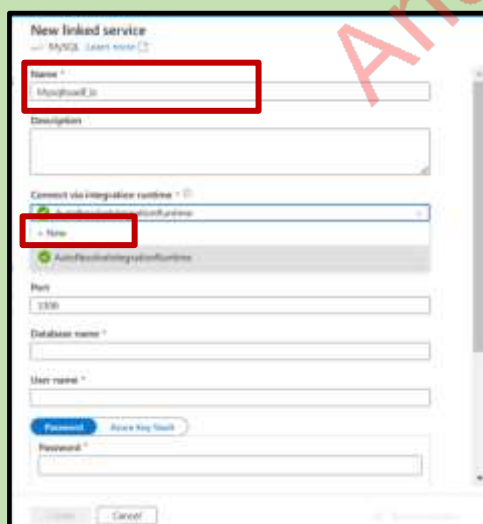
Click on mysql data base and click on continue.



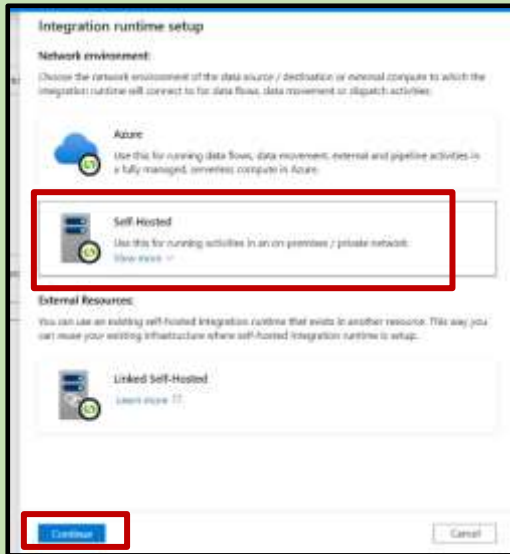
Give the name as onprempductdata and click new



Now give the linked service name and click on new for integration run time

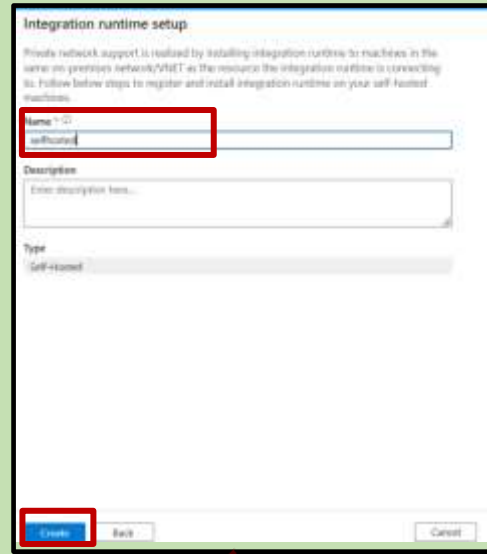


Click on selfhosted



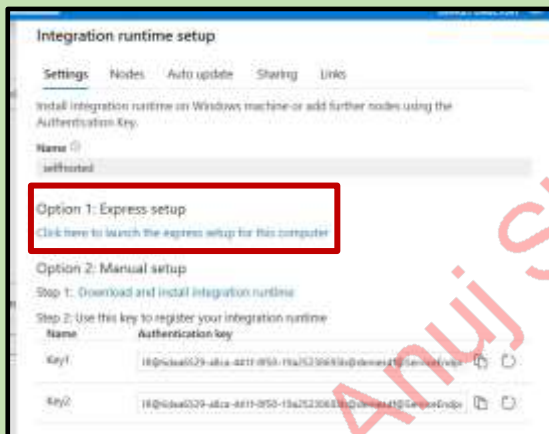
The screenshot shows the 'Integration runtime setup' window with the 'Network environment' tab selected. Under 'Choose the network environment of the data source / destination or external compute to which the integration runtime will connect to for data flows, data movement or dispatch activities', the 'Self-Hosted' option is highlighted with a red box. Below it, the 'External Resources' section is visible. At the bottom, the 'Continue' button is also highlighted with a red box.

Give the name as selfhosted click on create



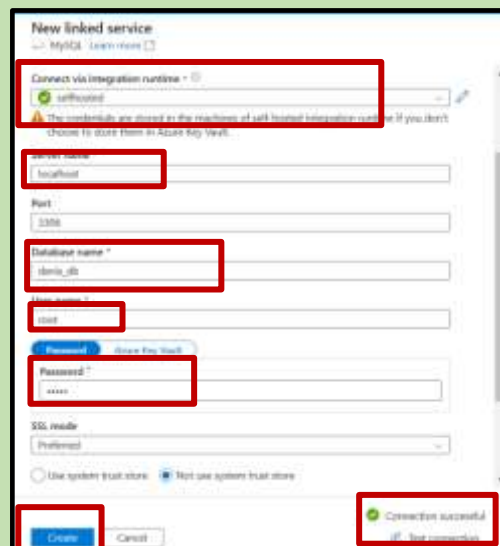
The screenshot shows the 'Integration runtime setup' window with the 'Name' field set to 'selfhosted' and the 'Type' set to 'Self-hosted'. The 'Create' button at the bottom is highlighted with a red box.

Click on express setup and download Integration Runtime



The screenshot shows the 'Integration runtime setup' window with the 'Express setup' option highlighted by a red box. The 'Authentication key' section is also visible, showing two keys.

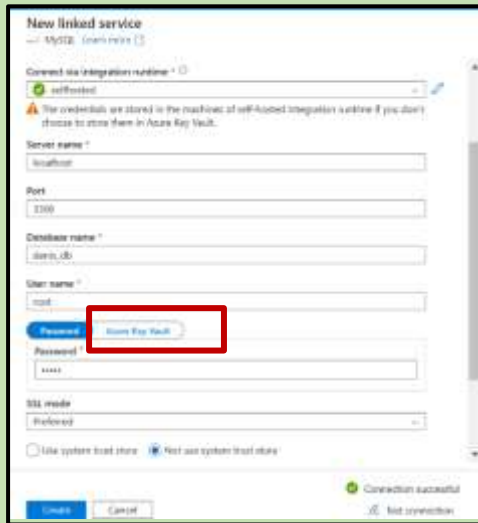
Now selfhosted is selected we need to give the server name database name username and password



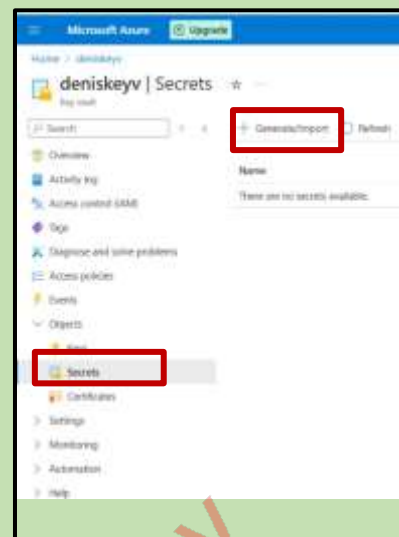
The screenshot shows the 'New linked service' window for MySQL. The 'Connect via integration runtime' dropdown is set to 'selfhosted'. The 'Server name' field is set to 'localhost', the 'Database name' is 'demo_db', the 'Username' is 'root', and the 'Password' is 'xxxx'. The 'SSL mode' is set to 'Preferred'. The 'Create' button at the bottom is highlighted with a red box. A 'Connection successful' message is also visible at the bottom right.

Method 2:

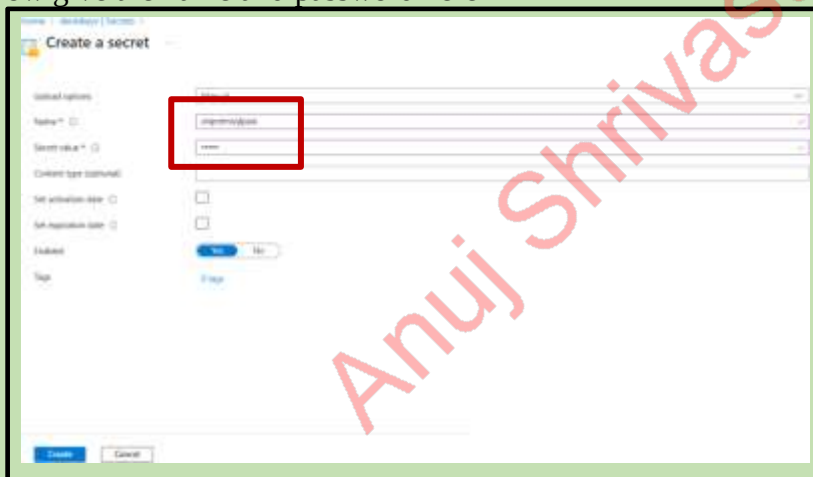
You need to select azure key vault



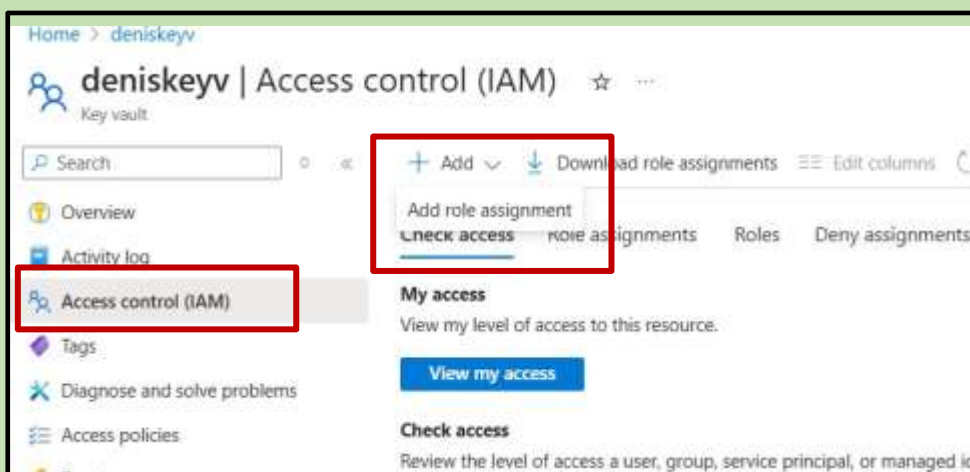
Open the keyvault select secret , generate



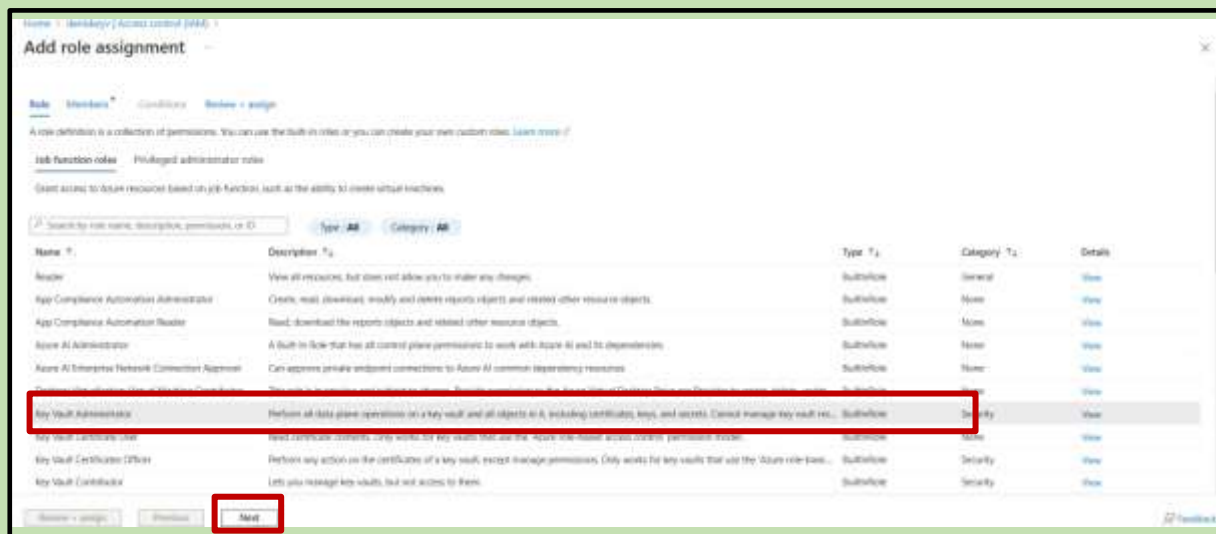
Now give the name and password here



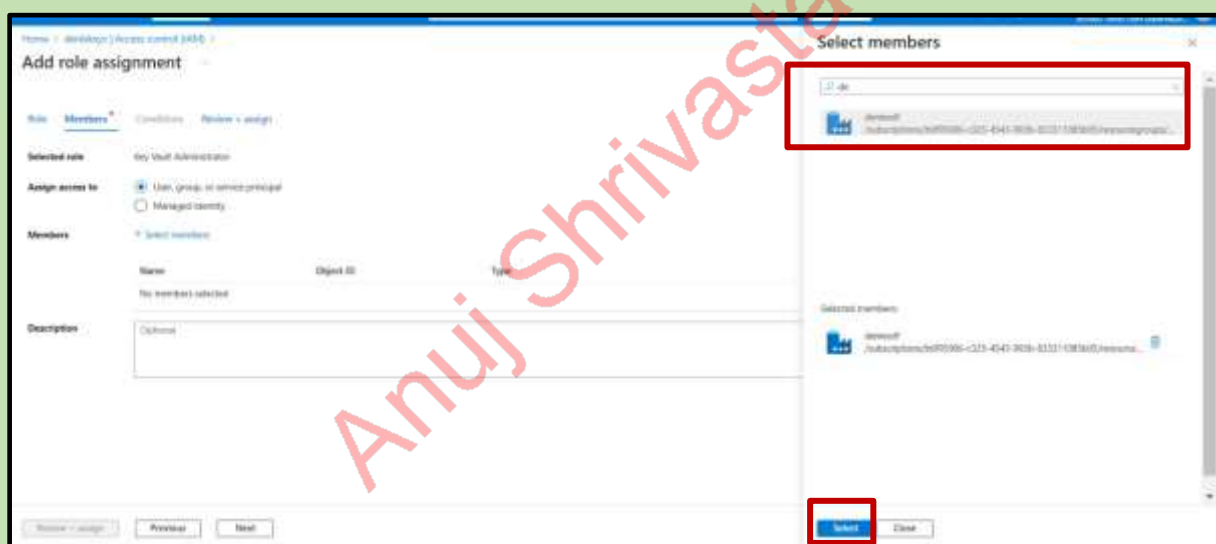
Click on access Control and add role assignment



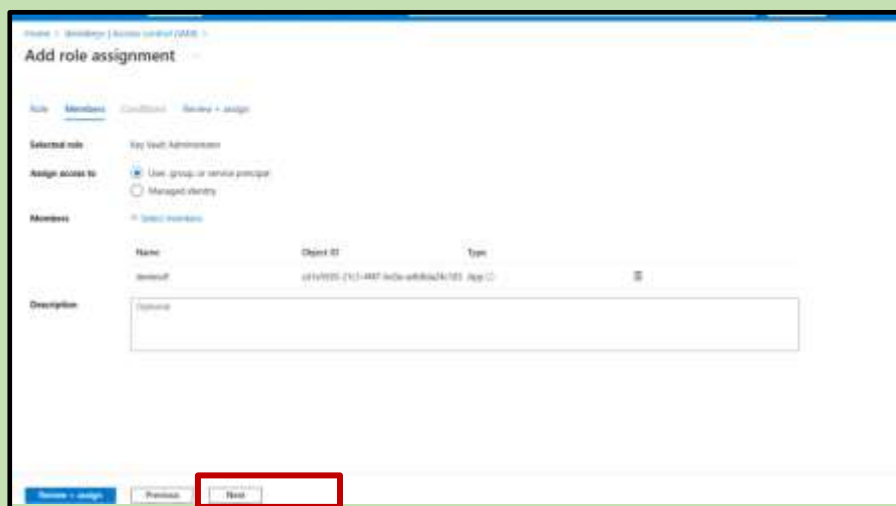
Add key vault administration and click on next



Add azure data factory click on select



Click next



Click on review and assign

The screenshot shows the 'Review + assign' tab of the Azure Key Vault access policy configuration. The 'Review + assign' button is highlighted with a red box. The 'Previous' and 'Next' buttons are also visible.

Access policies → create

The screenshot shows the 'Access policies' page in the Azure Key Vault console. The 'Create' button is highlighted with a red box. The 'Access policies' link in the left sidebar is also highlighted with a red box.

Click on below permissions click next

The screenshot shows the 'Create an access policy' page in the Azure Key Vault console. The 'Next' button is highlighted with a red box. The 'Key permissions', 'Secret permissions', and 'Certificate permissions' sections are also highlighted with red boxes.

Give a\Azure data factory click next

Click next

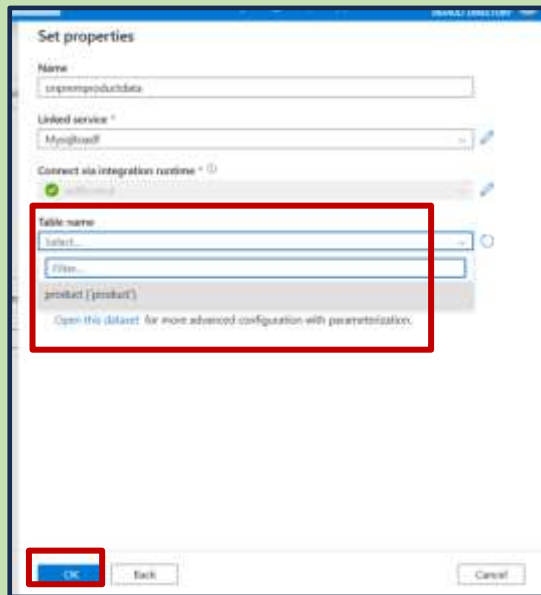
Create→

New

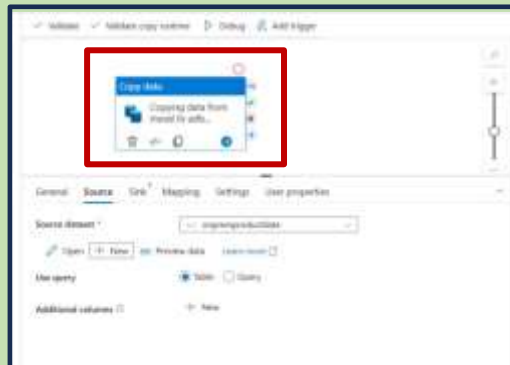
Give the details

create

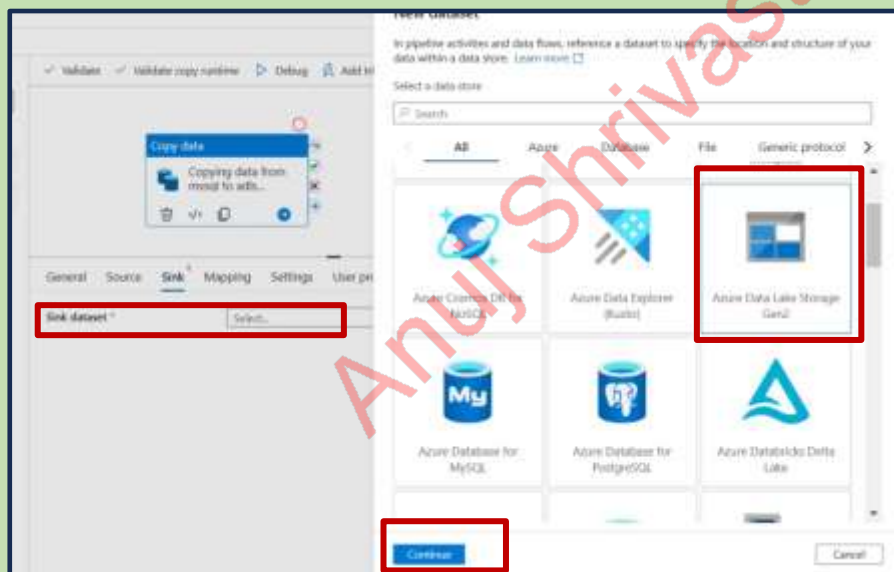
Select → Table



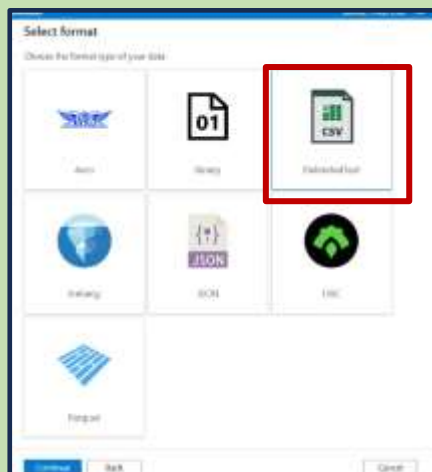
Give the name of source



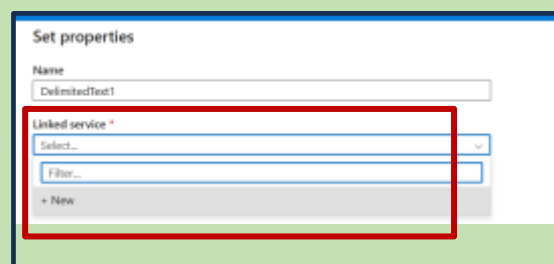
Now select the sink data set



Select the csv format



Select the new linked service



Give details → create linked service

New linked service

Azure Data Lake Storage (Gen2) Learn more

Name *

adftoadls_ls

Description

Connect via integration runtime *

AutoResolveIntegrationRuntime

Authentication type

Account key

Account selection method *

From Azure subscription Enter manually

Azure subscription *

Free Trial (b6f95906-c325-4543-993b-833311085b05)

Storage account name *

denissadls

Test connection *

To linked service To file path

Connection successful

Test connection

Create Cancel

Select the file name

Set properties

Name

DelimitedText1

Linked service *

adftoadls_ls

File path

File system / Directory / File name

First row as header

Import schema

From connection/store From sample file None

Advanced

select bronze

Browse

Select a file or folder.

Root folder > denisproject

Bronze

Gold

Silver

Now click on ok

Set properties

Name

DelimitedText1

Linked service *

adftoadls_ls

File path

denisproject / Bronze / File name

First row as header

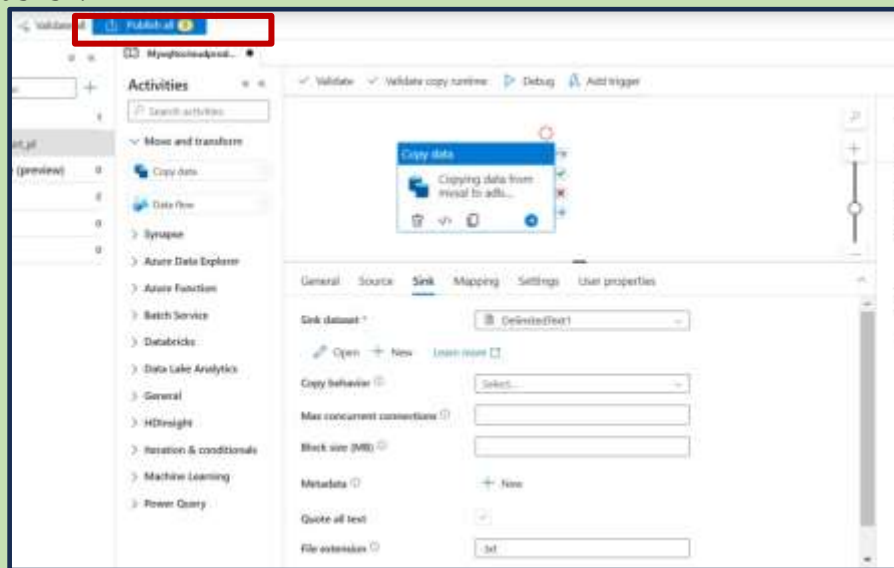
Import schema

From connection/store From sample file None

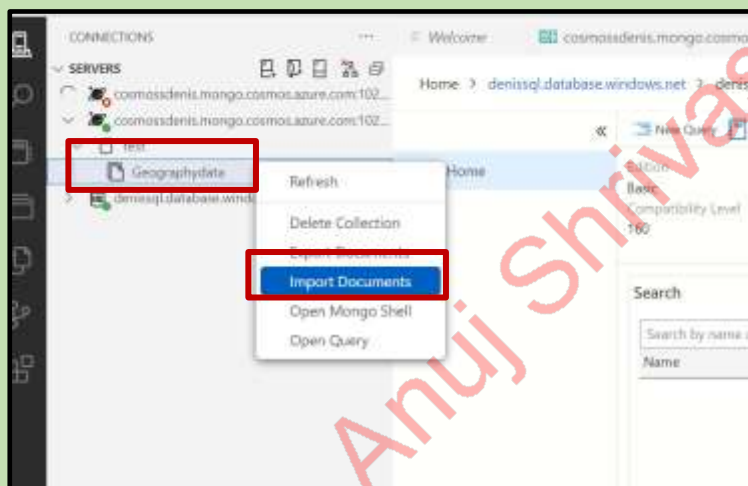
Advanced

Ok Back Cancel

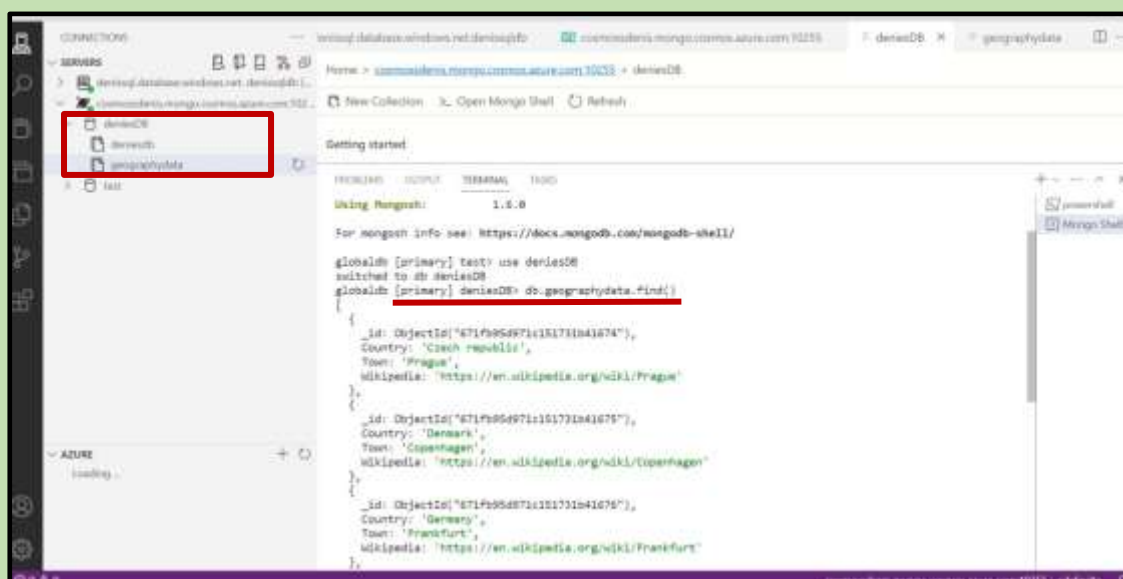
Publish:



Now open cosmos db create geography data and import the Json file



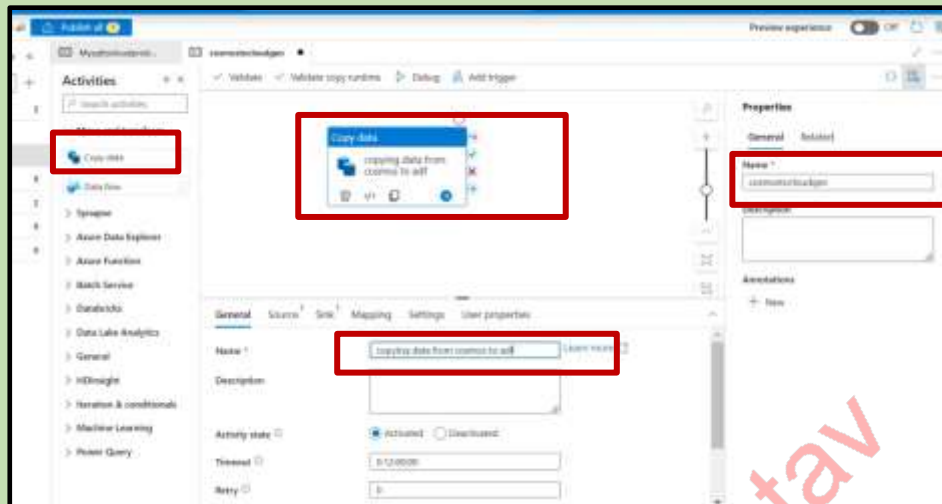
Now we can see the data



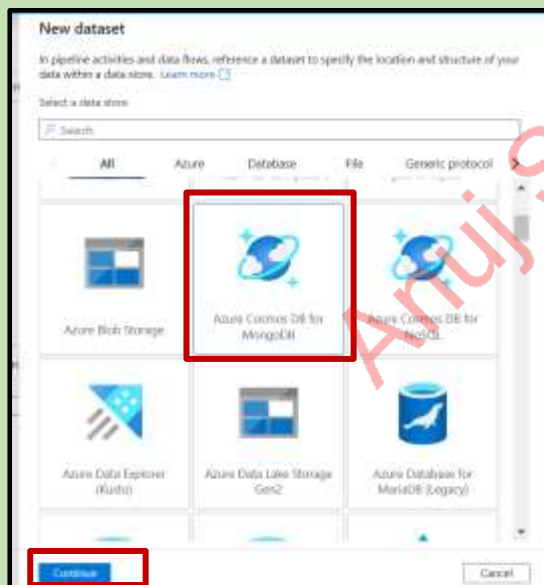
Pipeline 2:

Cosmos: Azure data lake storage

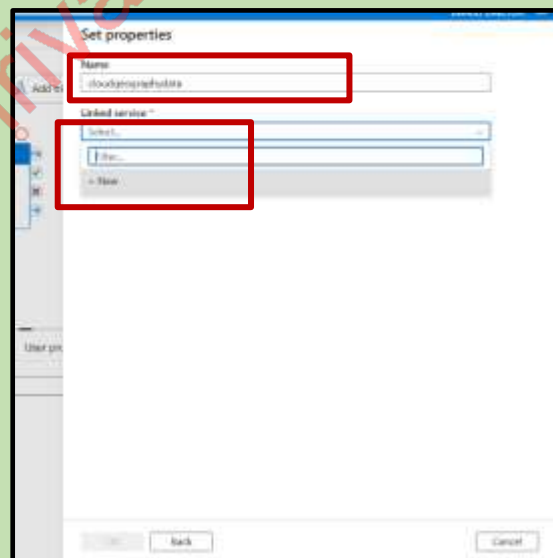
Create one new pipeline → give name



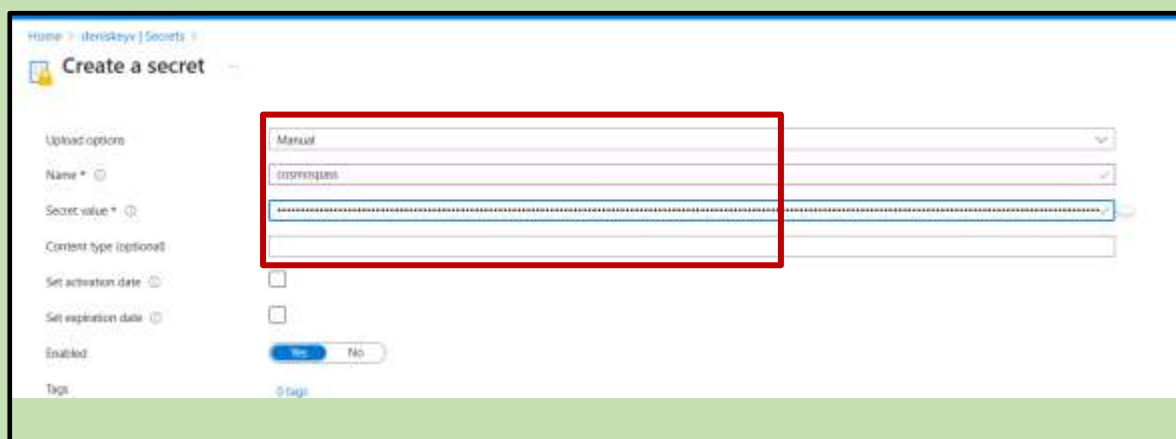
Now select the data set



give the name and link service



Give the details primary password



Provide details:

New linked service
Azure Cosmos DB for MongoDB [Learn more](#)

Name *
CosmosDbMongoDb1

Description

Connect via integration runtime *
AutoResolveIntegrationRuntime

Connection string [Azure Key Vault](#)

Server version ⓘ
☐ Above 3.2 ☒ 3.2

AKV linked service * ⓘ
AzureKeyVaulttoadff

Secret name * ⓘ
cosmospass
[Edit](#)

Secret version ⓘ
8e02043ca4554892a66ee1d834bdd858 (Current version)
[Edit](#)

[Create](#) [Cancel](#)

Connection successful
[Test connection](#)

Select the collection name

Set properties

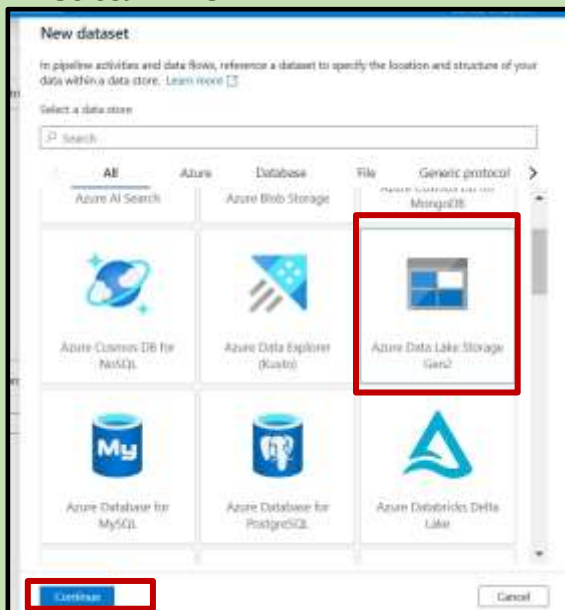
Name
cloudgeographydata

Linked service *
CosmosDbMongoDb1

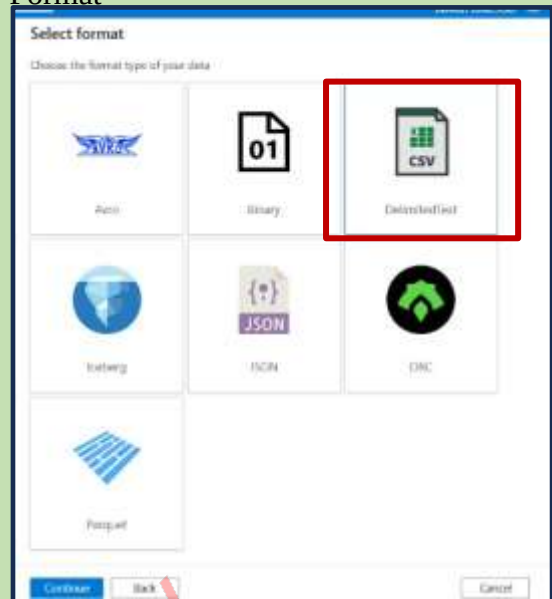
Collection name
deniesdb
☐ Enter manually
[Advanced](#)

[OK](#) [Back](#) [Cancel](#)

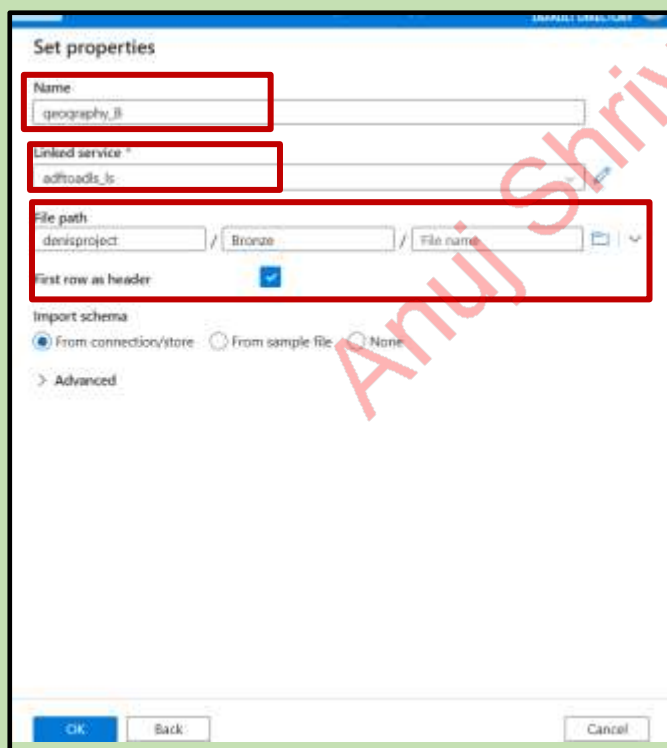
Select ADLS



Format



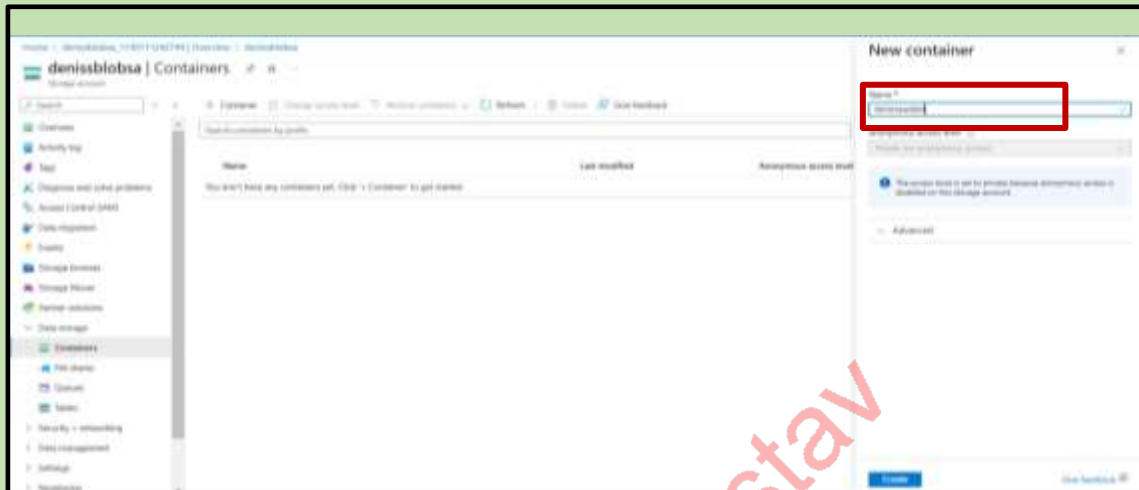
Select the file folder and publish



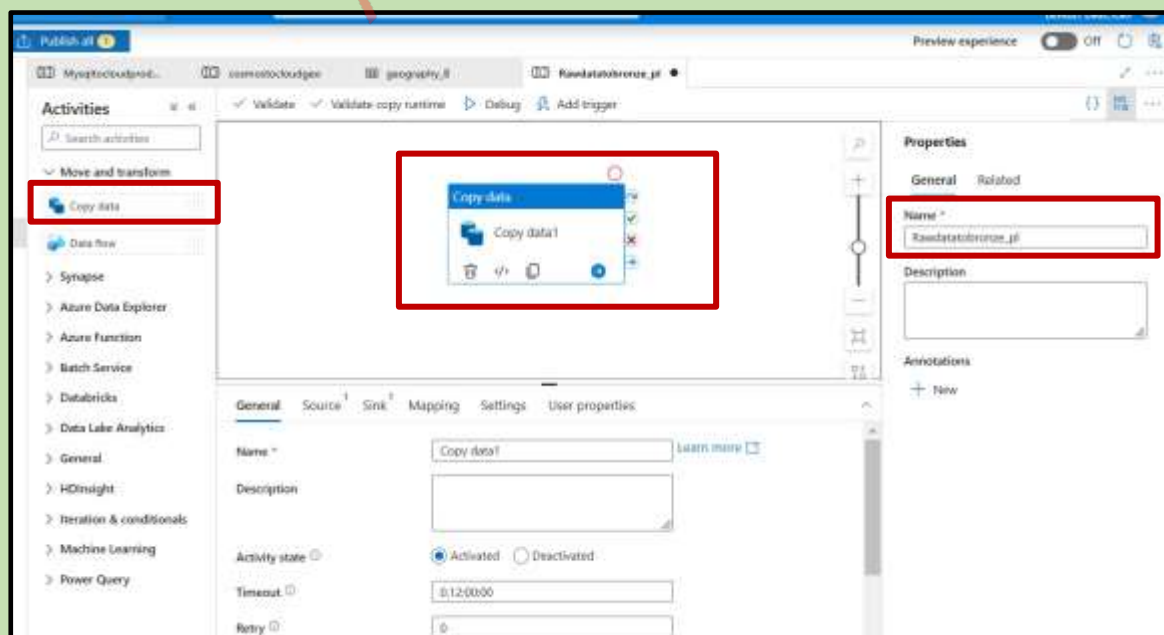
Pipeline 3:

Azure blob storage → Azure data lake storage → Bronze

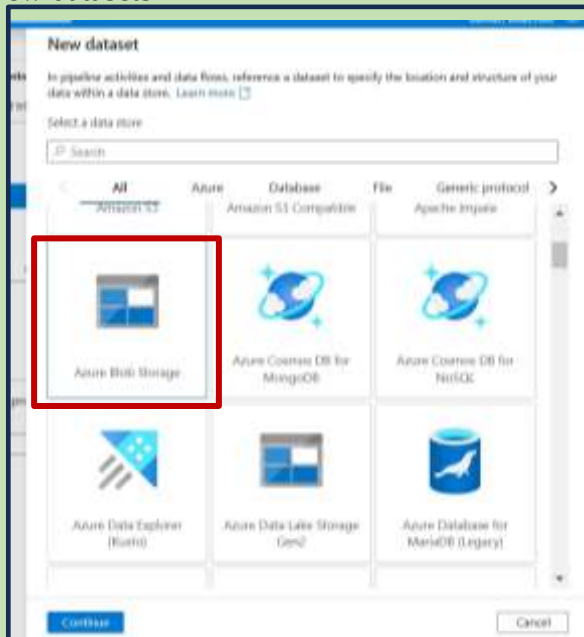
Open Blob storage create container and upload the raw files



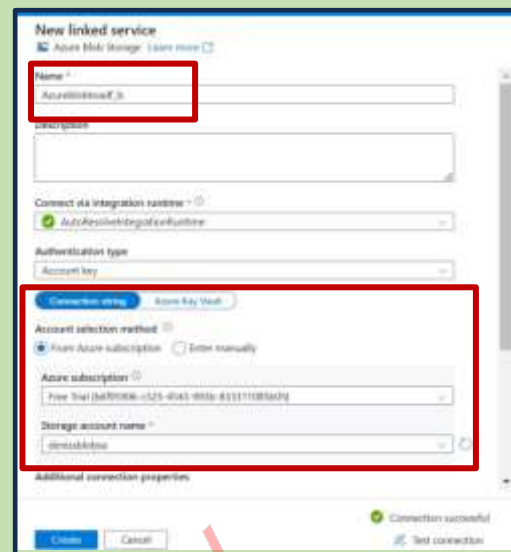
Give the pipeline name



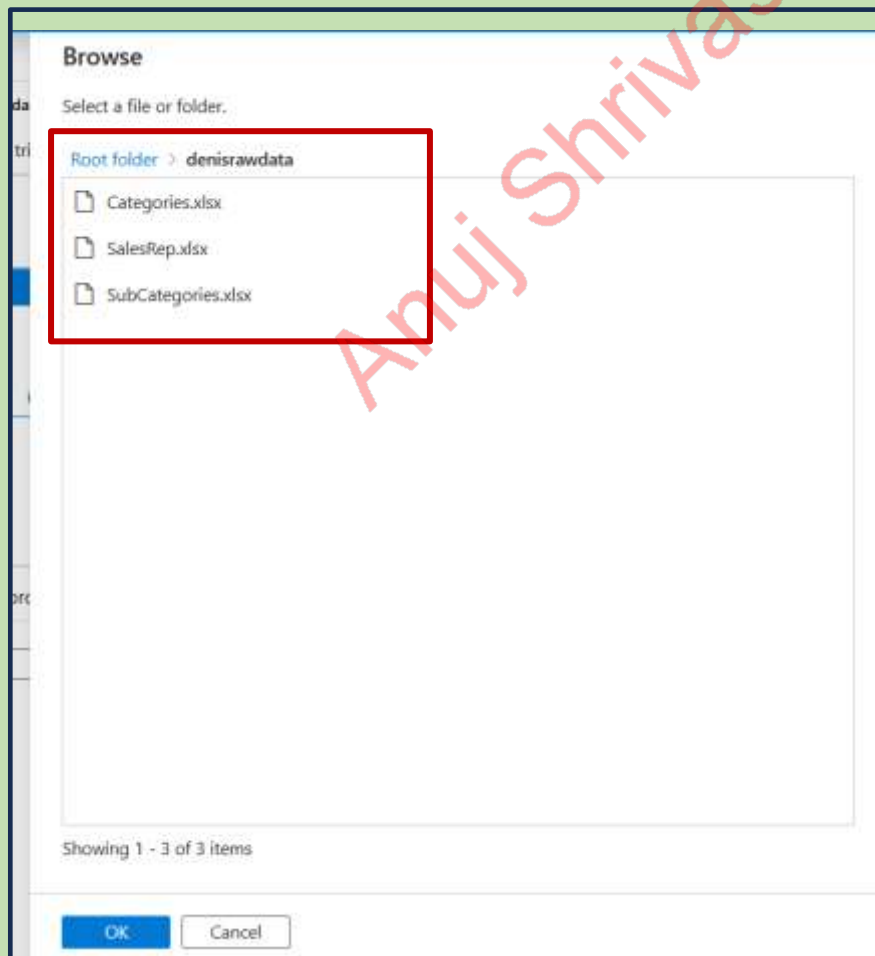
New datasets



Give details



Select the source data



Name → linked service

Set properties

Name: Rawfiles

Linked service: Azureblobadfs

File path: denisrawdata / Directory: / File name:

First row as header: ☒

Import schema: ☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK Back Cancel

Source is done

Activities: Search activities

Move and transform: Copy data, Data flow

Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, Data Lake Analytics, General, HDInsight, Iteration & conditionals, Machine Learning, Power Query

Copy data: copying data from blob to adls

General Source Sink Mapping Settings User properties

Source dataset: Rawfiles

File path type: ☐ File path in dataset ☐ Prefix ☒ Wildcard file path ☐ List of files

Wildcard paths: denisrawdata / Wildcard folder path

Filter by last modified: ☐

Start time (UTC): End time (UTC):

OK Back Cancel

Set properties

Name
Rawfiles

Linked service *
Azureblobload1.js

File path
denisrawdata / Directory / File name

First row as header ☒

Import schema
☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK Back Cancel

Source is done

Activities

Search activities

Move and transform

Copy data

Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

Data Lake Analytics

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Validate Validate copy runtime Debug Add trigger

Copy data
copying data from blob to adls

General Source Sink Mapping Settings User properties

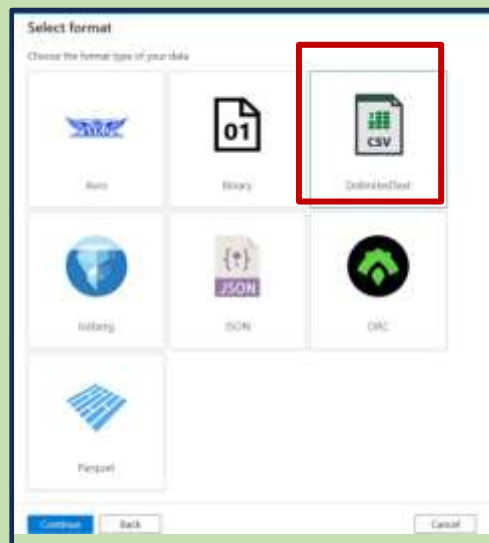
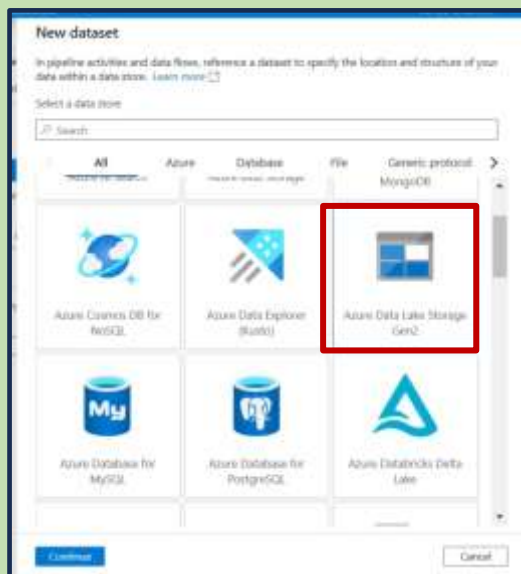
Source dataset *
Rawfiles

File path type
☐ File path in dataset ☐ Prefix ☒ Wildcard file path ☐ List of files

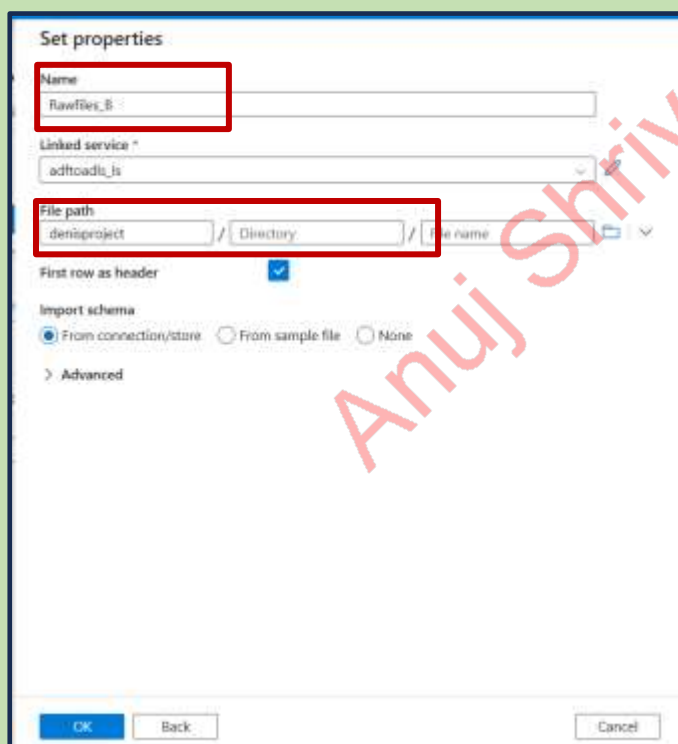
Wildcard paths
denisrawdata / Wildcard folder path / *

Filter by last modified

Start time (UTC) End time (UTC)



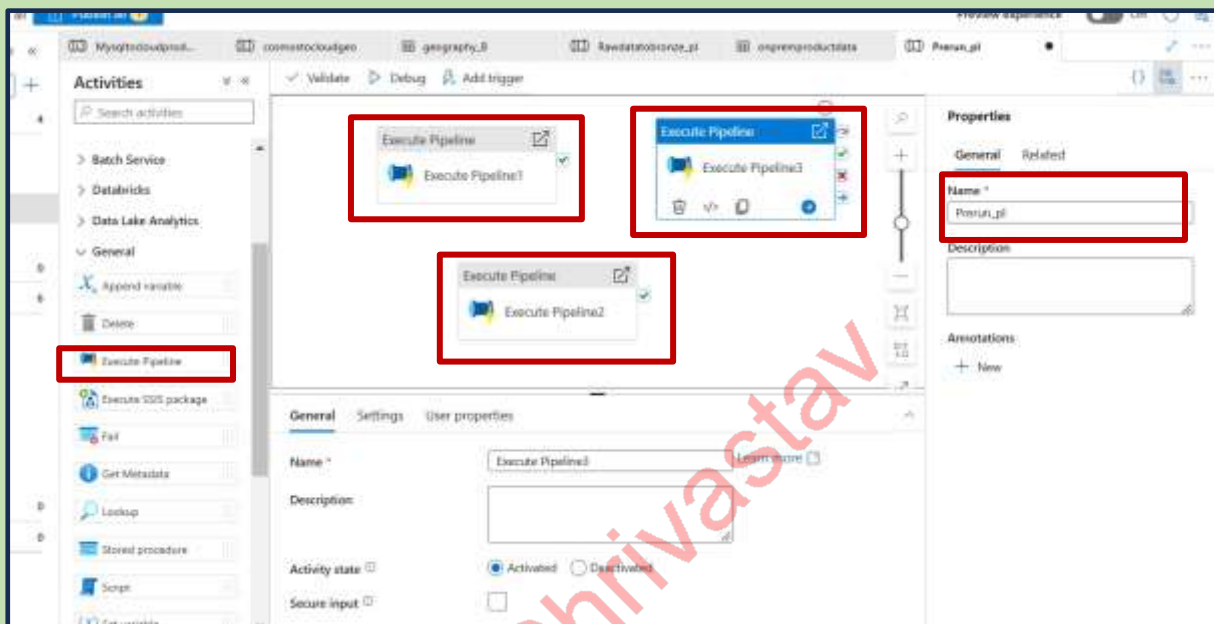
Give the name and select the file path and linked service



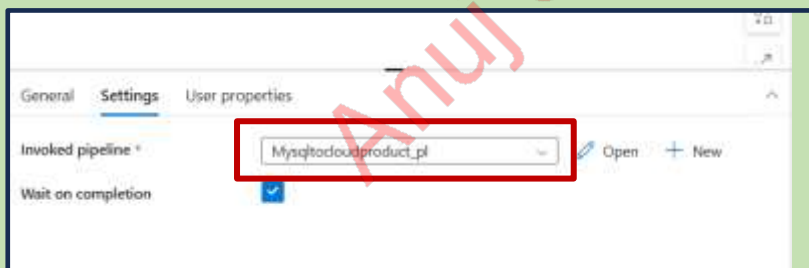
Pipeline 4:

Azure blob storage Prerunpipeline → Prerunpipeline → Mysqldcloud →
cosmostocloud → rawdatatobronze

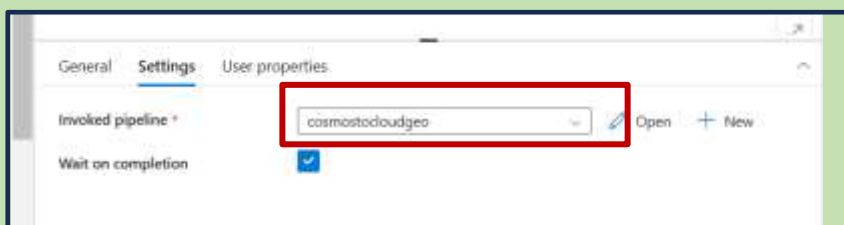
Now we need to run all the 3 pipelines together for that we need to create new pipeline Need to move execute pipeline and give the name of the pipeline



Select the 1st pipeline



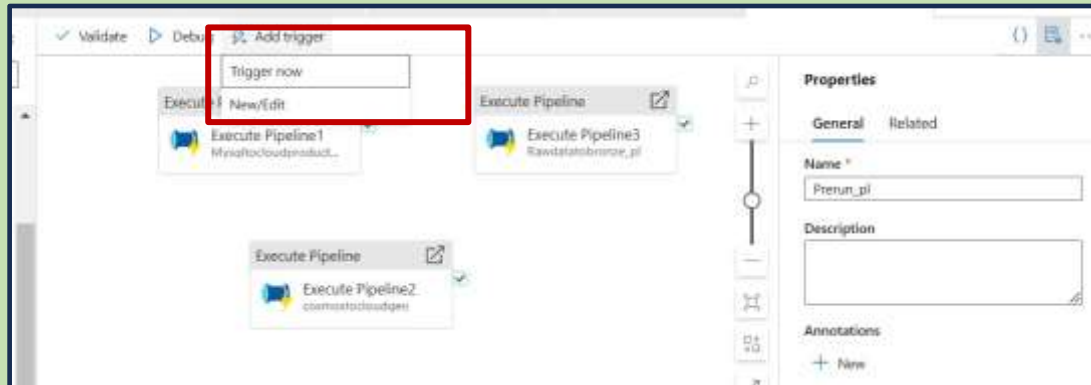
Select the 2nd pipeline



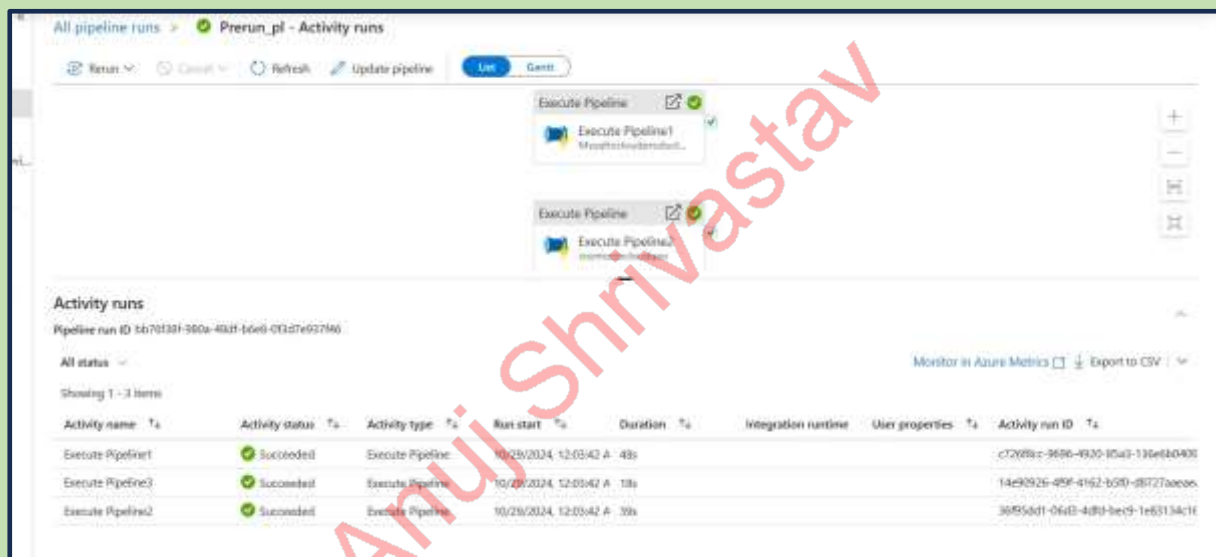
select the 3rd pipeline



Publish → trigger pipelines

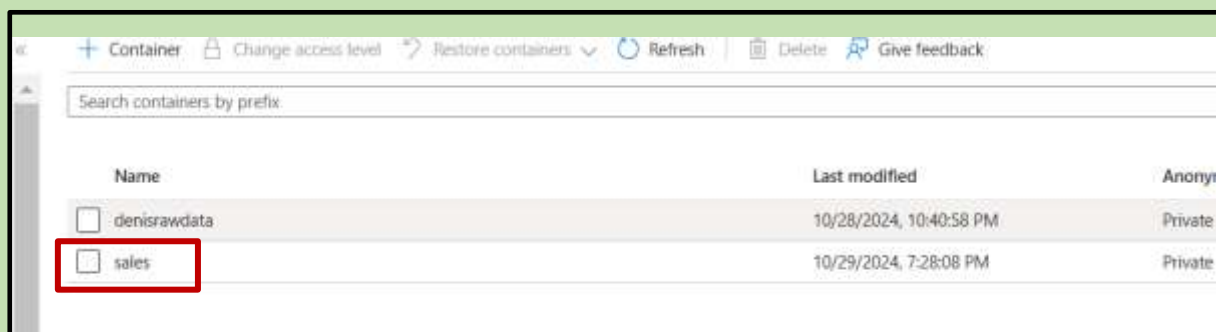


Pipeline runs successfully

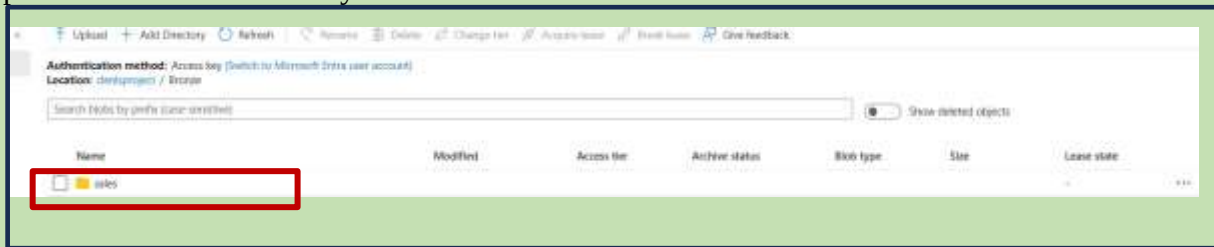


Pipeline 5:

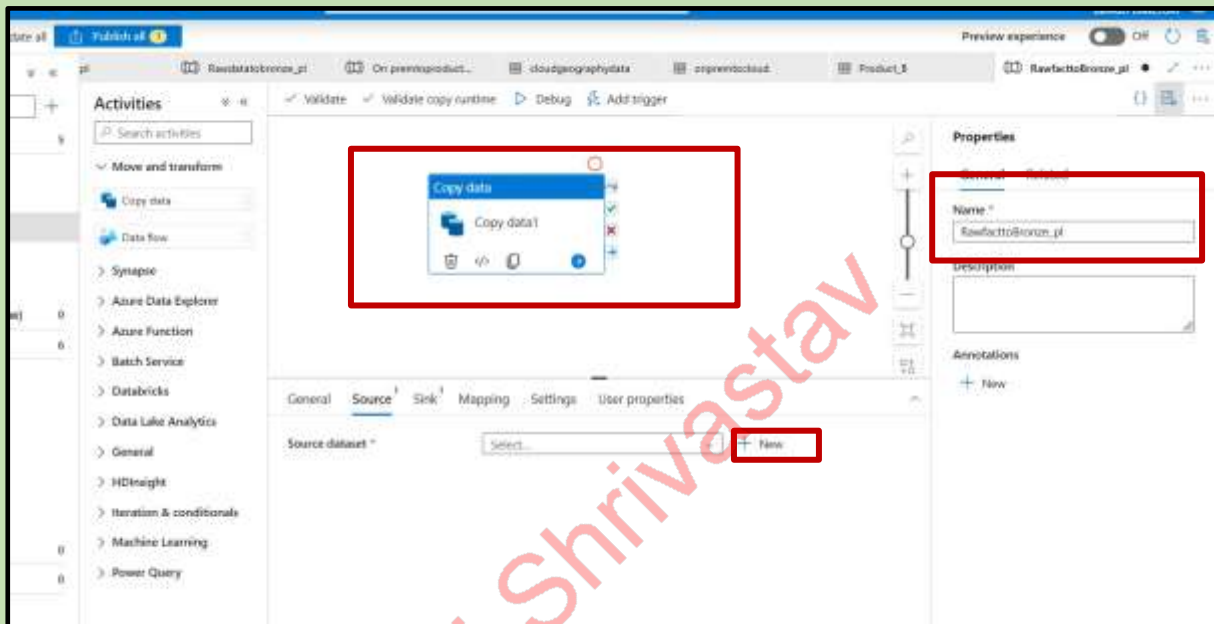
Now we need to dump the sales data in azure blob storage



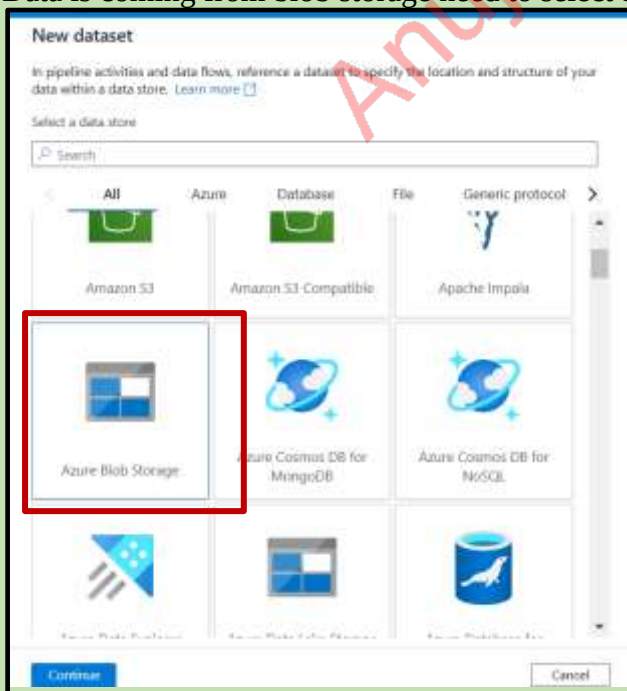
Open ads → create directory → sales



Now we need to create one more pipeline give the name of the pipeline and click on new



Data is coming from blob storage need to select blob and click on continue



Click→ format

select →ls→one file name

Select format
Choose the format type of your data

Avro Binary DelimitedText
Excel JSON ODBC
Parquet XML

Continue Back Cancel

Set properties

Name: RawFactfiles

Linked service: Azureblobtoadlf_ls

File path: demo\rawdata / Directory / Sales 2014.csv

First row as header: ☒

Import schema: ☒ From connection/store ☐ From sample file ☐ None

Advanced

OK Back Cancel

Name ls→ file path

Set properties

Name: RawFactfiles

Linked service: Azureblobtoadlf_ls

File path: sales / Directory / File name

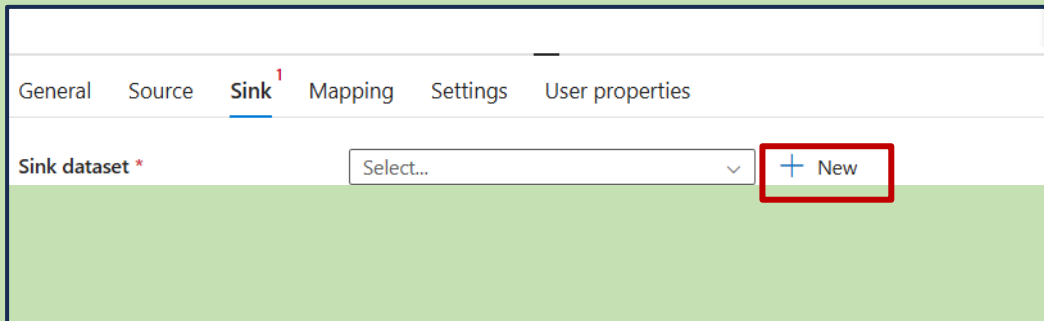
First row as header: ☒

Import schema: ☒ From connection/store ☐ From sample file ☐ None

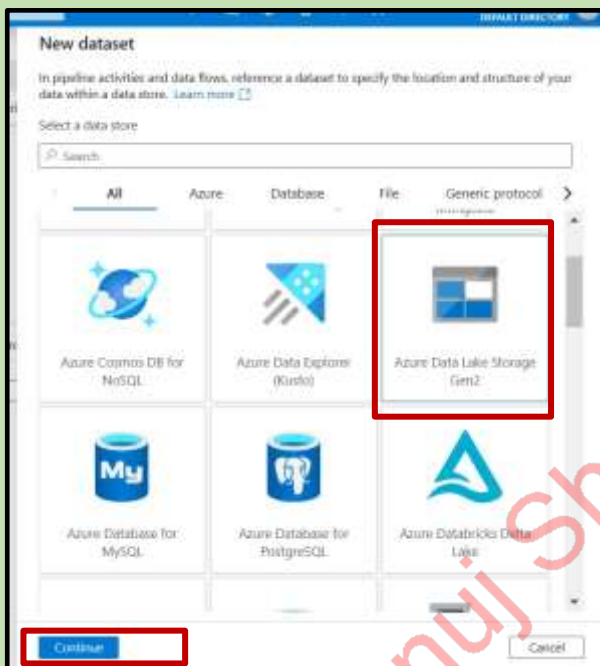
Advanced

OK Back Cancel

Sink → new



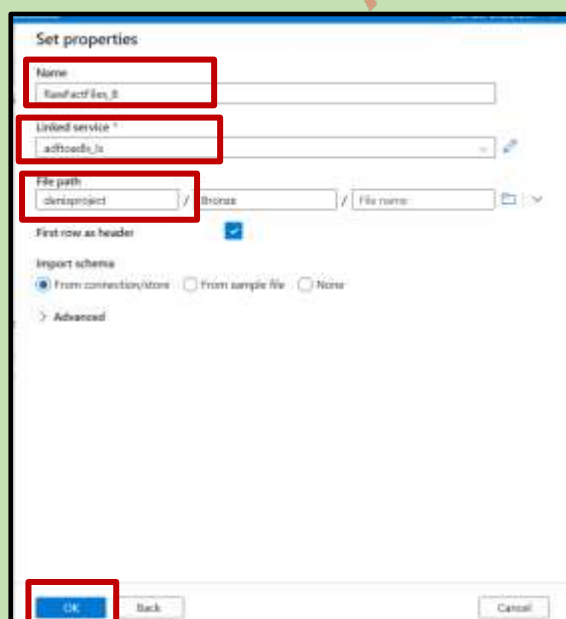
Click on new data set



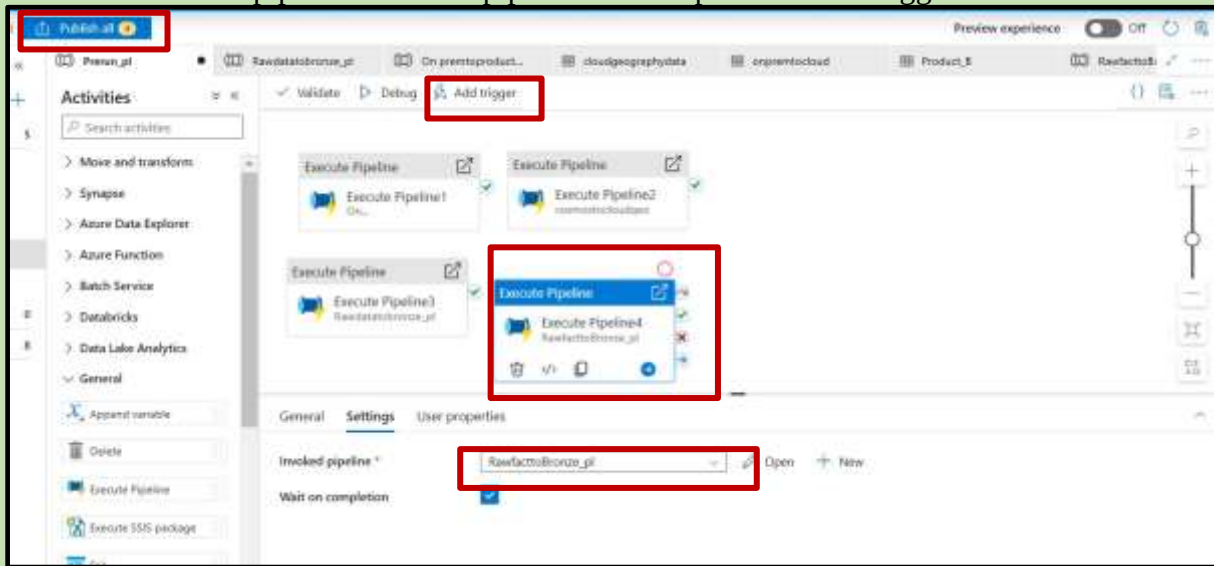
Select the format



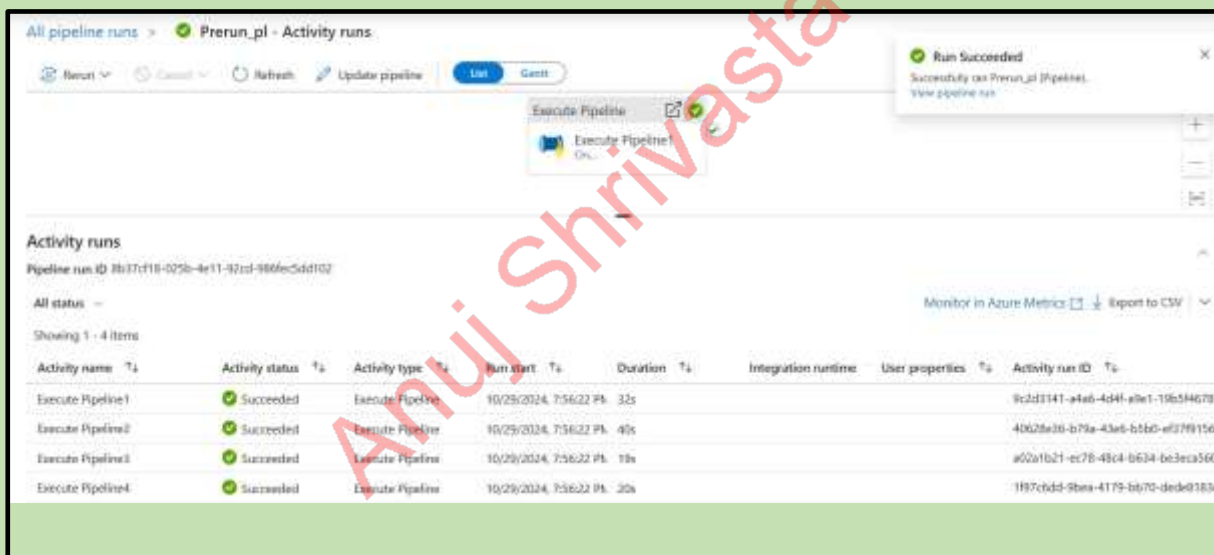
Now we can give the name ls and filepath



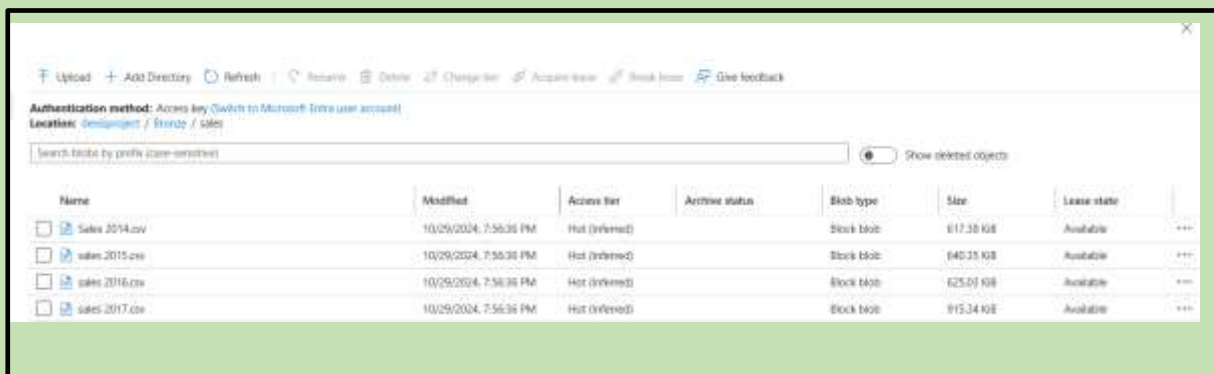
General → execution pipeline → invoke pipeline name → publish → run trigger



Pipelines got run successfully



As we can see the data moved to adls--->bronze--->Sales



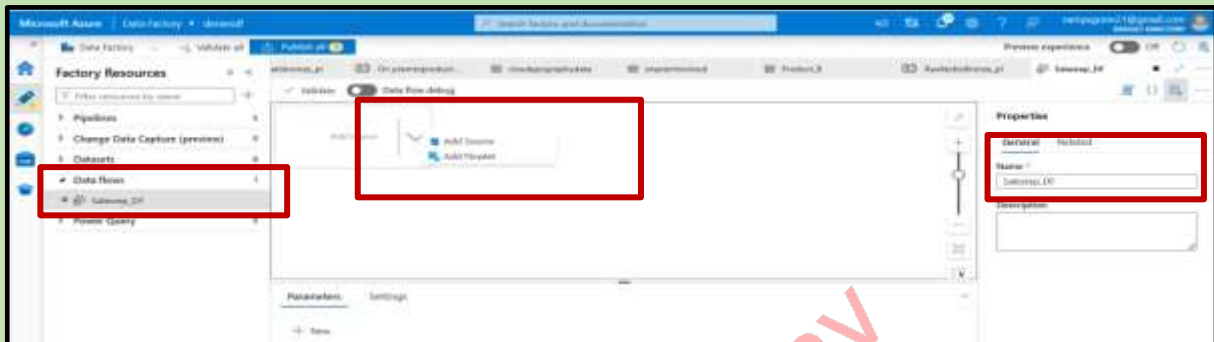
DATA TRANSFORMATION:

Data Flow 1:

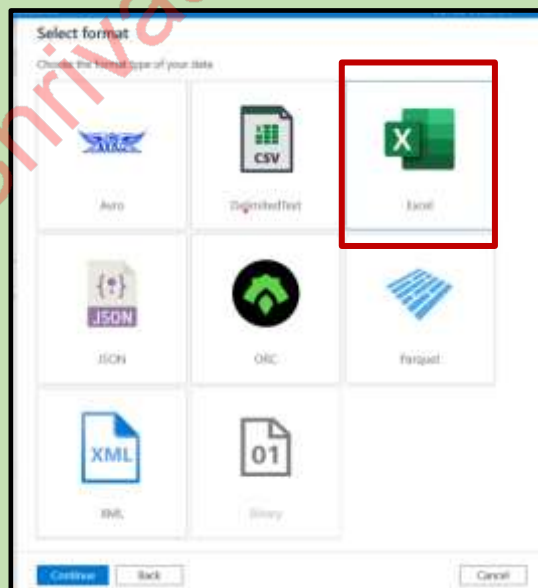
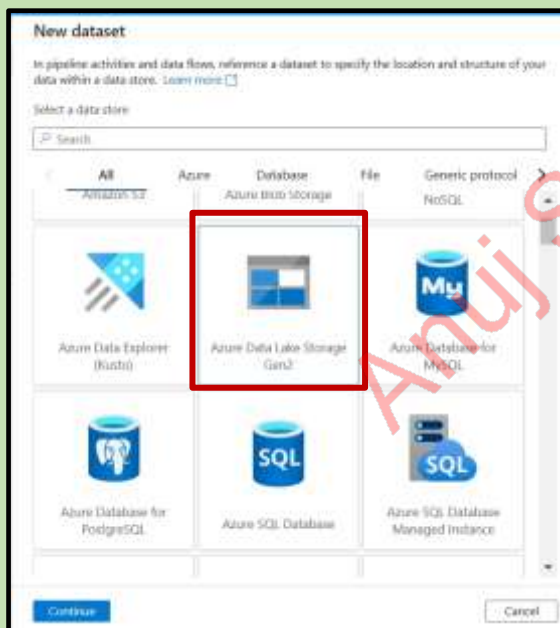
- Salesrepdataflow

Click on Data flows --->New data flow --->Addsource

Give name



Datasets details



Set properties

Name: SalesRep_ID

Linked service: adfcloudfs_ls

File path: denaliproject / Bronze / SalesRepData

Worksheet mode:
☒ Name ☐ Index

Sheet name: Sales rep

First row as header: ☒

Import schema:
☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK Back Cancel

Data preview

Data preview

Id	SalesRepID	Sales Rep Name
1	10-0	San Francisco
2	10-1	John Wiley
3	10-2	Ben-Brook
4	10-3	Mark-Spencer
5	10-4	Edo-Edo
6	10-5	Edo-Edo
7	10-6	Edo-Edo
8	10-7	Edo-Edo
9	10-8	Edo-Edo
10	10-9	Edo-Edo

Requirement: remove 'ID

Derive column → give expression

Derived column settings

Output stream name: derivedColumn1

Description: Creating/updating the column: SalesRepID, Sales Rep Name, SalesRepID

Input stream: source1

Columns:

Column	Expression
SalesRepID	regexExtract(SalesRepID, '10-0+')

Derived column's settings Optimize Inspect Data preview			
Number of rows INSERT 7 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL			
Refresh Typecast Modify Map drifted Statistics Remove Export to CSV			
SalesRepID	Sales Rep Name	ISalesRepID	
ID - 6	Jan Novotny	6	
ID - 7	John White	7	
ID - 5	Ellen Woody	5	
ID - 3	Mark Spancer	3	
ID - 1	Ellie Gill	1	
ID - 2	Bill Muray	2	
ID - 4	El Bob	4	

Now click on plus and add select and delete sales rep

The screenshot shows the configuration for a 'select' node. The 'Input columns' section is highlighted with a red box, showing three mappings: 'derivedColumn1's column' to 'SalesRepID', 'Sales Rep Name' to 'Sales Rep Name', and 'ISalesRepID' to 'ISalesRepID'. The 'Output stream name' is 'select1'.

The screenshot shows the configuration for a 'select' node. The 'Input columns' section shows two mappings: 'SalesRepID' to 'SalesRepID' and 'Sales Rep Name' to 'Sales Rep Name'. The 'Output stream name' is 'select1'.

After clean→

Validate Data flow debug Debug Settings

Select settings Optimize Inspect **Data preview** Previous Next

Number of rows **INSERT 7** **UPDATE 0** **DELETE 0** **UPSERT 0** **LOOKUP 0** **ERROR 0** **TOTAL 7**

Refresh Typecast Change the type of a column Statistics Remove Export to CSV

↑↓ SalesRepID	↑↓ Sales Rep Name
6	Jan Novotny
7	John White
5	Ellen Woody
3	Mark Spancer
1	Ellie Gill
2	Bill Murray
4	El Bob

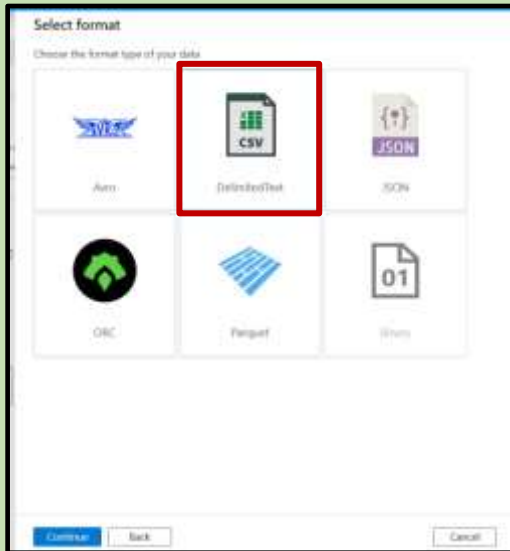
Select sink

The screenshot displays the Databricks workspace interface. At the top, there are tabs for 'Product_0', 'cloudgeographydata', 'openrowcloud', 'Product_0', 'Rawfact1', and 'Rawfact2'. Below the tabs, there are buttons for 'Validate', 'Data flow debug', and 'Debug Settings'. The main area shows a data pipeline with three stages: 'source1' (Import data from Salesreg_0), 'derivedColumn1' (Creating/Updating the column: SalesRepID, Sales Rep Name, SalesRepID), and 'select1' (Columns: 2 total). Below the pipeline, there is a 'Data preview' section showing a table with columns 'SalesRepID' and 'Sales Rep Name'. The table contains six rows of data. On the right side, there is a sidebar with various tools and functions, including 'Cast', 'Flatten', 'Parse', 'Bringify', 'Flow modifier', 'Filter', 'Sort', 'Alter flow', 'Account', 'Flatten', 'Flowlet', and 'Sink'. The 'Sink' option is highlighted with a red box.

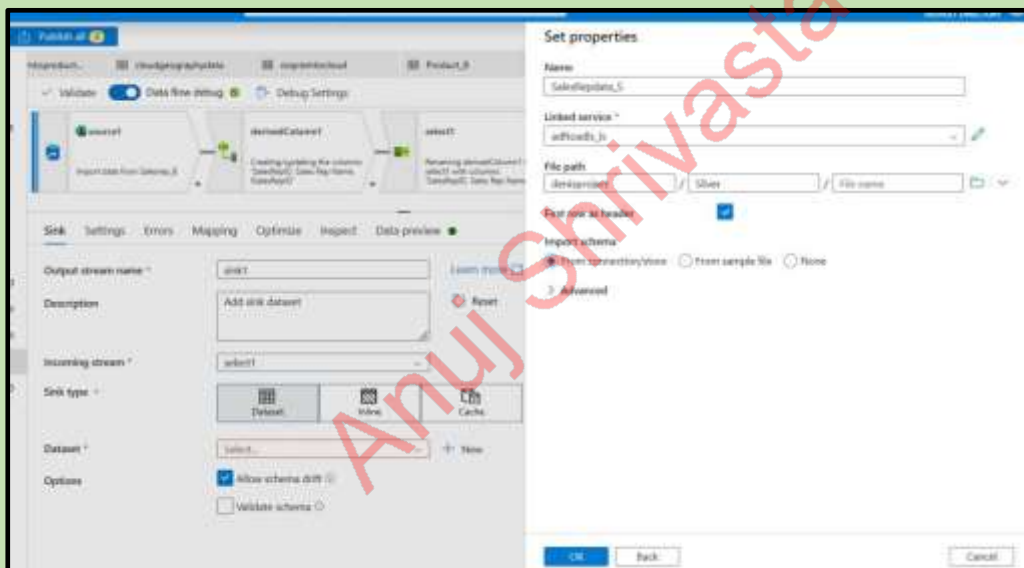
SalesRepID	Sales Rep Name
6	Alan Nowinski
7	John White
5	Ellen Woods
1	Mark Spencer
1	Ellie Gill
2	Bill Muter
4	El Bob

[illegible]

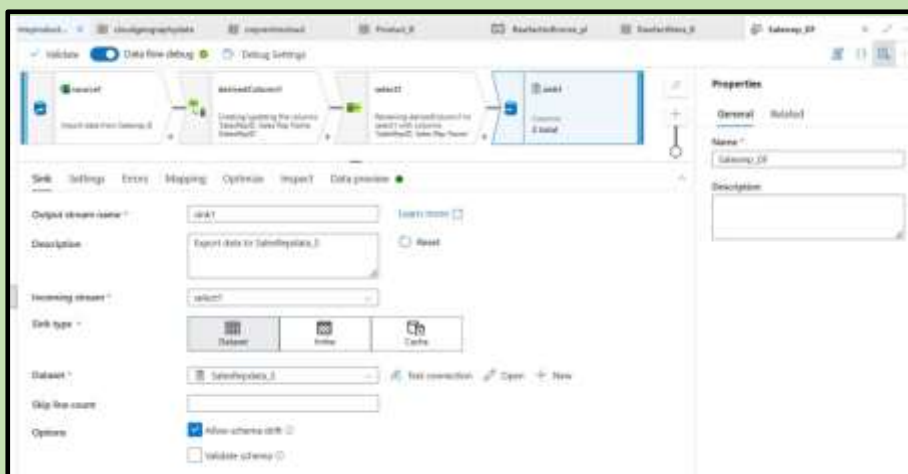
Select the format:



Set the properties



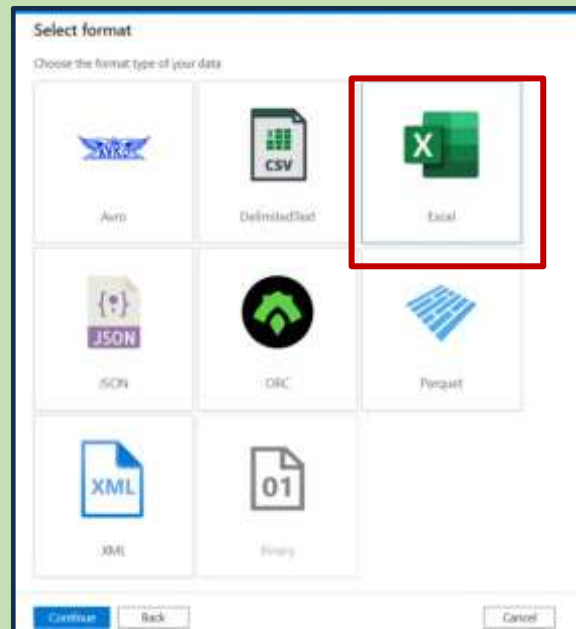
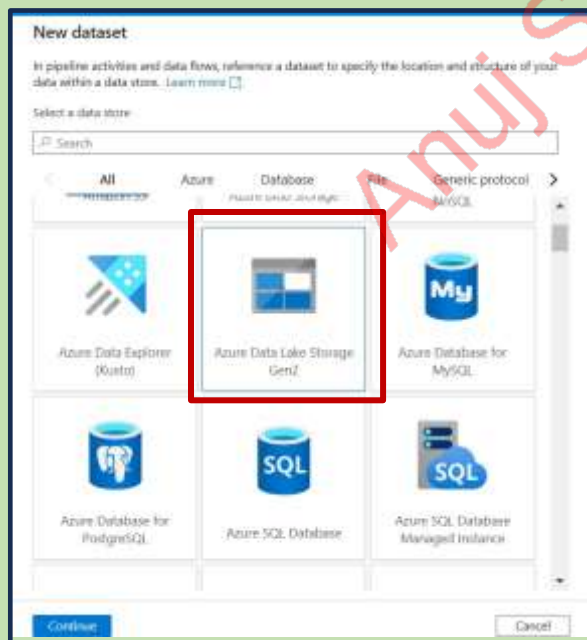
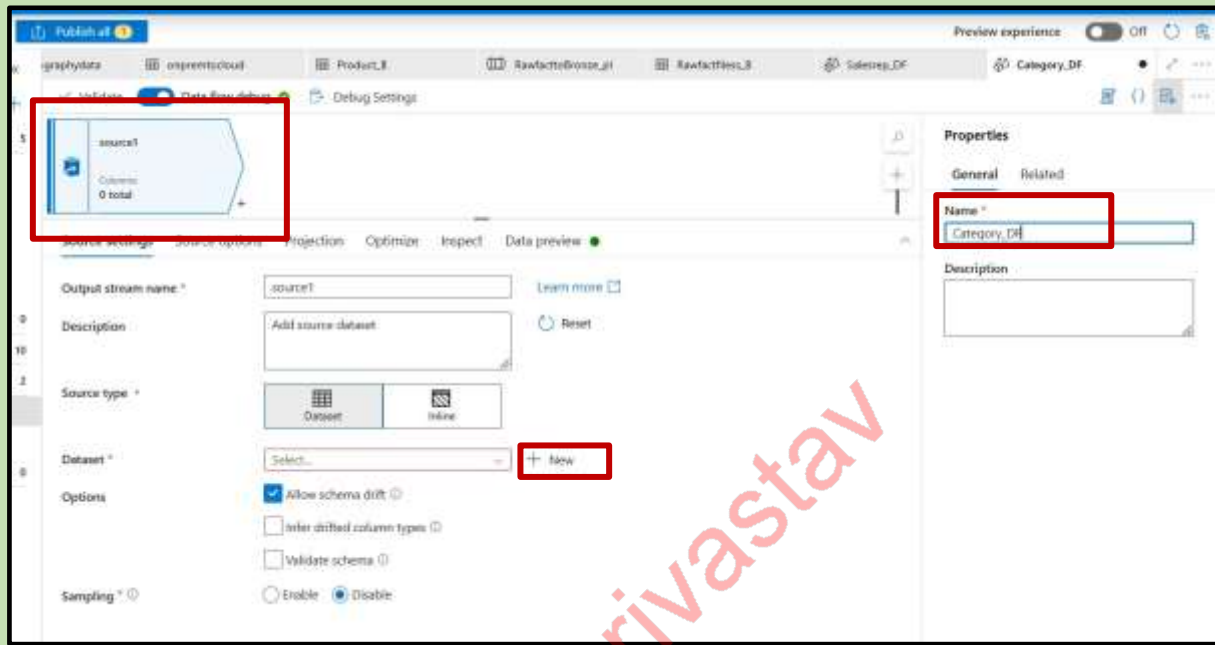
Data flow is created



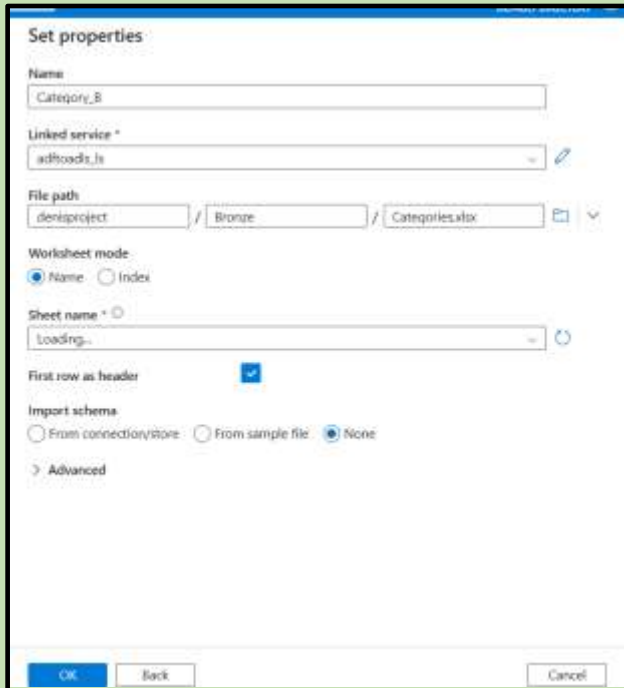
Data Flow2 :

- Categorydataflow

Create new data flow give the pipeline name and select new data set



Select properties:



Set properties

Name: Category_B

Linked service *: adhoods_ls

File path: denisproject / Bronze / Categories.xlsx

Worksheet mode: ☒ Name ☐ Index

Sheet name *: Loading...

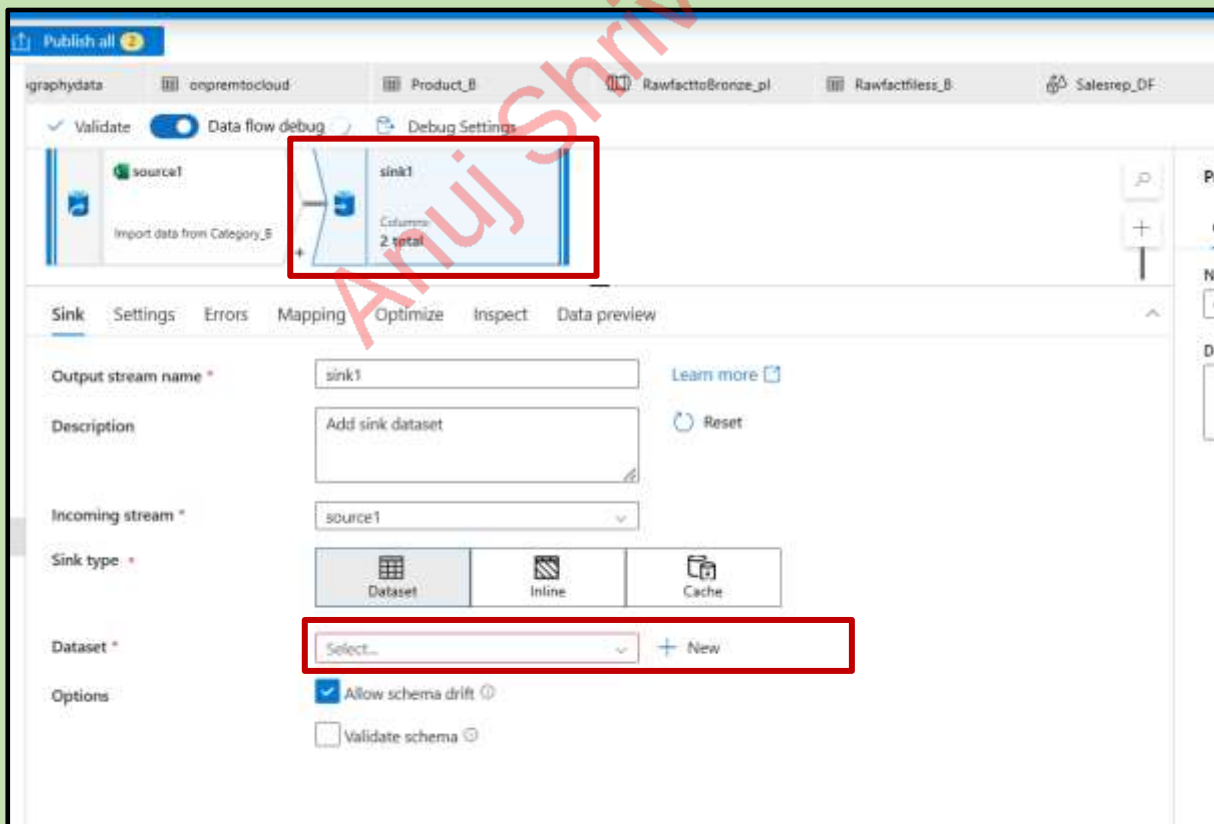
First row as header: ☒

Import schema: ☐ From connection/store ☐ From sample file ☒ None

> Advanced

OK Back Cancel

Now select sink because there is no need to clean data the data is good



Publish all

Validate Data flow debug Debug Settings

source1 Import data from Category_B

sink1 Columns: 2 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name *: sink1 [Learn more](#)

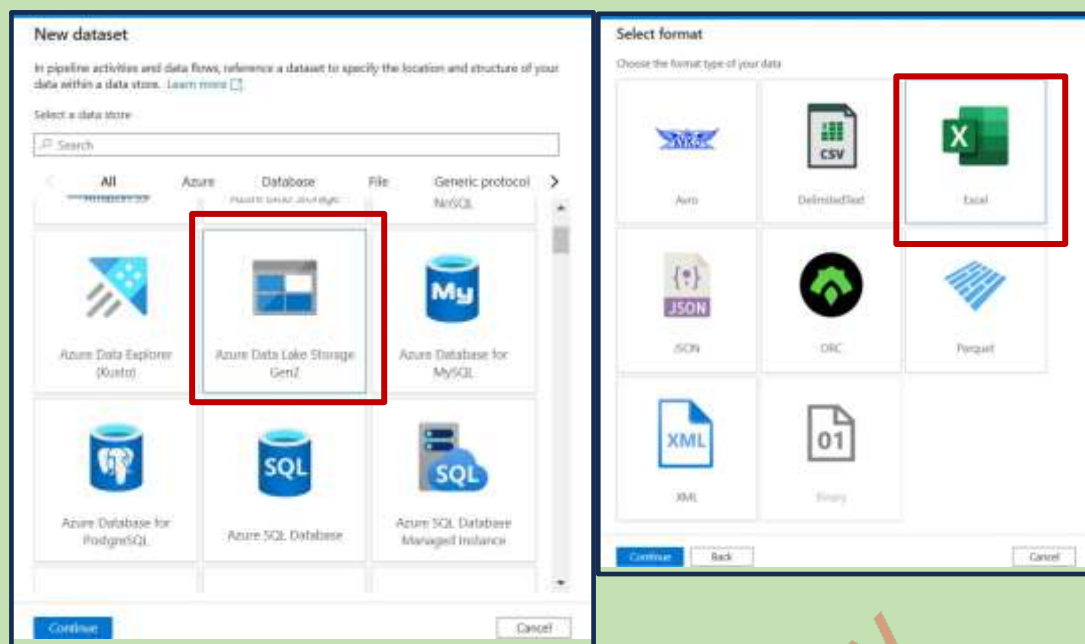
Description: Add sink dataset [Reset](#)

Incoming stream *: source1

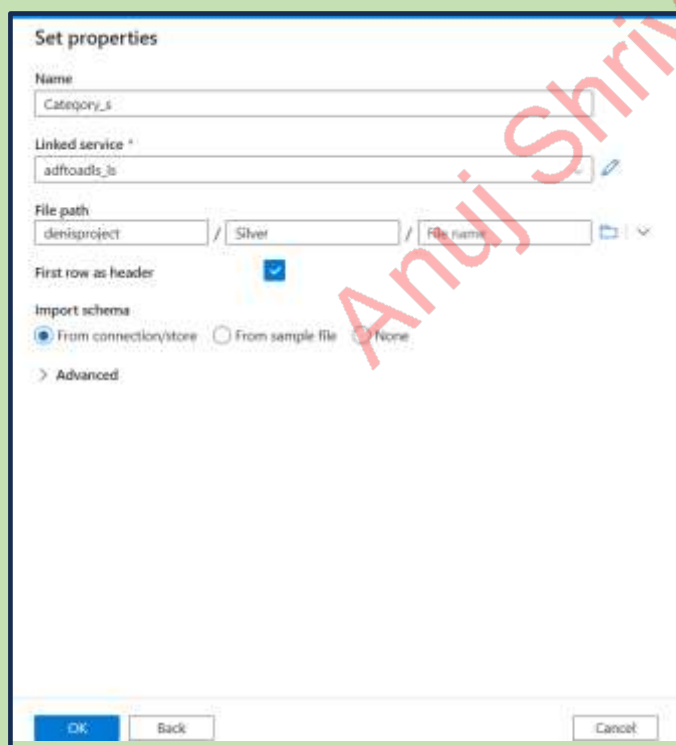
Sink type *: ☒ Dataset ☐ Inline ☐ Cache

Dataset *: Select... + New

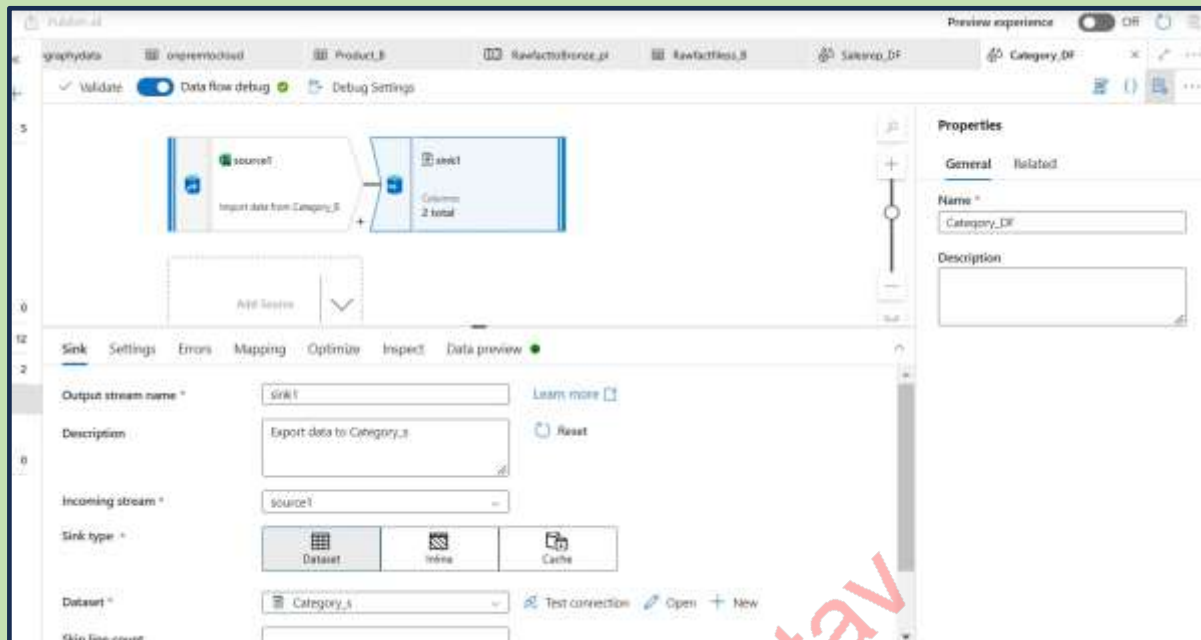
Options: ☒ Allow schema drift ☐ Validate schema



Set properties:



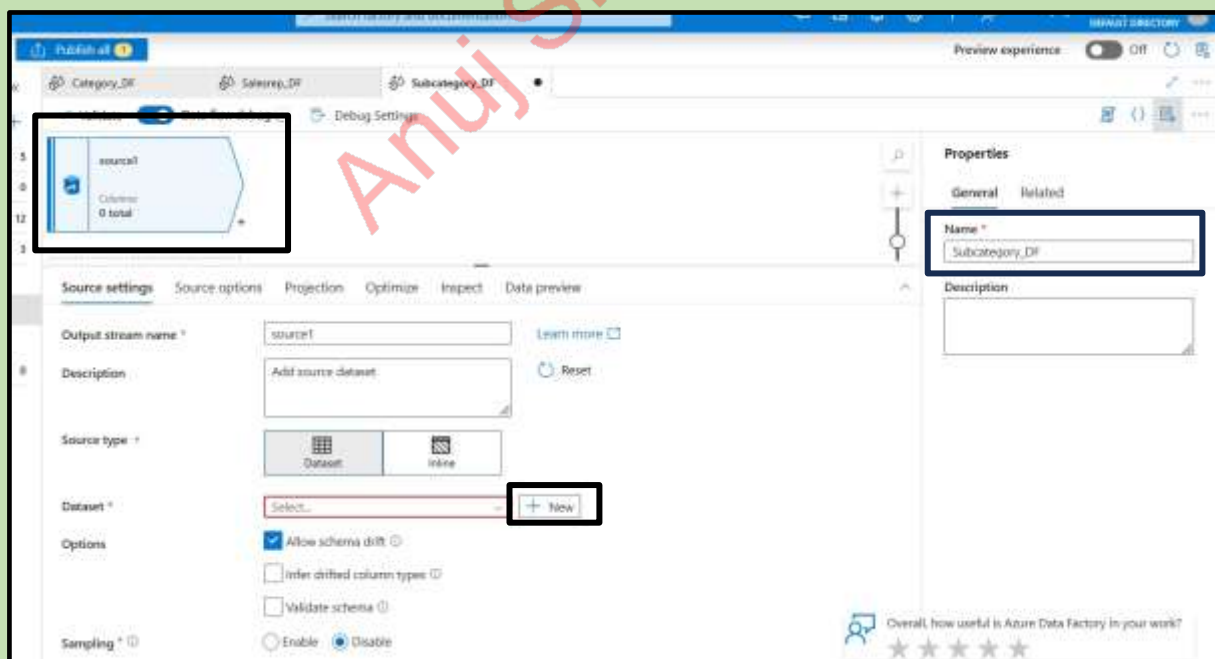
Dataflow created:

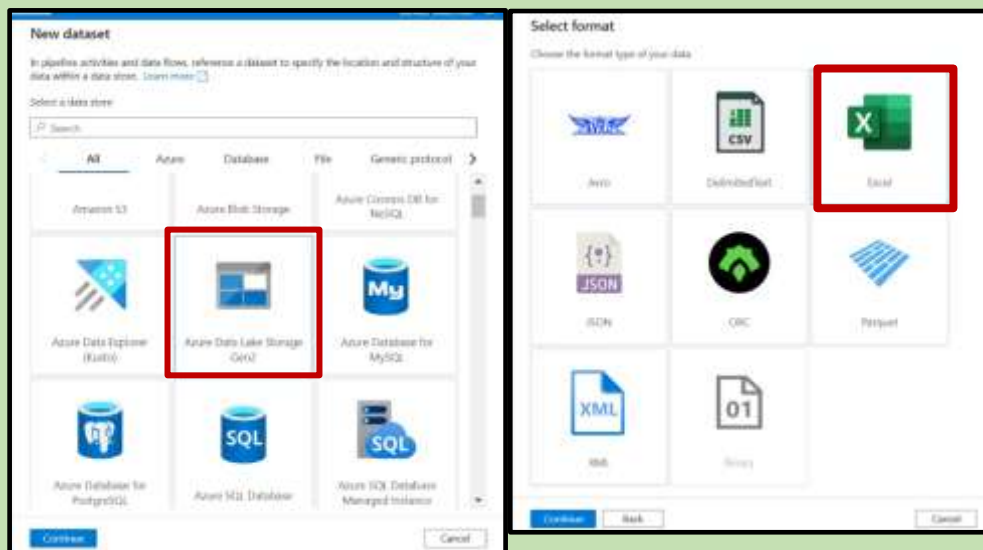


Data Flow3 :

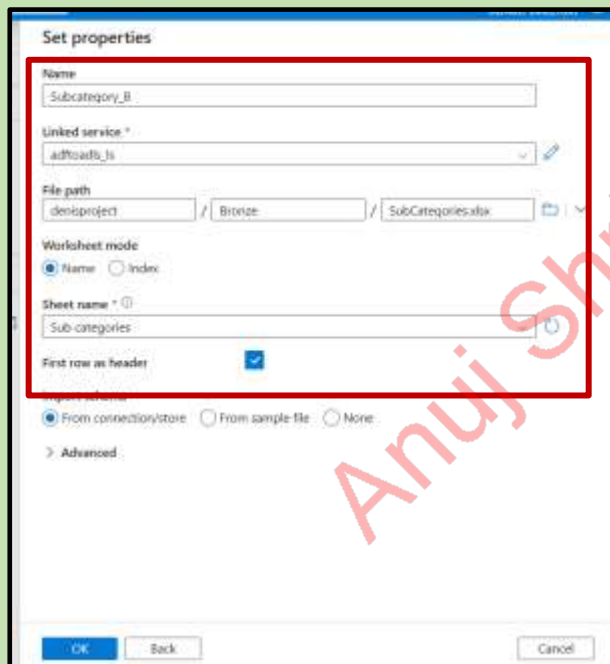
- Subcategorydataflow

Create new data flow give the name of the data flow and select new in the data set

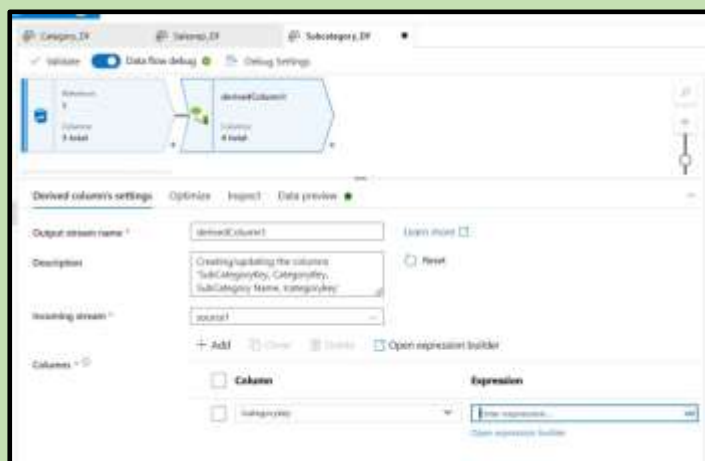




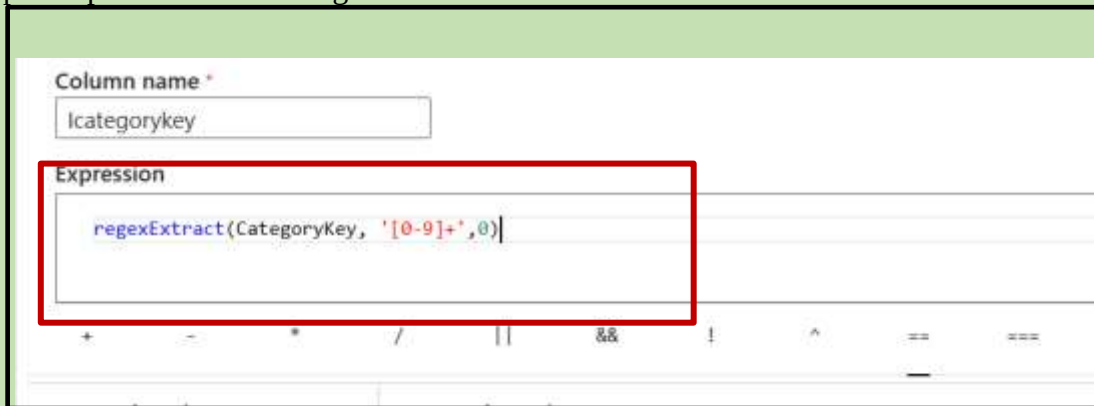
Set the properties



Now select the derived column and check the data preview



Open expression builder→ give command



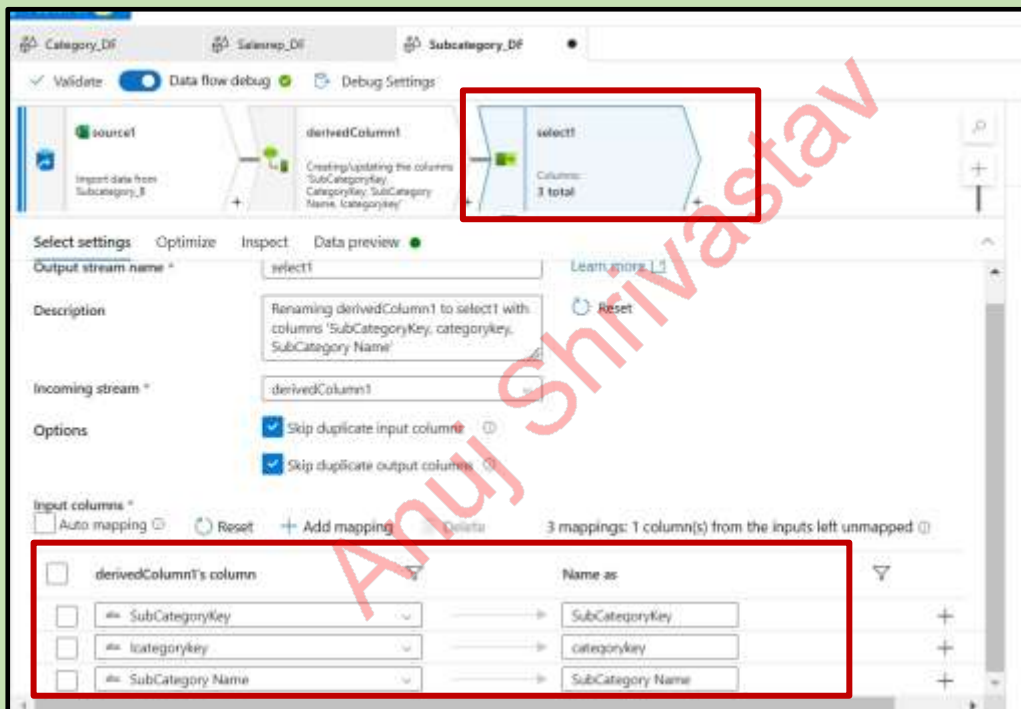
Column name *

lcategorykey

Expression

```
regexExtract(CategoryKey, '[0-9]+',0)
```

Now select one more column -> select and change the derived column name delete the previous derived column



Category_DF Salesrep_DF Subcategory_DF

Validate Data flow debug Debug Settings

source1

derivedColumn1

select1

Columns: 3 total

Select settings Optimize Inspect Data preview

Output stream name * select1

Description

Renaming derivedColumn1 to select1 with columns 'SubCategoryKey, categorykey, SubCategory Name'

Incoming stream * derivedColumn1

Options

☒ Skip duplicate input columns

☒ Skip duplicate output columns

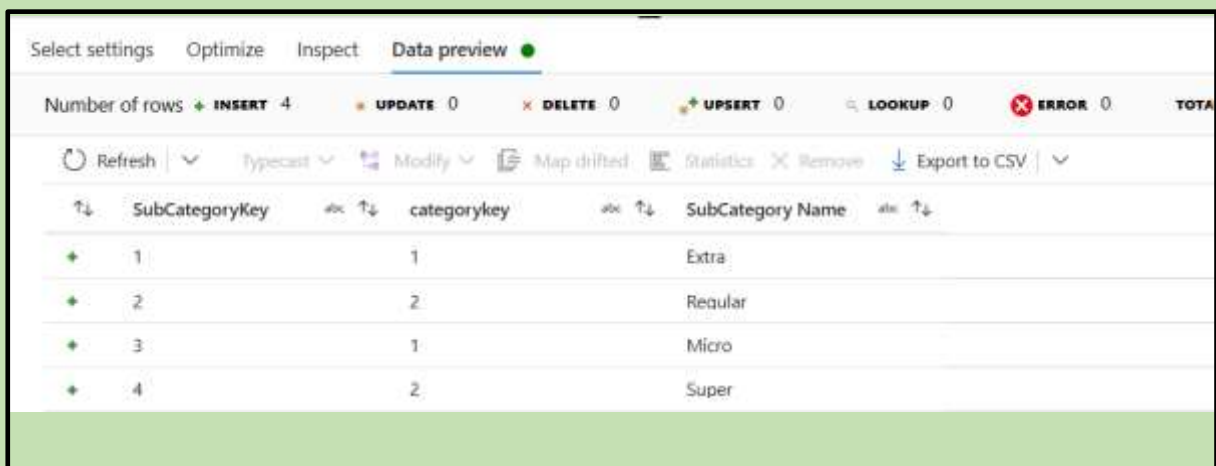
Input columns *

☐ Auto mapping

3 mappings: 1 column(s) from the inputs left unmapped

	derivedColumn1's column	Name as
☐	SubCategoryKey	SubCategoryKey
☐	lcategorykey	categorykey
☐	SubCategory Name	SubCategory Name

We can see the data preview data is been sorted



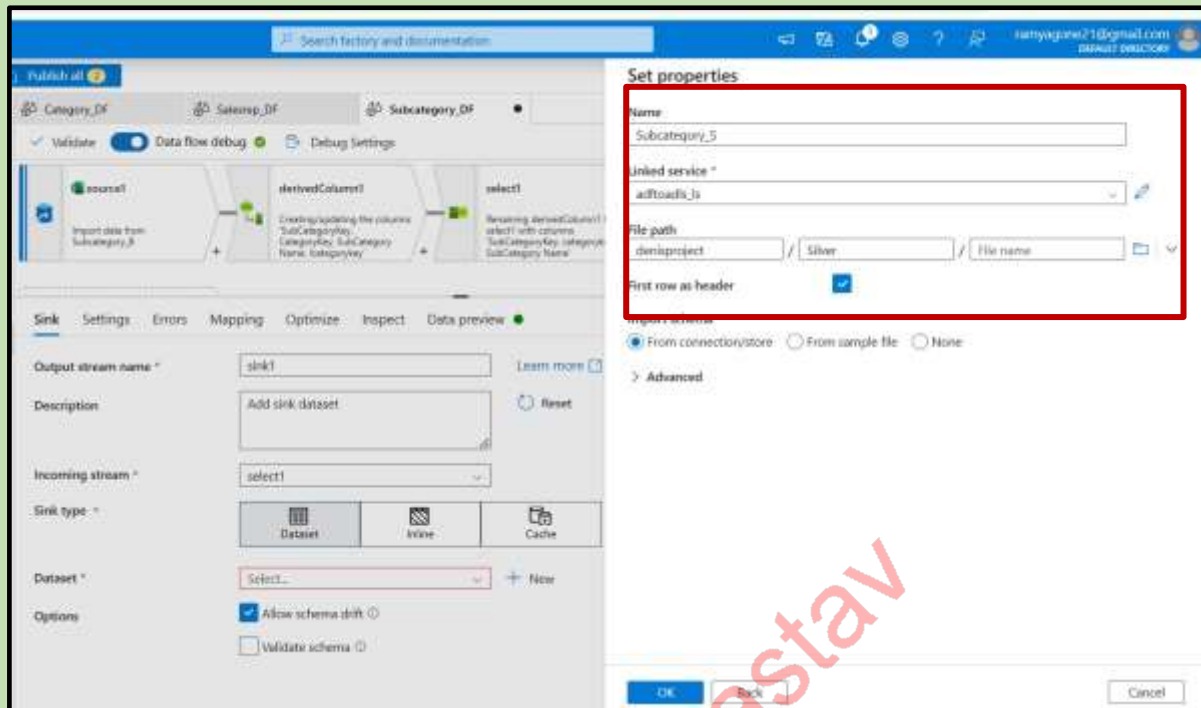
Select settings Optimize Inspect Data preview

Number of rows INSERT 4 UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

	SubCategoryKey	categorykey	SubCategory Name
+	1	1	Extra
+	2	2	Regular
+	3	1	Micro
+	4	2	Super

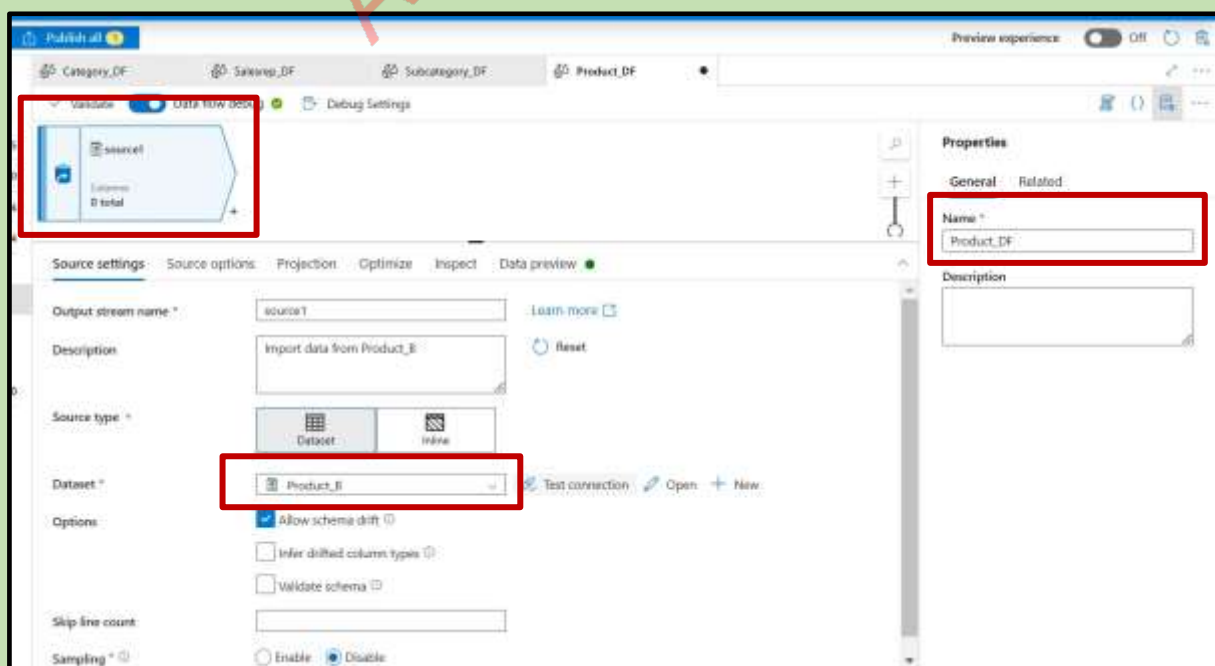
add more column to sink → set properties → publish



Data Flow4 :

- Productdataflow

Add new data set name it as product_DF and add the source we can see that the new data set already there because we have brought the data from blob to adls bronze.



Select one more column → aggregation → group by all columns

The screenshot shows the 'Aggregate' node configuration in Azure Data Studio. The 'Aggregate settings' pane is open, showing the 'Group by' section with a list of columns: ProductID, ProductName, Sub Category Key, Color, RetailPrice, and StandardCost. The 'Output stream name' is 'aggregate1' and the 'Incoming stream' is 'source1'.

Now we need to aggregate it

The screenshot shows the 'Aggregate settings' pane with the 'Expression' section expanded. A new column 'CountID' has been added with the expression 'count(ProductID)'.

New column is added

The screenshot shows the 'Data preview' pane with the aggregated data. The table has columns: ProductID, ProductName, Sub Category Key, Color, RetailPrice, StandardCost, and CountID. The 'CountID' column is highlighted with a red box.

ProductID	Product...	Sub Cate...	Color	RetailPrice	Standard...	CountID
1	Alder	3	Red	23.95	7.55	1
2	Linder	2	Blue	23.95	7.55	1
3	Magnum	2	Green	23.95	7.55	1
4	Quad	1	Red	43.95	13.75	1
5	Black Mo...	1	Blue	43.95	13.75	1
6	Quad	4	Green	43.95	13.75	1
7	Blind	1	Red	26.95	8.25	2
8	Verheulen	3	Blue	26.95	8.25	1
9	Magnum	1	Green	26.95	8.25	1
10	Carlota	1	Floresce...	29.95	9.15	2
11	Carlota	4	Floresce...	29.95	9.15	1

Add one more column → select → delete countID

The screenshot shows the Power Query Editor interface. At the top, there are tabs for 'Category_DF', 'Salesrep_DF', 'Subcategory_DF', and 'Product_DF'. The 'select' step is highlighted with a red box, showing 'Columns: 7 total'. Below the step list, the 'aggregate's column' section shows a list of columns being mapped to the output. The 'CountID' column is highlighted with a red box, indicating it is being added to the output columns.

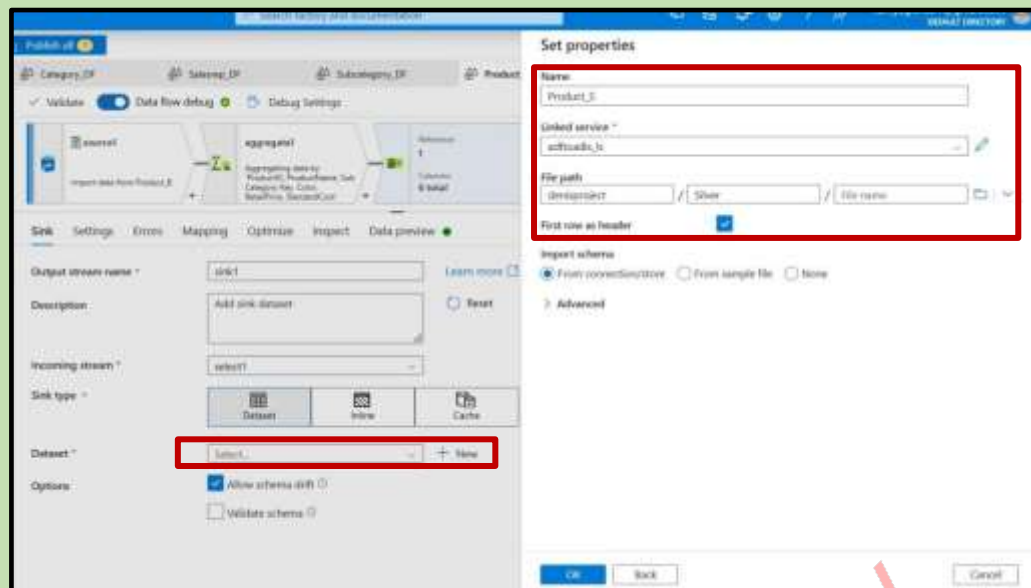
aggregate's column	Name as
1m ProductID	ProductID
4m ProductName	ProductName
1m Sub Category Key	Sub Category Key
4m Color	Color
12 RetailPrice	RetailPrice
12 StandardCost	StandardCost
1m CountID	CountID

Successfully removed the duplicate values

The screenshot shows the 'Data preview' tab in Power Query Editor. The table displays 11 rows of product data. The 'CountID' column is visible, showing unique values for each product.

ProductID	ProductName	Sub Category Key	Color	RetailPrice	StandardCost
1	Alder	3	Red	23.95	7.55
2	Linder	2	Blue	23.95	7.55
3	Magnum	2	Green	23.95	7.55
4	Quad	1	Red	43.95	13.75
5	Black Monk	1	Blue	43.95	13.75
6	Quad	4	Green	43.95	13.75
7	Bling	1	Red	26.95	8.25
8	VaniHelen	3	Blue	26.95	8.25
9	Magnum	1	Green	26.95	8.25
10	Carlota	1	Florescen...	29.95	9.15
11	Carlota	4	Florescen...	29.95	9.15

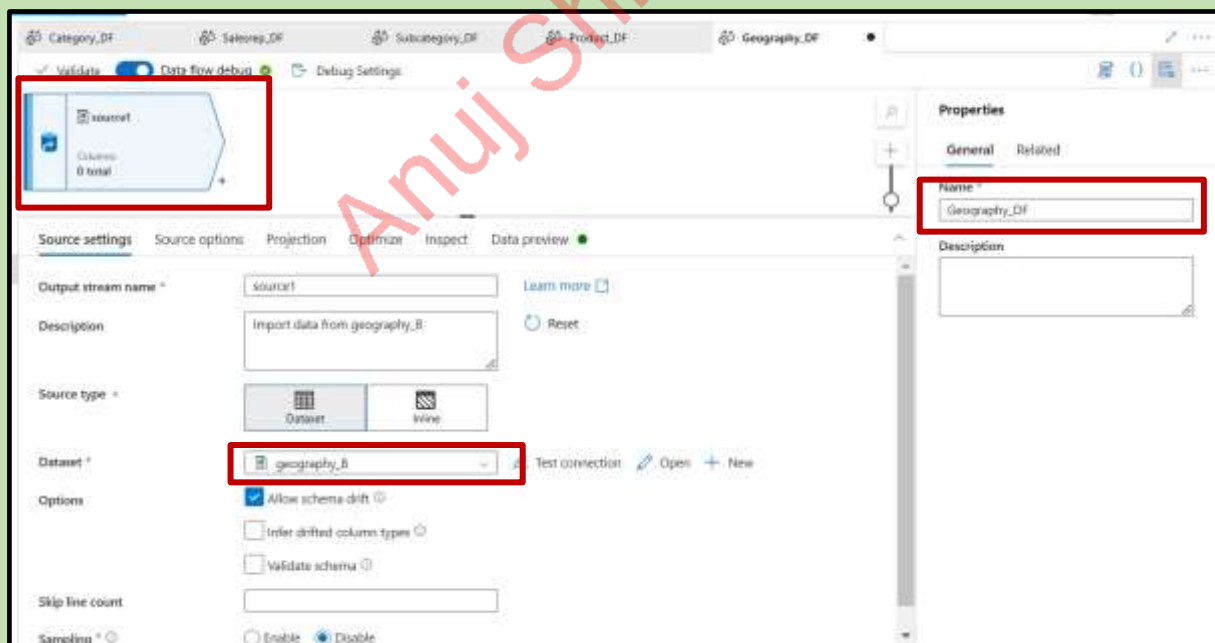
Add sink add data set give the set propertied details and publish



Data Flow5 :

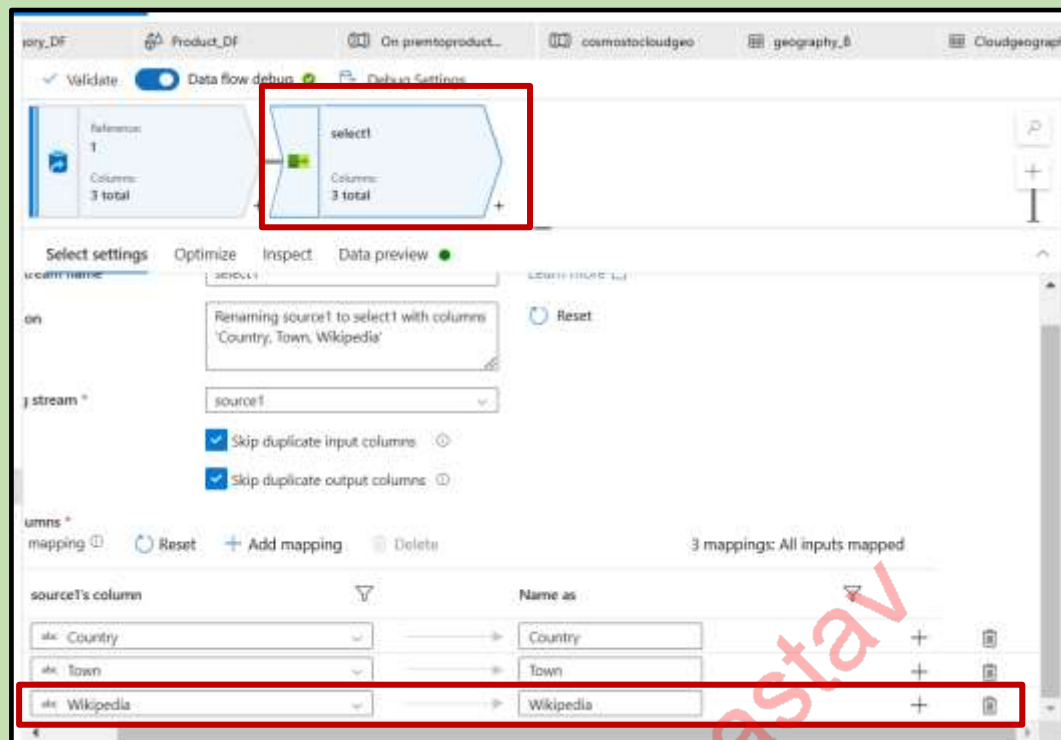
- Geographydataflow

Add source and select the data set

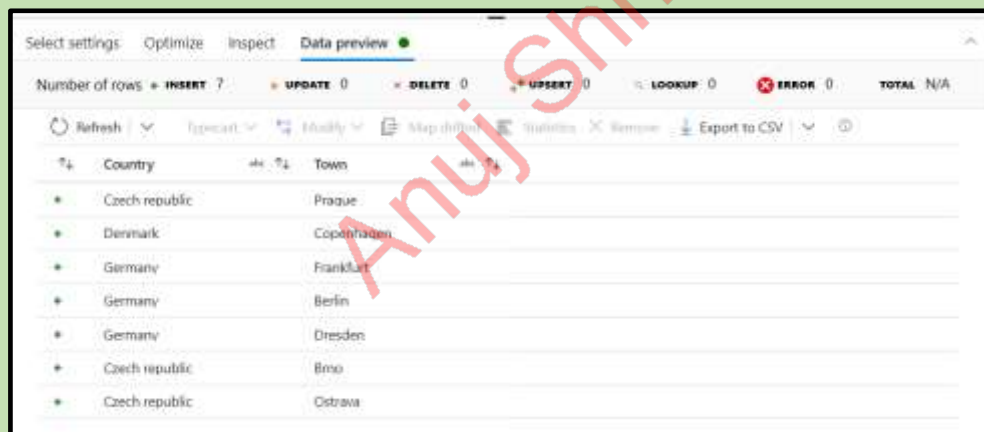


Requirement: Create unique key (Geo Key) and geography table

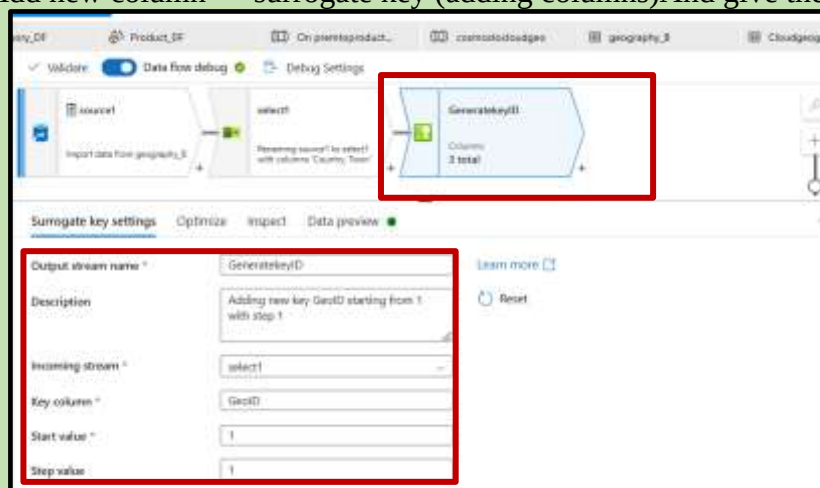
Add one more column --> select , delete the wikipedia column



As we can see the column is deleted



Add new column--> surrogate key (adding columns) And give the details



New column added

	Country	Town	GeoID	
+	Czech republic	Prague	1	
+	Denmark	Copenhagen	2	
+	Germany	Frankfurt	3	
+	Germany	Berlin	4	
+	Germany	Dresden	5	
+	Czech republic	Brno	6	
+	Czech republic	Ostrava	7	

Changing order of columns

New column is added ---> Select and keep the columns in order

select2

Reversing GenerateKeyID to select2 with columns: GeoID, Country, Town

Columns: 3 total

Output stream name: select2

Description: Reversing GenerateKeyID to select2 with columns: GeoID, Country, Town

Incoming stream: GenerateKeyID

Options: ☒ Skip duplicate input columns, ☒ Skip duplicate output columns

Input columns: ☐ GenerateKeyID's column, ☐ GeoID, ☐ Country, ☐ Town

Name as: GeoID, Country, Town

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All | Azure | Database | File | Generic protocol

Amazon S3 | Azure Blob Storage | Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto) | Azure Data Lake Storage Gen2 | Azure Database for MySQL

Azure Database for PostgreSQL | Azure SQL Database | Azure SQL Database Managed Instance

Continue | Cancel

Select format

Choose the format type of your data

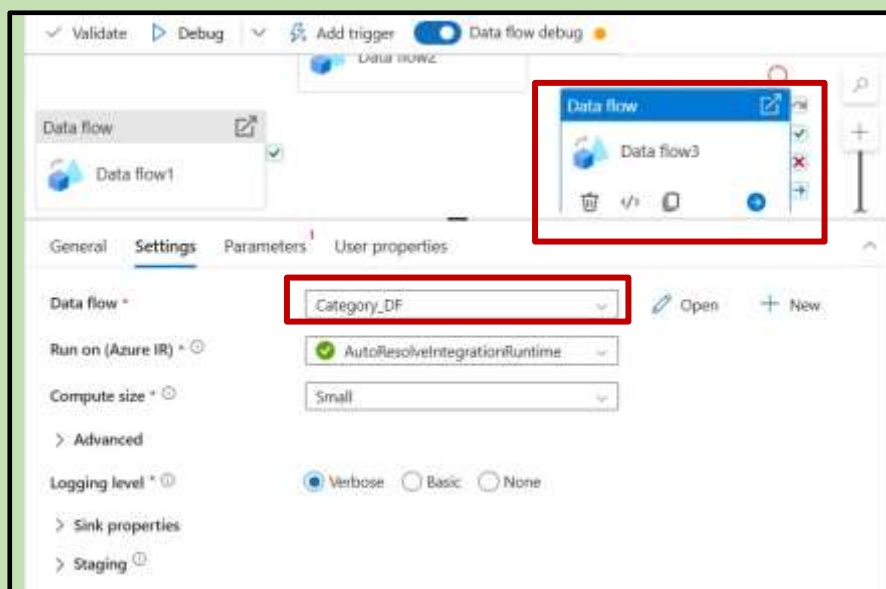
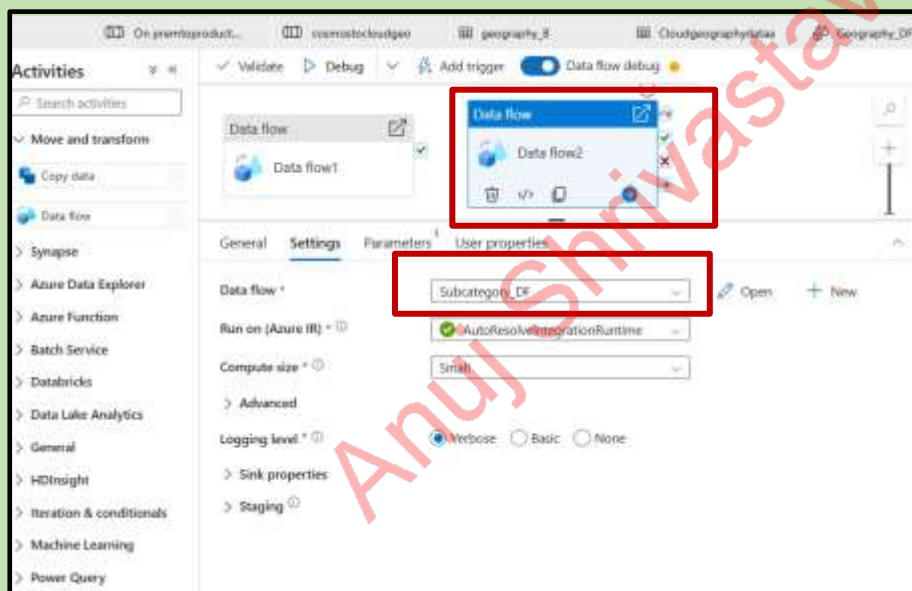
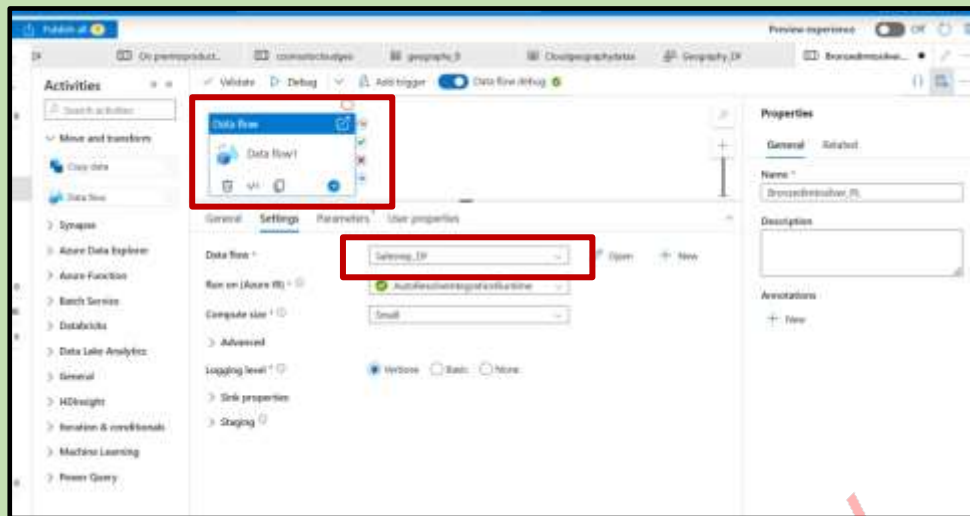
Avro | DelimitedText | JSON

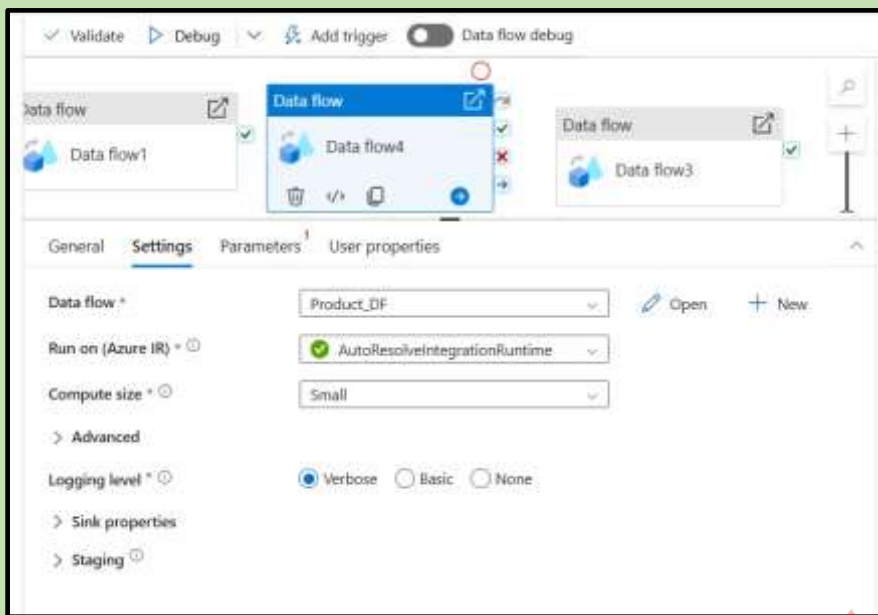
ORC | Parquet | Snowflake

Continue | Back | Cancel

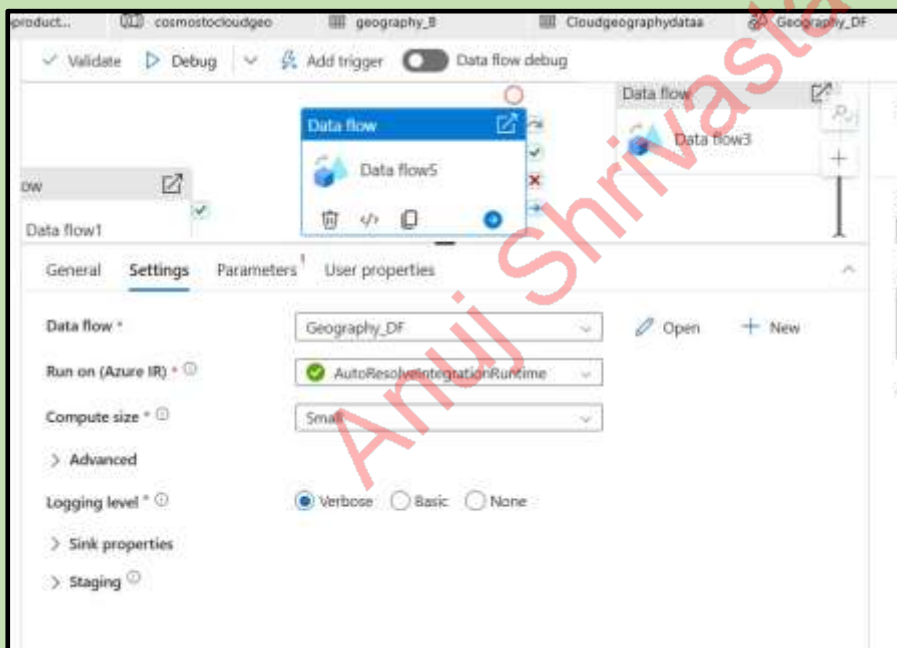
Pipeline:

Create new pipeline add all dataflow in the pipeline and give the pipeline name



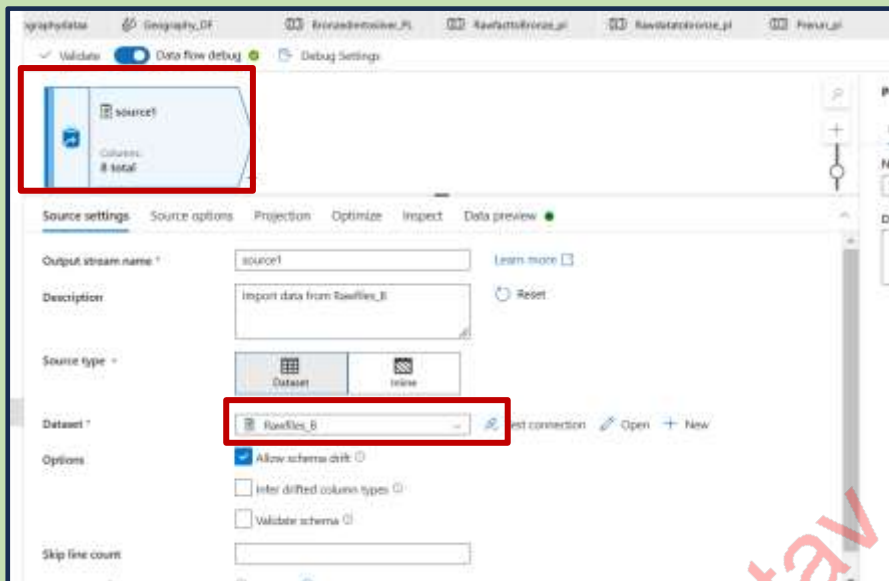


Five Data flows created and publish it

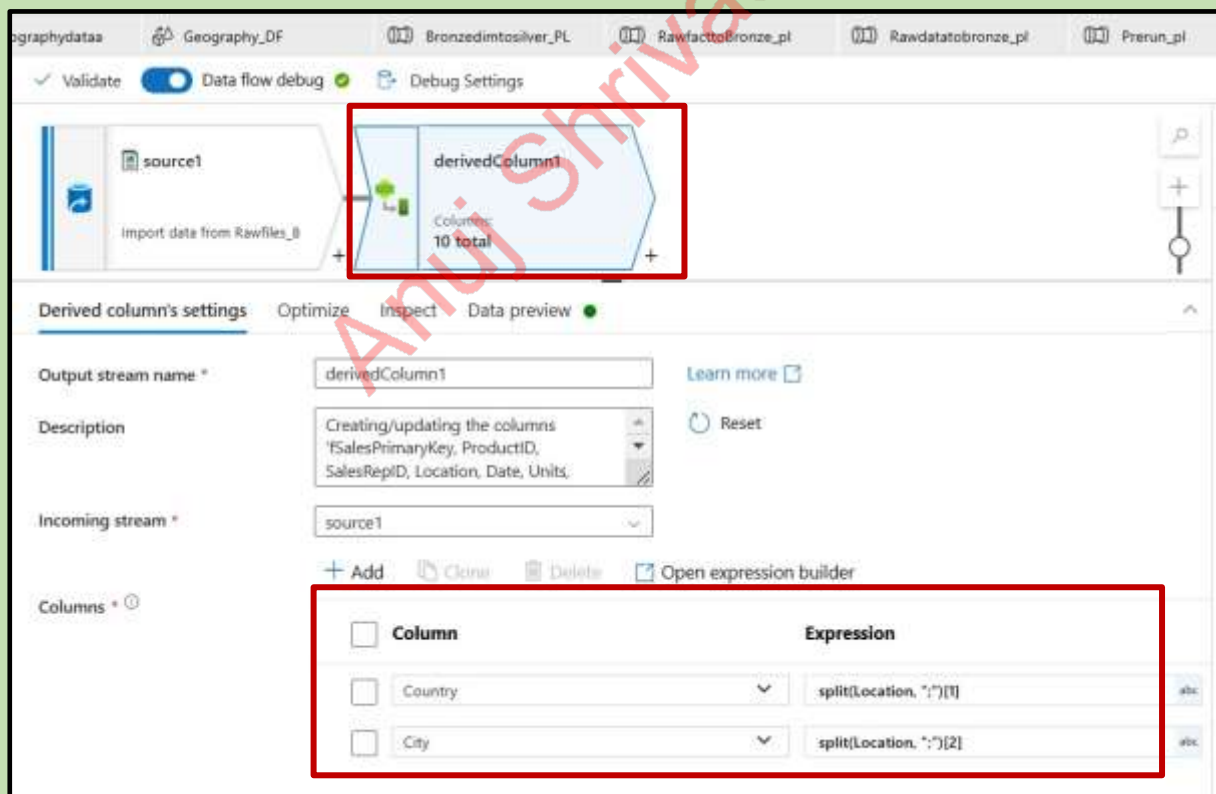


Data Flow6 :

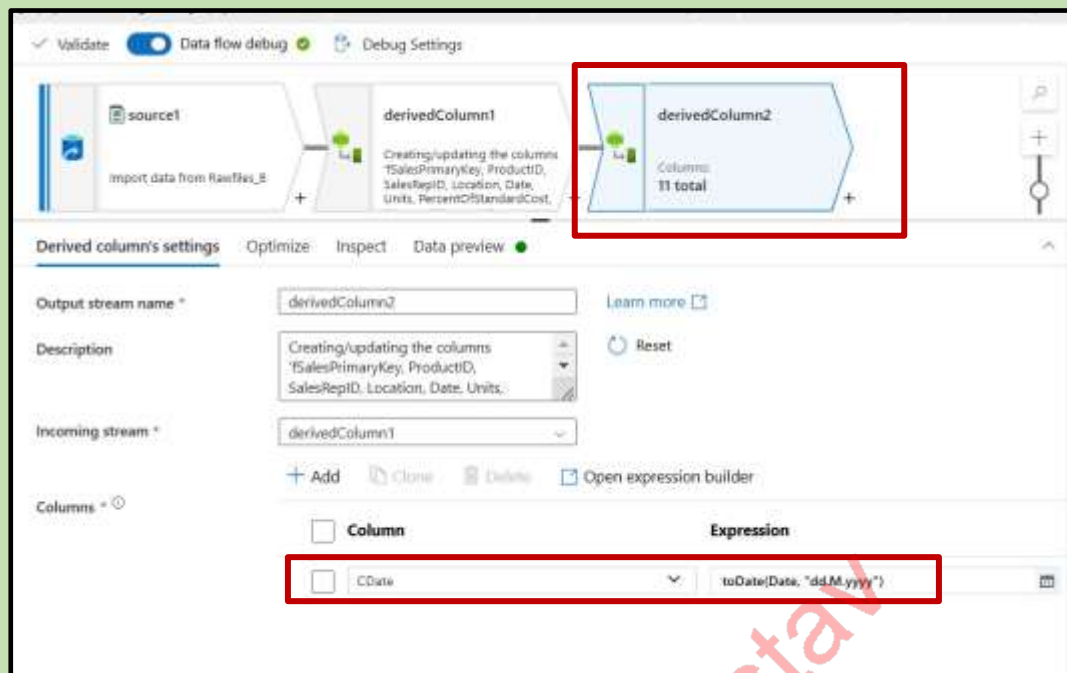
New data flow is created add source and give the data set



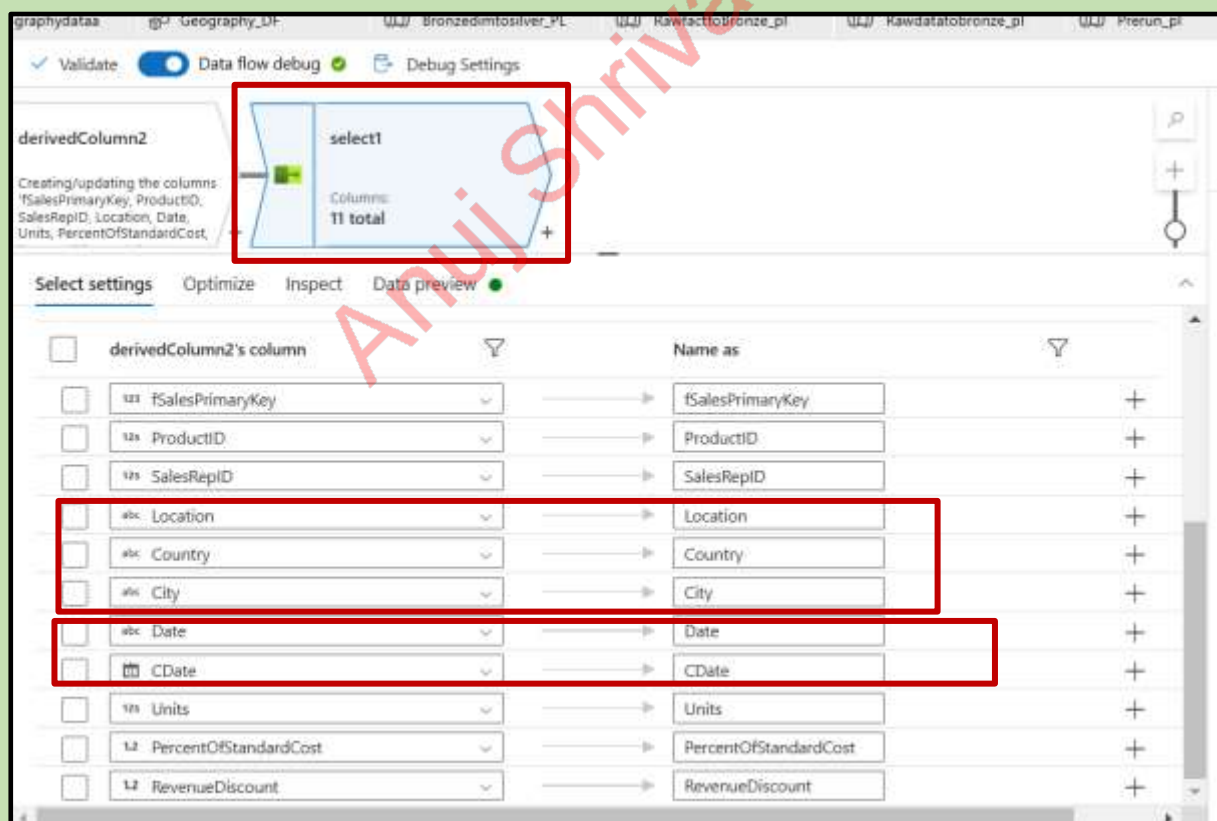
Add derived column

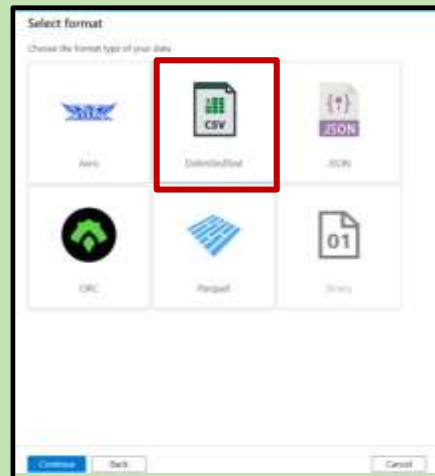
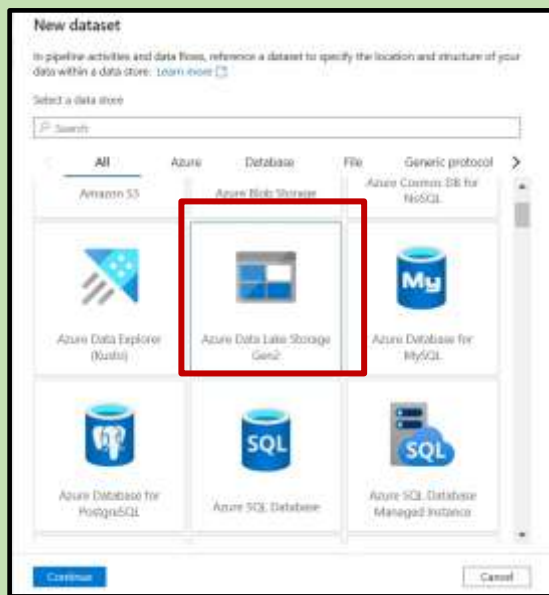


Add one more derived column

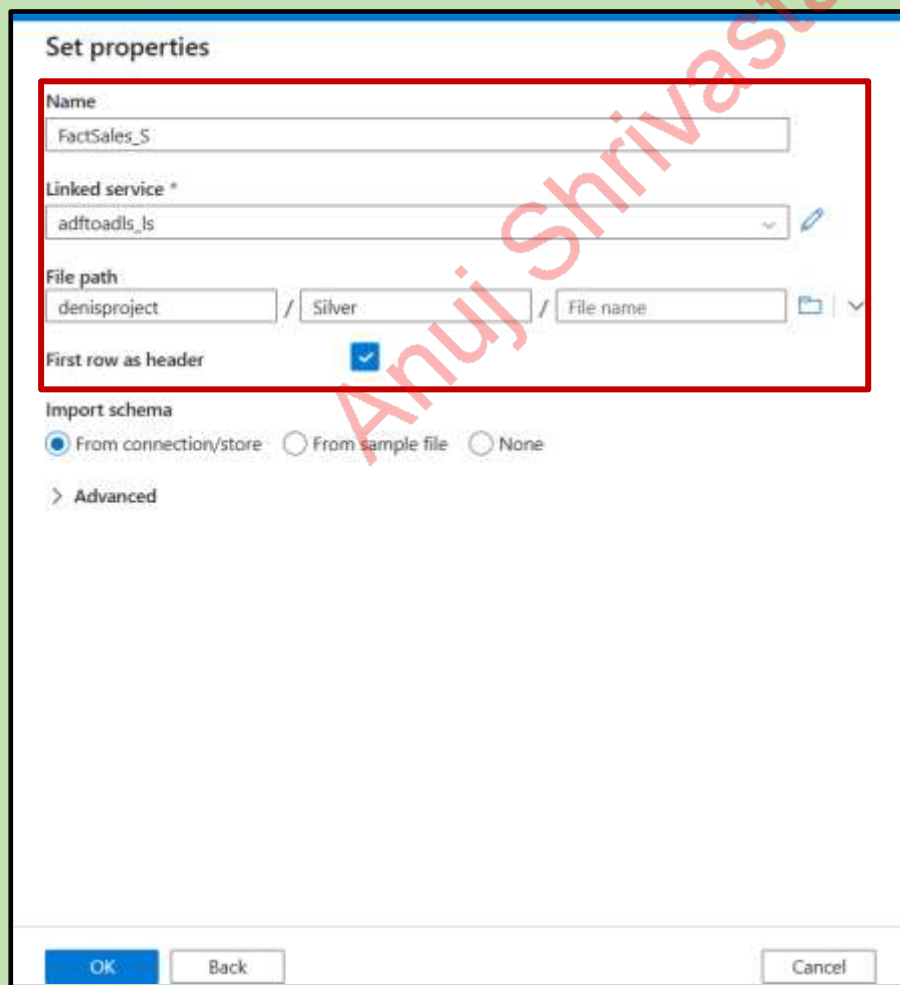


Add select column remove location and drag Country, City remove date and drag CDate



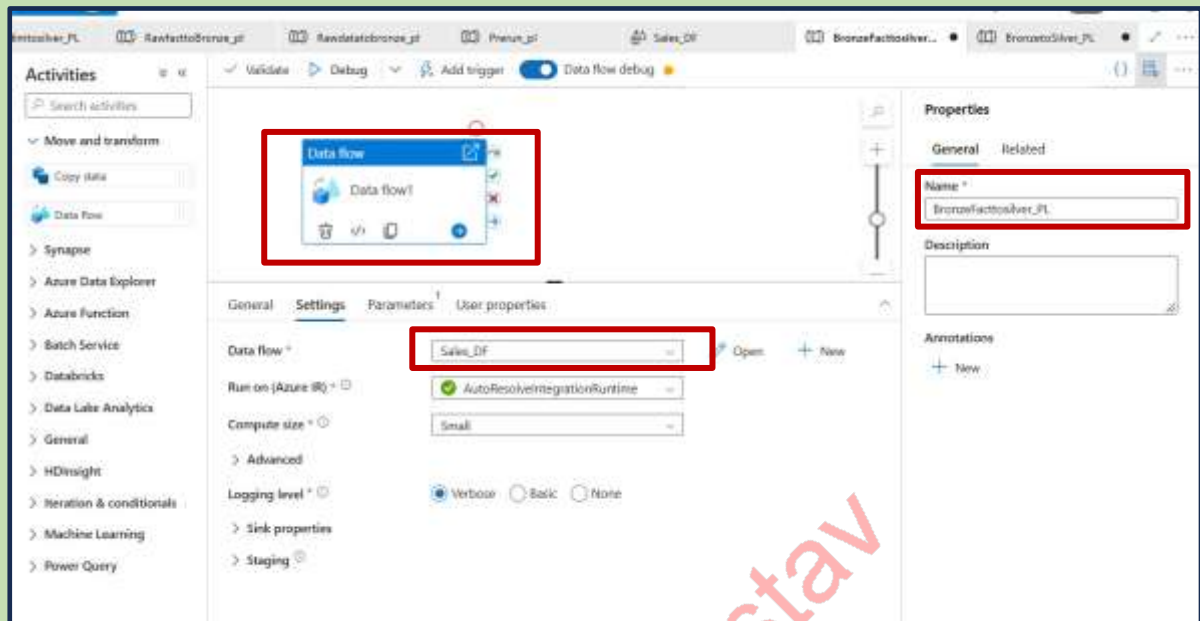


Set the properties and publish

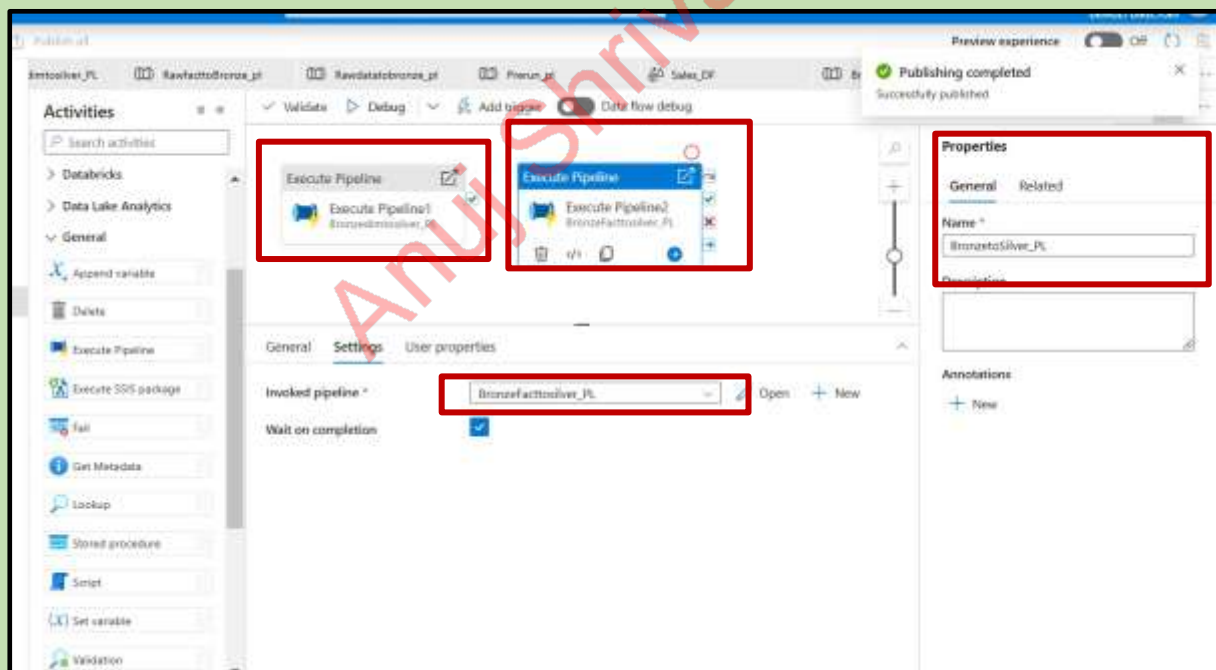


Pipeline:

Create new pipeline move the data flow add the name and data flow

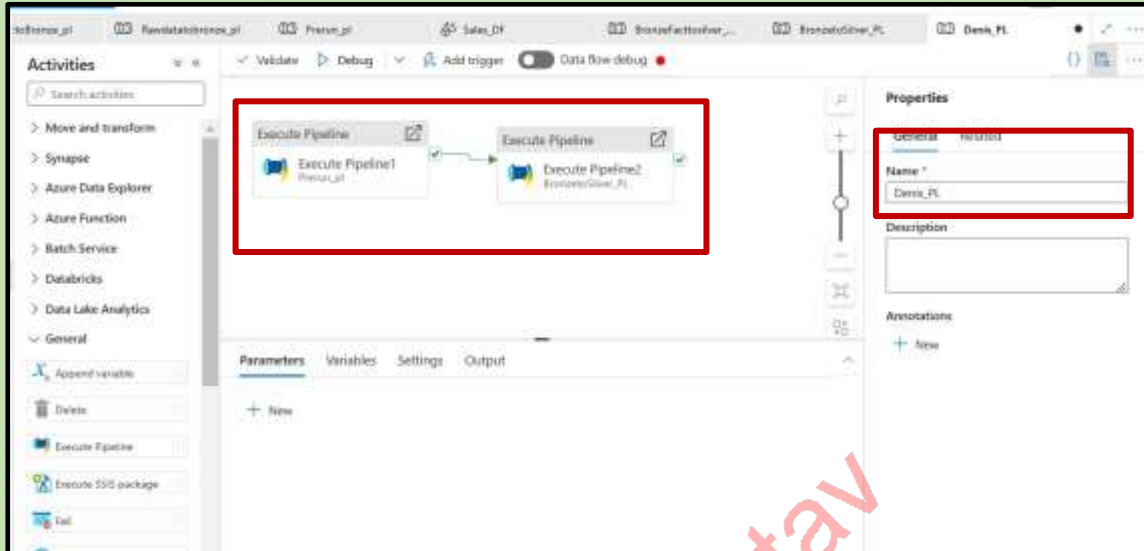


Now drag 2 execute pipeline and invoke two pipelines from bronze to Silver and publish



Final Pipeline:

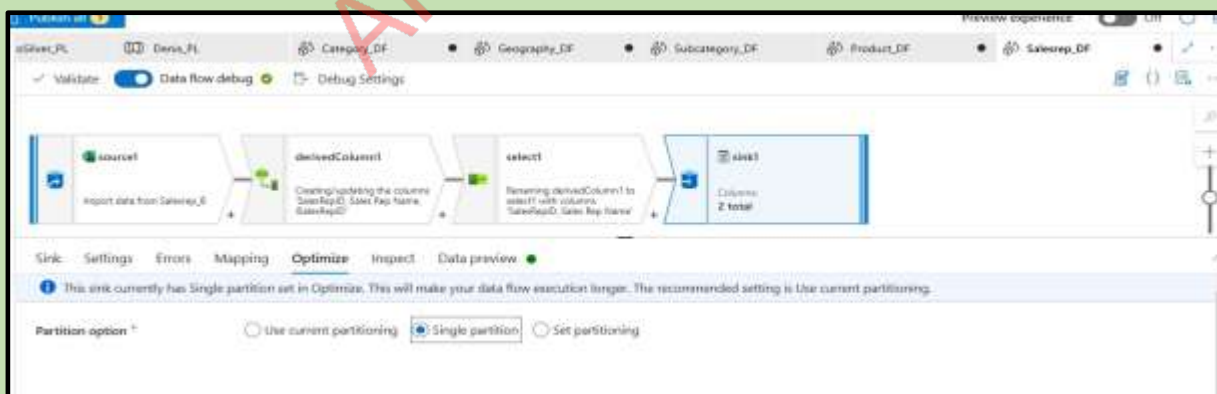
Now we need to add two execute pipelines --->Prerun , Bronze to silver and publish it and run it



After successful pipeline run if we get the files in this format

Name	Modified	Access tier	Archive status	Blob type	Size	Access state
LI						---
Sales5-00001.csv	11/2/2024, 9:22:16 AM	Hot (Inferred)		Block blob	916.08 KB	Available
Sales5-00002.csv	11/2/2024, 9:22:17 AM	Hot (Inferred)		Block blob	642.53 KB	Available
Sales5-00003.csv	11/2/2024, 9:22:17 AM	Hot (Inferred)		Block blob	627.3 KB	Available
Sales5-00004.csv	11/2/2024, 9:22:17 AM	Hot (Inferred)		Block blob	619.59 KB	Available

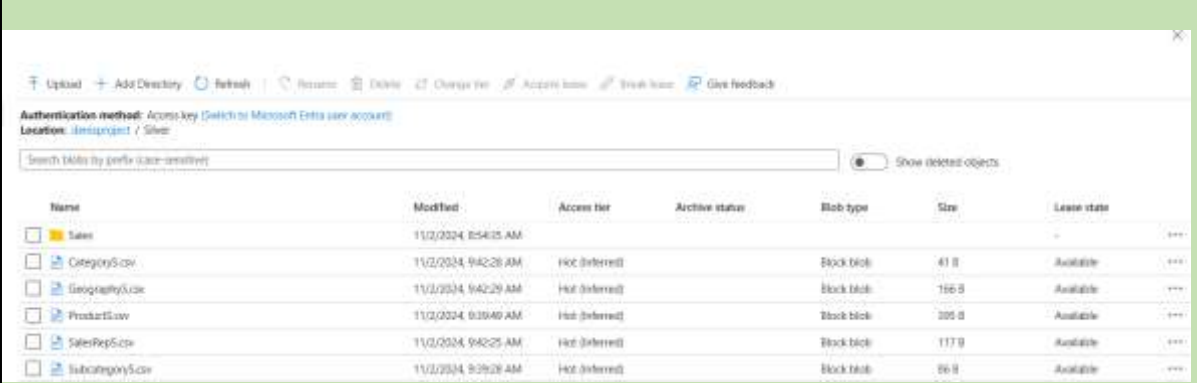
Go to the data flows and give it as single partition in the sink side



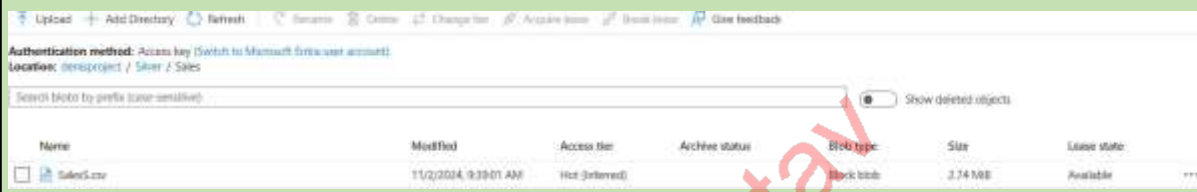
Pipeline run successfully

Activity runs							
Pipeline run ID: 645d11bf-c5b5-4453-a7cb-dff86c58416							
All status: --							
Showing 1 - 2 items							
Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Execute Pipeline2	Succeeded	Execute Pipeline	11/2/2024, 9:39:50 AM	3m 32s			0dd1f172-d759-4c77-b17b-785124e89
Execute Pipeline1	Succeeded	Execute Pipeline	11/2/2024, 9:34:51 AM	4m 18s			d1a00d7-132b-4325-9979-ad1a8bdc

Now we can check in adls→ silver



Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
Sales	11/2/2024, 8:54:05 AM					
Category.csv	11/2/2024, 9:42:28 AM	Hot (Inferred)		Block blob	41 B	Available
Geography.csv	11/2/2024, 9:42:29 AM	Hot (Inferred)		Block blob	160 B	Available
Product.csv	11/2/2024, 9:39:40 AM	Hot (Inferred)		Block blob	305 B	Available
SalesRep.csv	11/2/2024, 9:42:25 AM	Hot (Inferred)		Block blob	117 B	Available
Subcategory.csv	11/2/2024, 9:39:28 AM	Hot (Inferred)		Block blob	86 B	Available



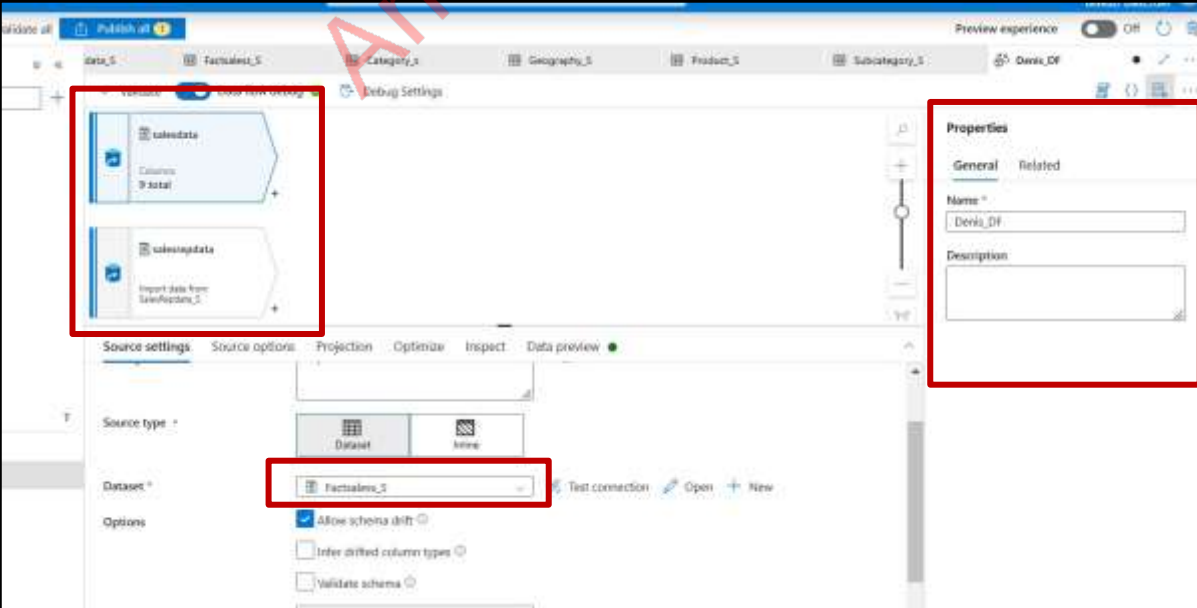
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
Sales.csv	11/2/2024, 9:39:01 AM	Hot (Inferred)		Block blob	3.74 MB	Available

Final Data Flow :

Name the data flow as Denis_DF

Add the source and add the data set salesrep

Add one more source add the salesrep dataset-filename-salesrep



The screenshot shows the Microsoft Power BI Desktop Dataflow Designer interface. The canvas displays two data sources: 'salesdata' and 'salesrepdata'. The 'salesrepdata' source is highlighted with a red box. The 'Properties' pane on the right shows the dataflow named 'Denis_DF'. The 'Source settings' pane at the bottom shows the 'Dataset' selected for 'FactSales_S'.

Add join and give details

Join settings

Left stream *
salesdata

Right stream *
salesrepdata

Join type *

Full outer Inner **Left outer** Right outer Custom (cross)

Use fuzzy matching ☐

Join conditions *

Left: salesdata's column
SalesRepID

Right: salesrepdata's column
SalesRepID

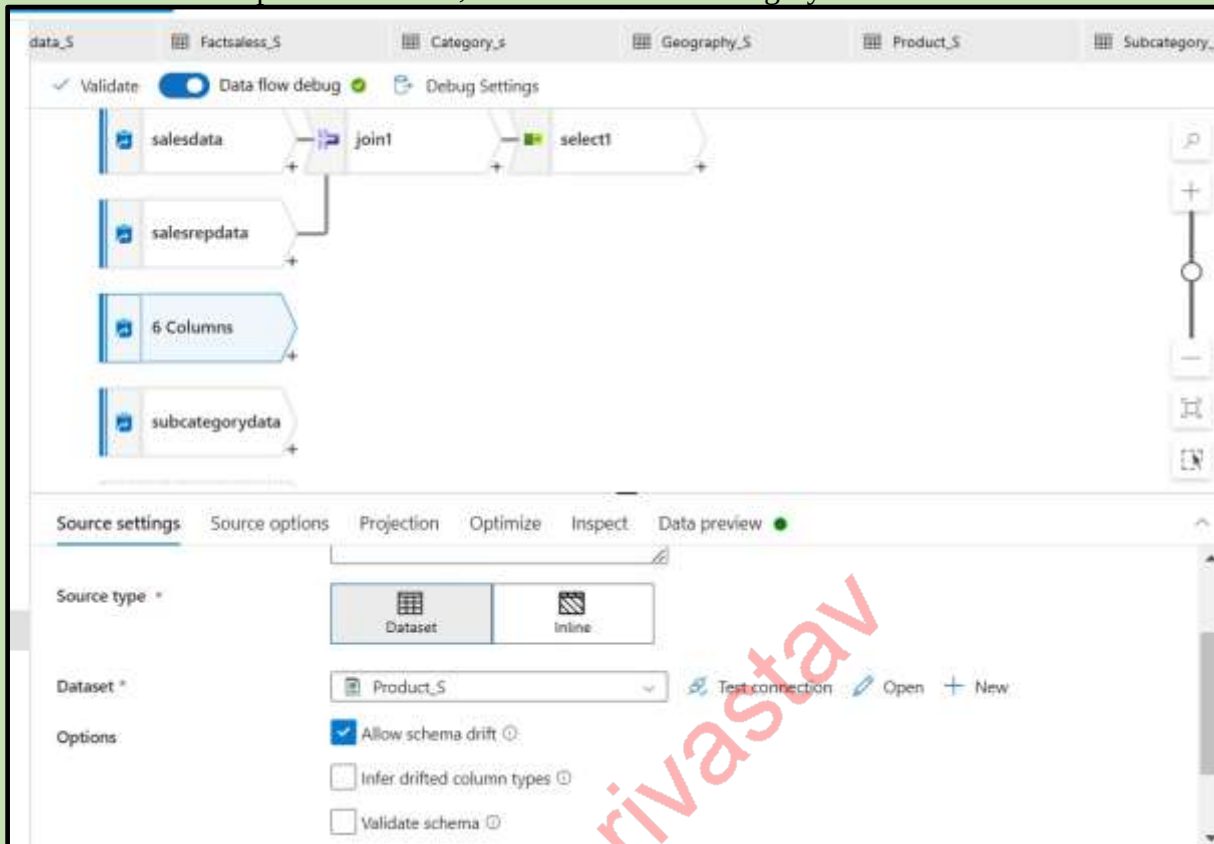
Add select and delete one sales rep id

Select settings

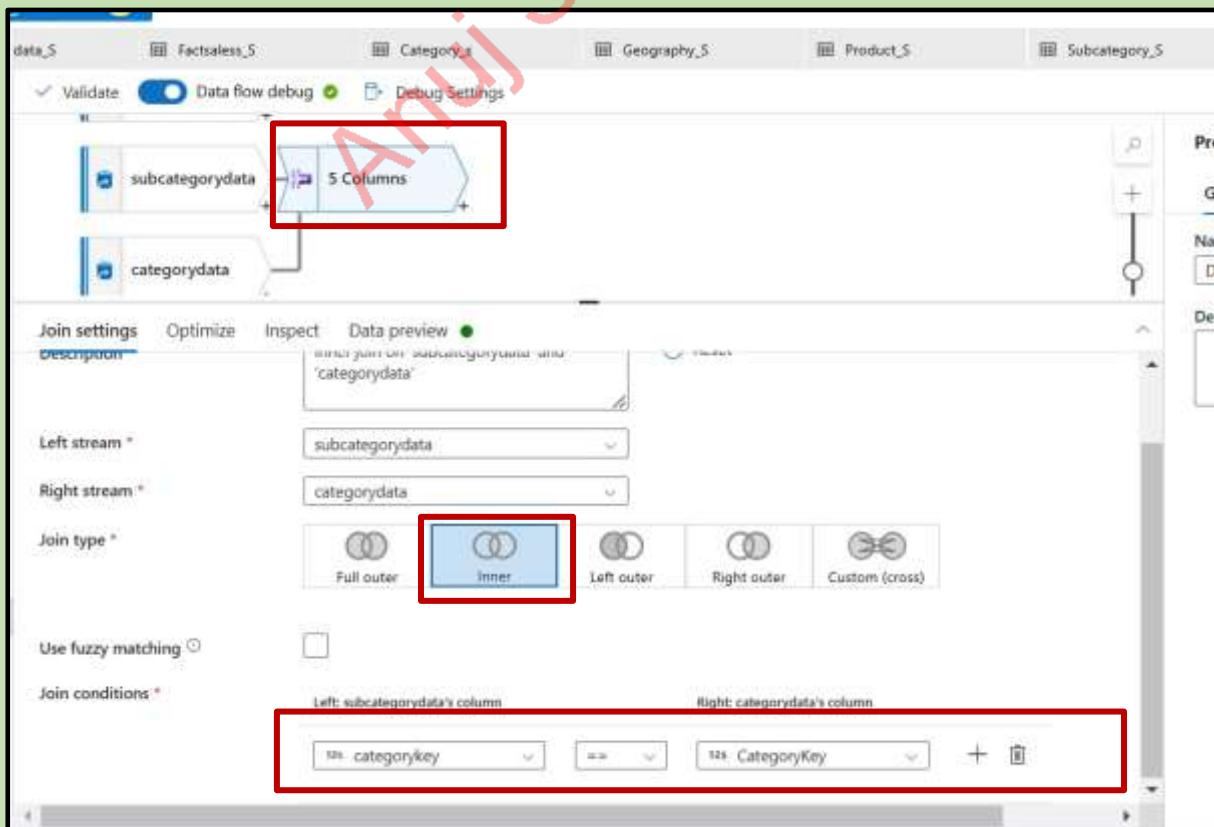
joint's column

Column	Name as
123 SalesPrimaryKey	SalesPrimaryKey
123 ProductID	ProductID
123 salesdata@SalesRepID	SalesRepID
123 salesrepdata@SalesRepID	SalesRepID
400 Country	Country
400 City	City
400 Date	Date
120 Units	Units
12 PercentOfStandardCost	PercentOfStandardCost
12 RevenueDiscount	RevenueDiscount
400 Sales Rep Name	Sales Rep Name

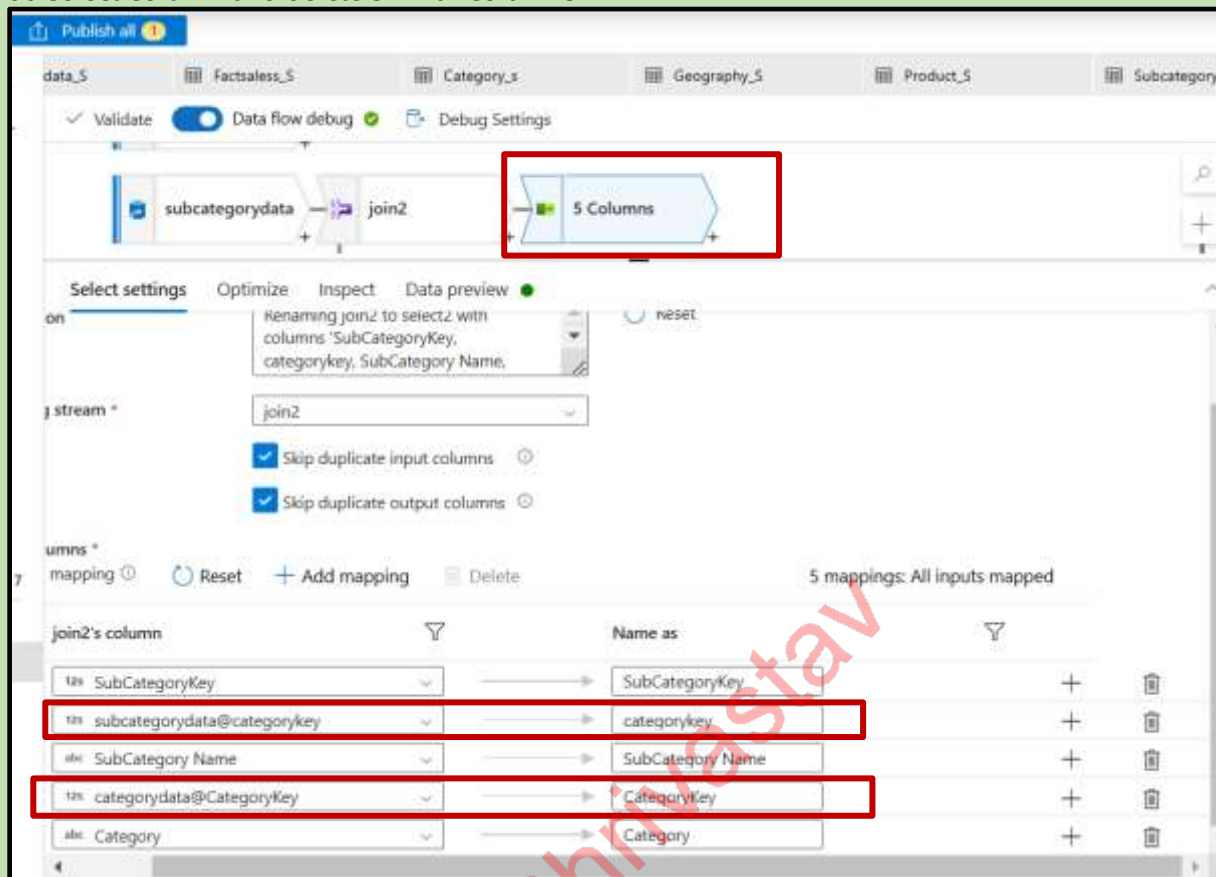
Now add source ---> product dataset , add source --> subcategory data



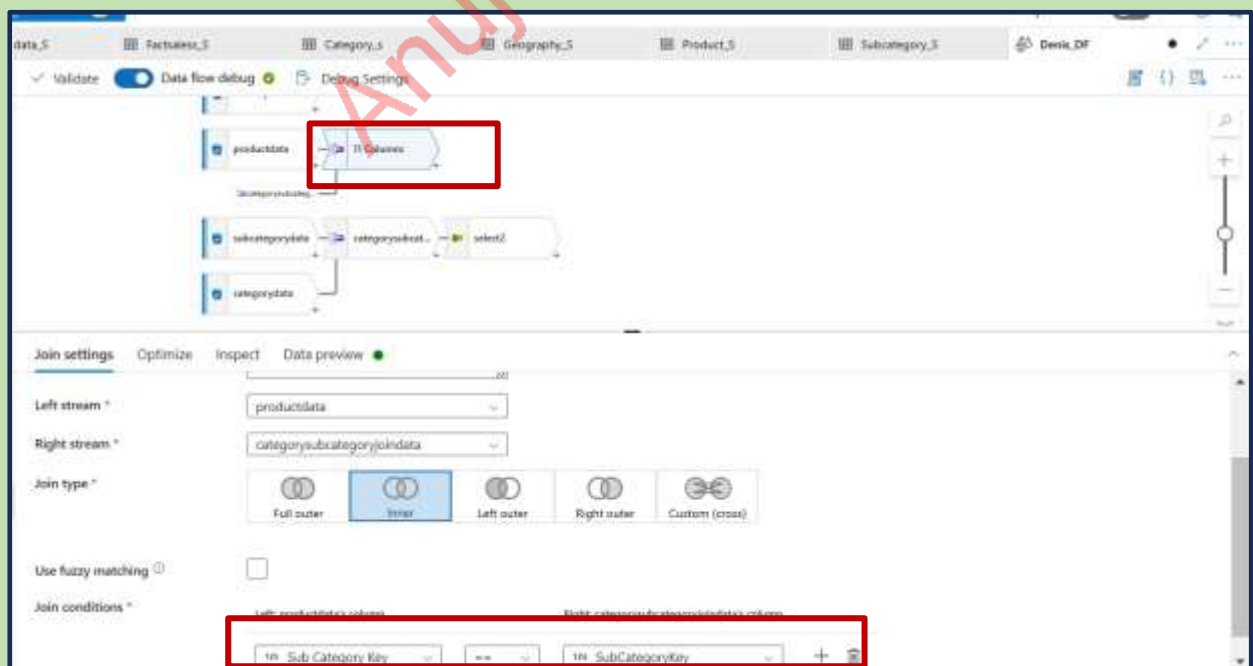
Add source -->Category data --> add join for subcategory data and join with category data and add the name as categorysubcategoryjoindata



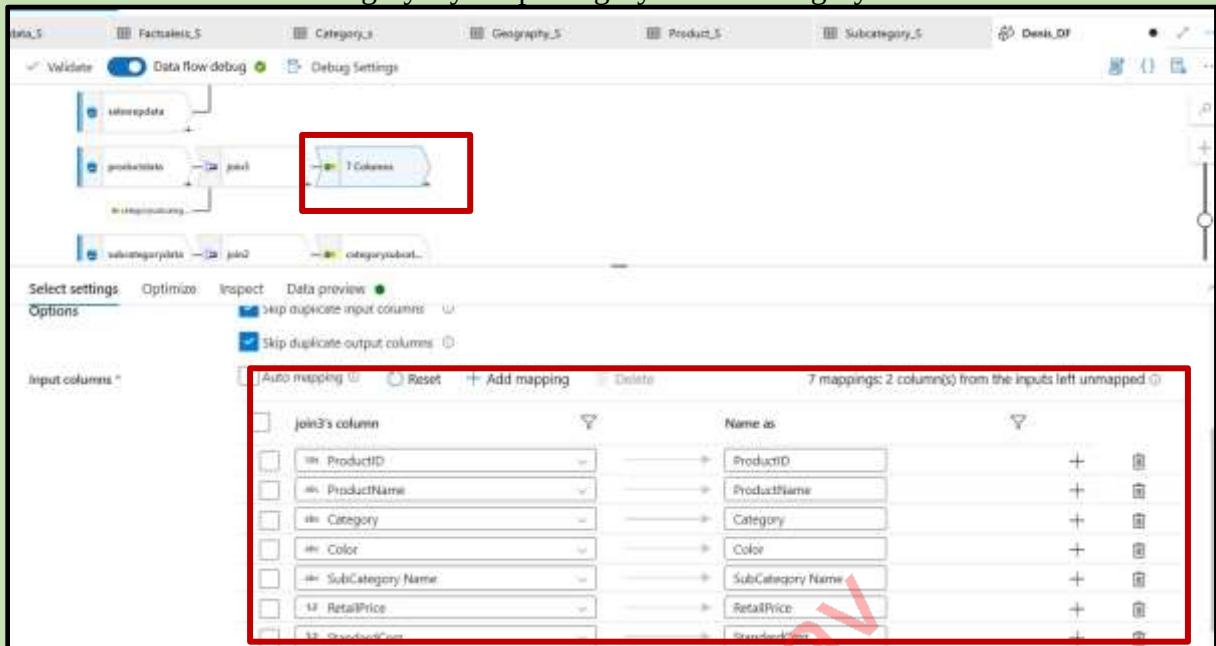
Add select column and delete similar columns



Now add join to product data join it with categorysubcategoryjoindata and give the conditions check the data preview



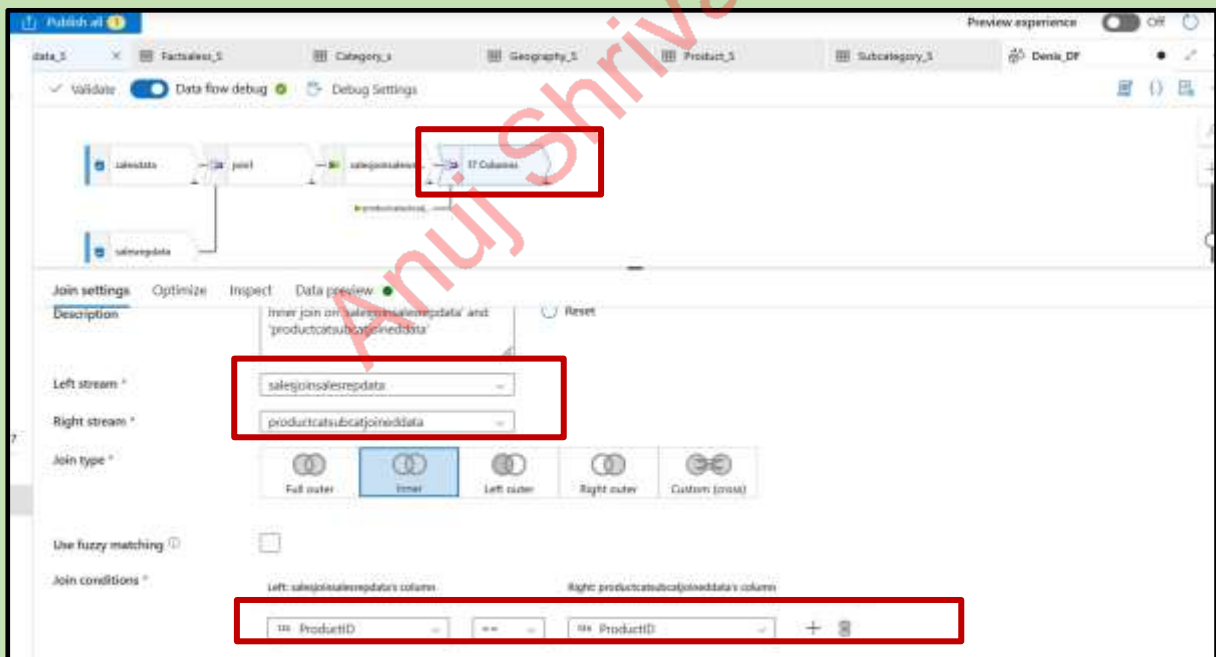
Add select and delete subcategorykey keep category and subcategory name in order



Input columns:

Input column	Name as
ProductID	ProductID
ProductName	ProductName
Category	Category
Color	Color
SubCategory Name	SubCategory Name
RetailPrice	RetailPrice
StandardCost	StandardCost

Now salesjoinsalesrepdata add join with productcatsubcatjoindata Give the condition and preview the data



Join settings:

Description: Inner join on salesjoinsalesrepdata and productcatsubcatjoindata

Left stream: salesjoinsalesrepdata

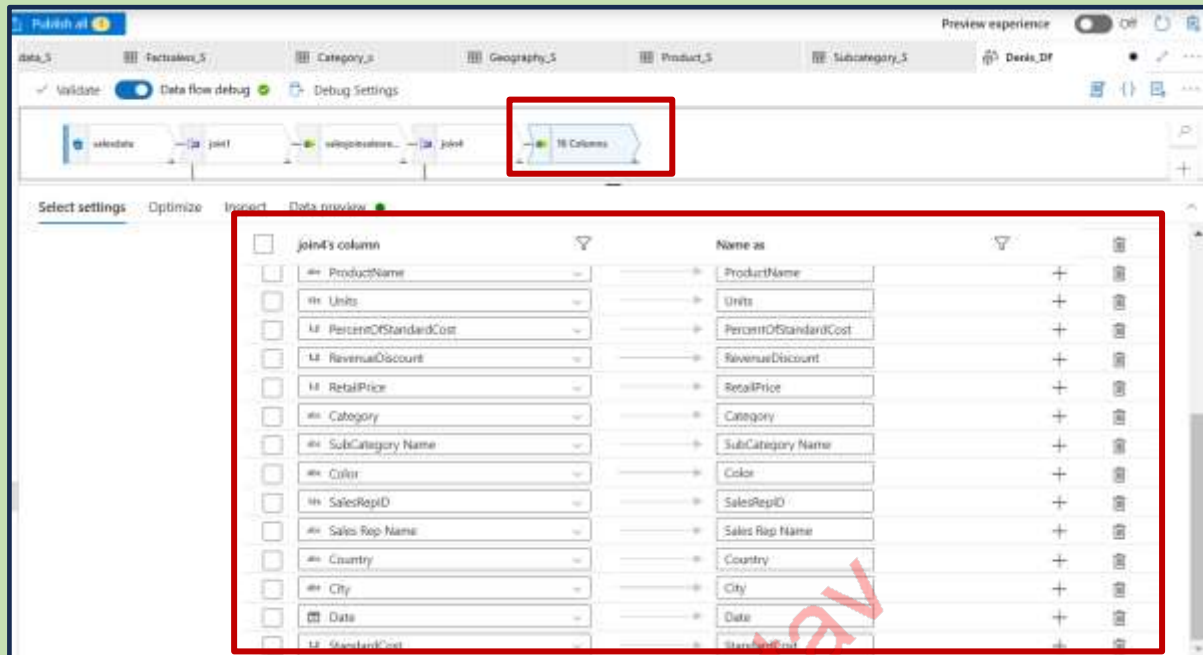
Right stream: productcatsubcatjoindata

Join type: Inner

Join conditions:

Left: salesjoinsalesrepdata's column	Operator	Right: productcatsubcatjoindata's column
ProductID	=	ProductID

Add one select column to join 4 and delete similar columns product ID arrange it in serial order

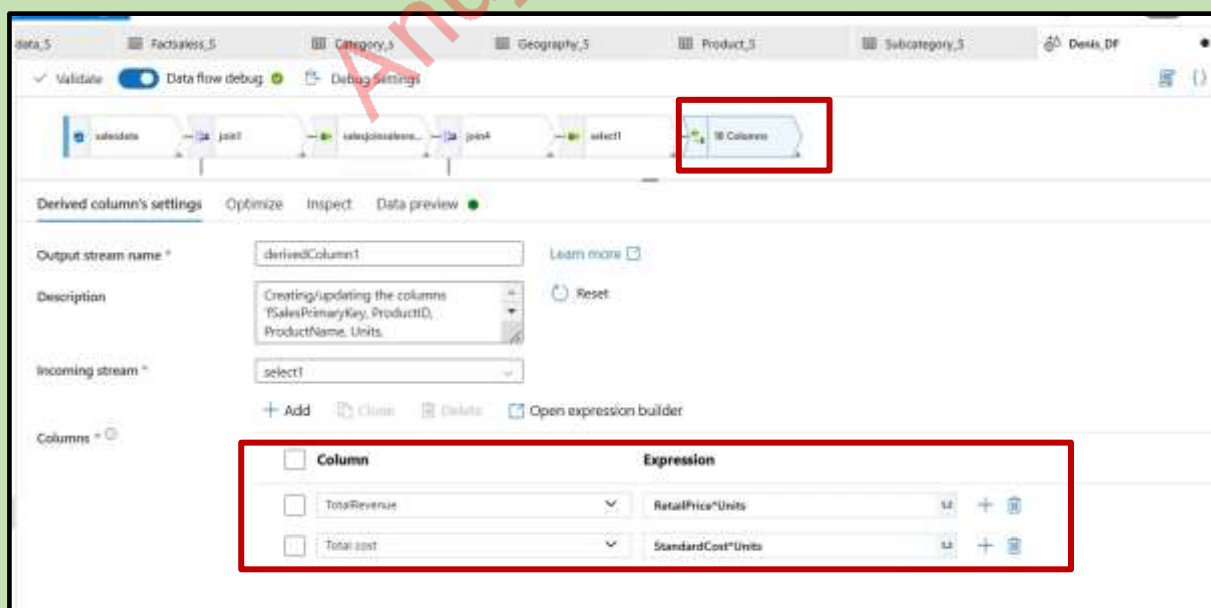


BUISINESS REQUIREMENT:

Calculate **Total Revenue** in sales table using product's Retail Price and multiplying it by the units

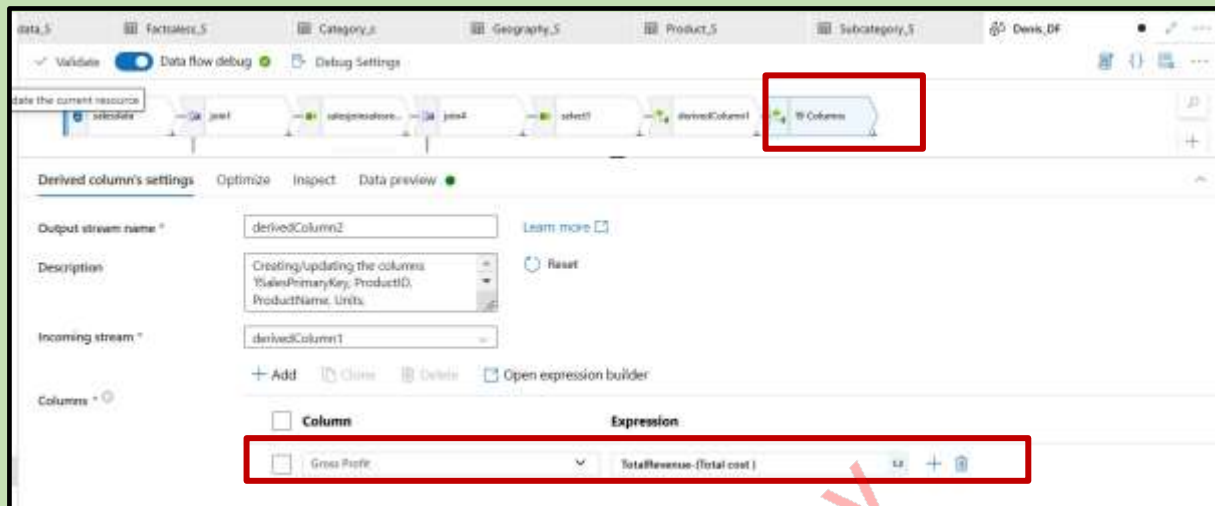
Calculate **Total Cost** in Sales Table, using Product's standard cost, and multiplying it by the units

Add the derived column and give the conditions



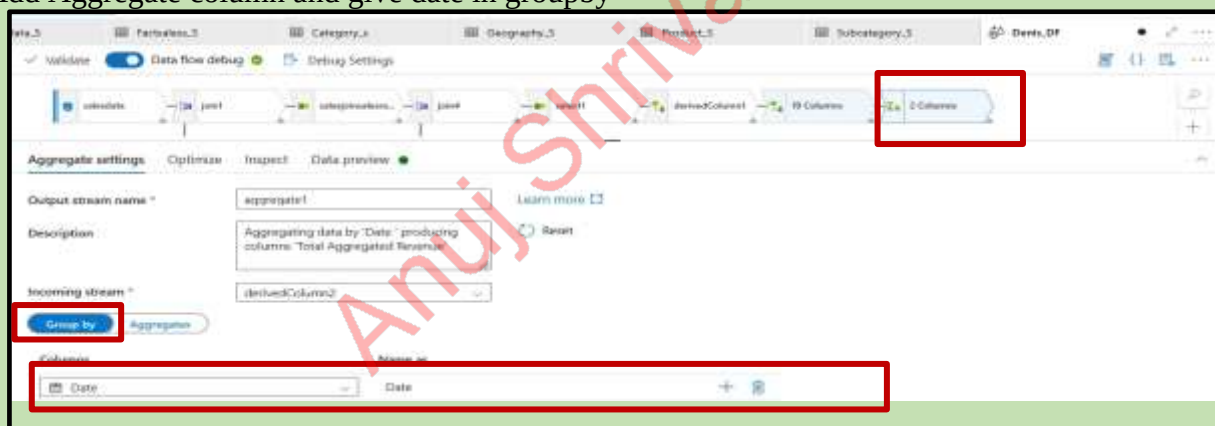
Calculate **Gross Profit** in Sales: Total Revenue – Total cost

Add one more derived column and give the condition

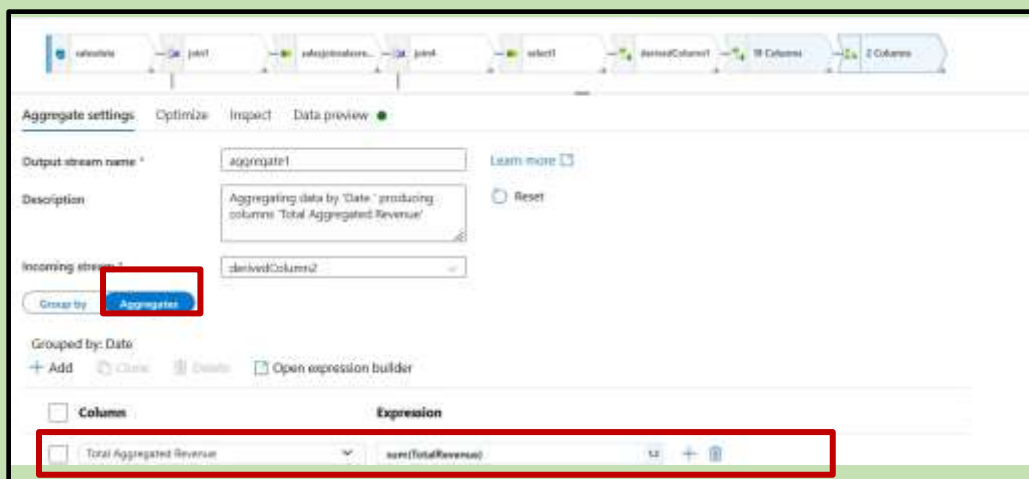


Calculate a measure for a **AVG Sales per day** – This is average sum of Total Revenue per day based on the date of actual sales

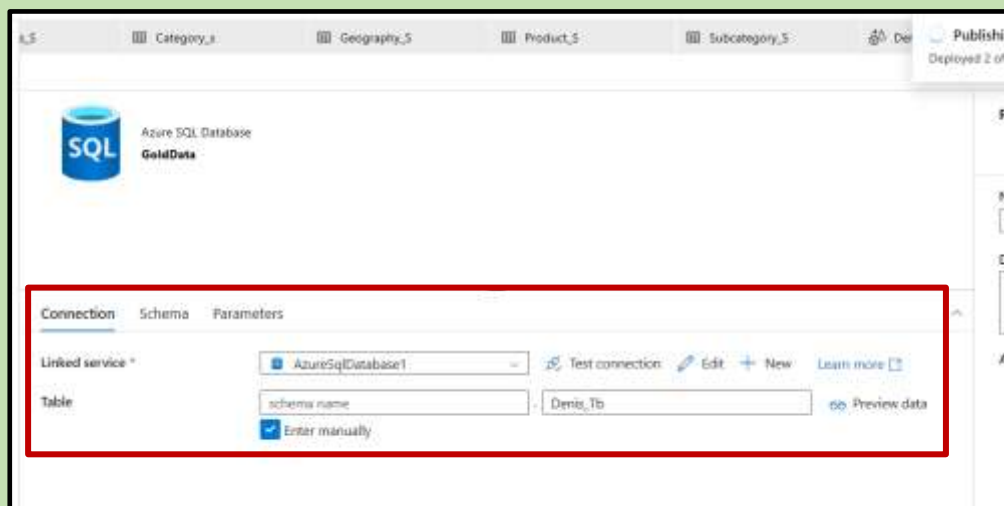
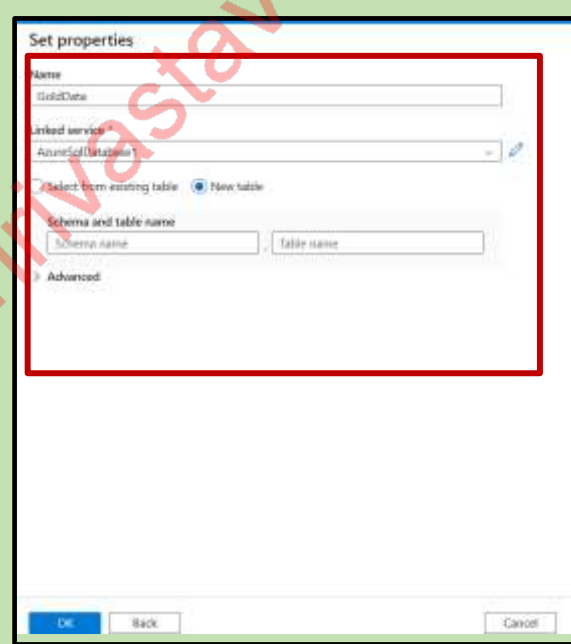
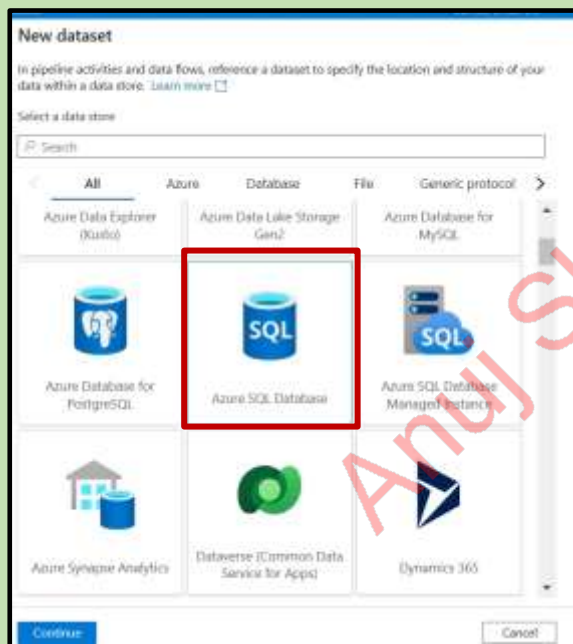
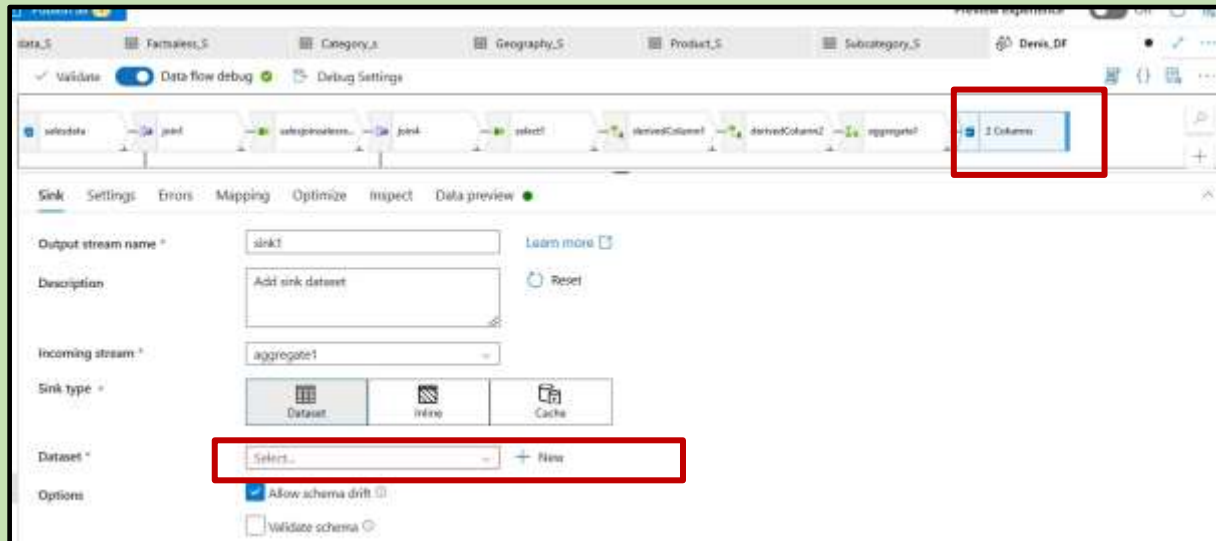
Add Aggregate column and give date in groupby



Give Total Aggregated Revenue sum(TotalRevenue) in Aggregation

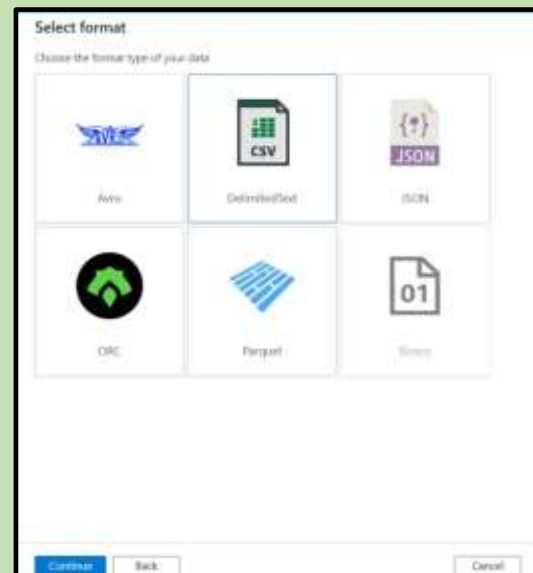
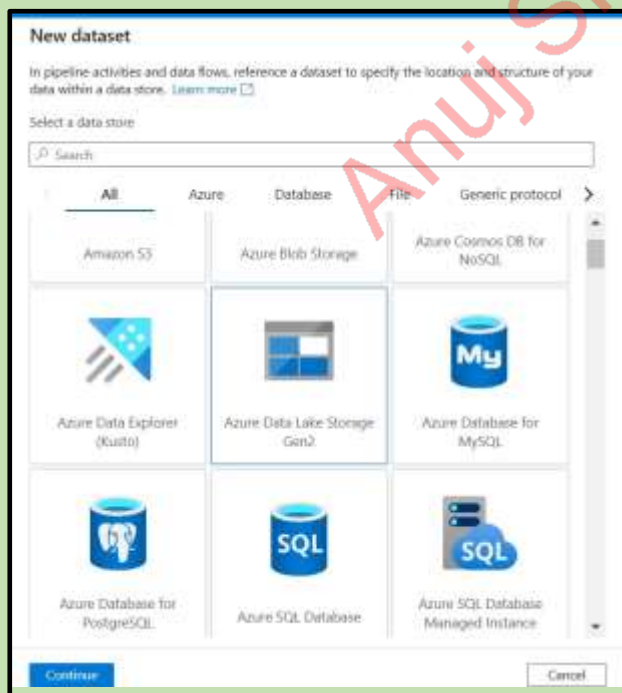
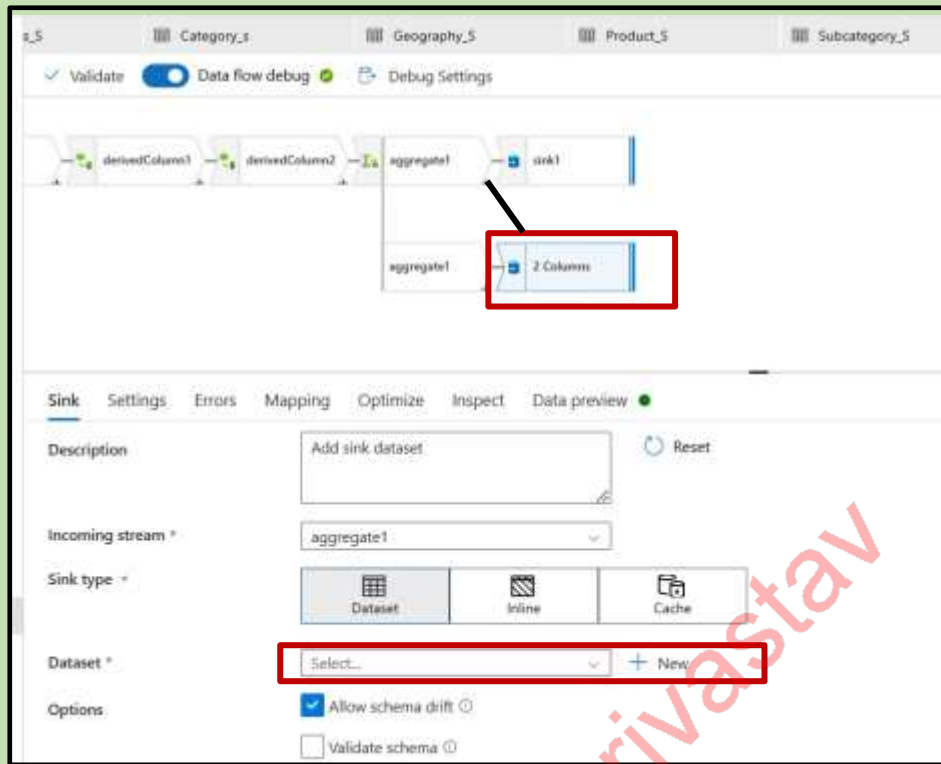


Now add the sink and add the data set

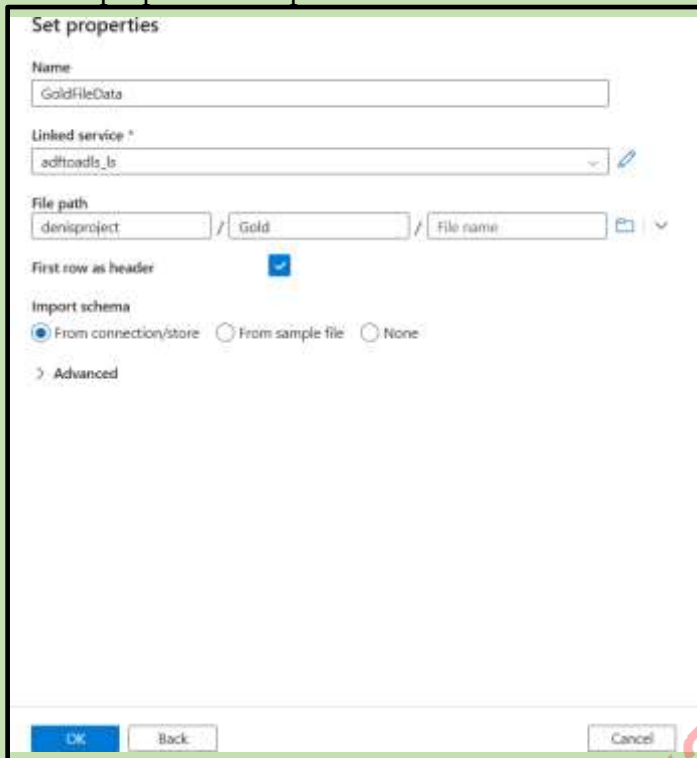


We can also store one copy in gold

Add one more sink



Give the properties and provide linked service



Set properties

Name: GoldFileData

Linked service *: adftoadls_ls

File path: denisproject / Gold / File name

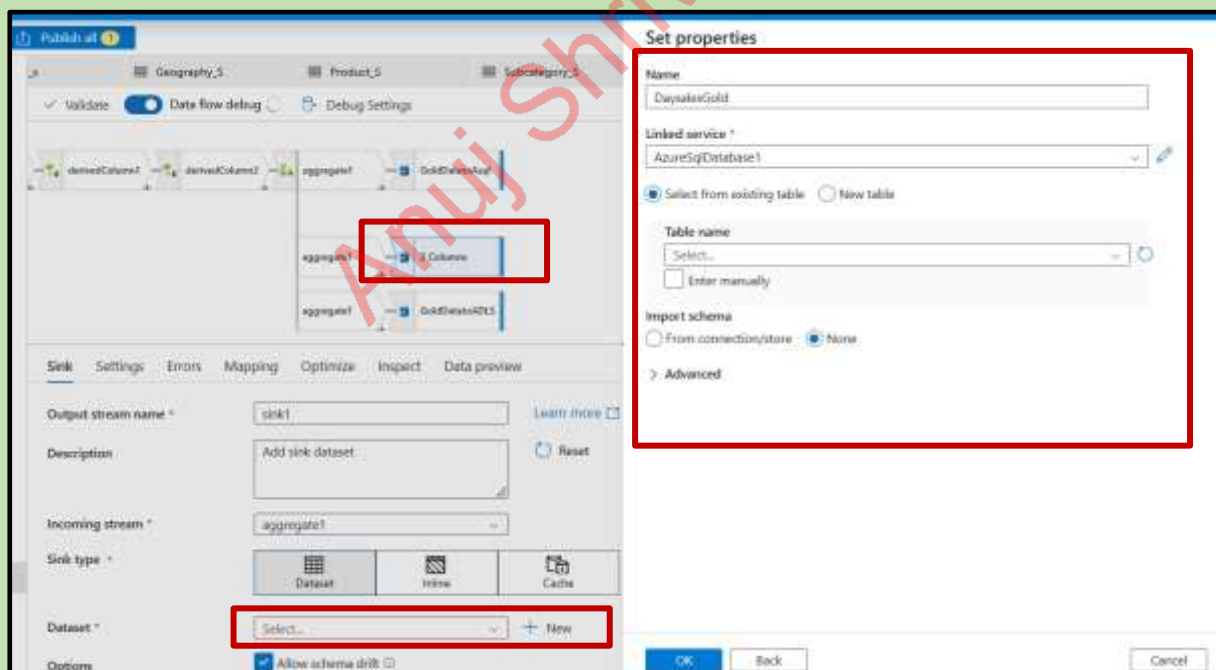
First row as header: ☒

Import schema: ☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK Back Cancel

Add one more sink after aggregation to sql db



The screenshot shows the Azure Data Factory interface with a data flow named 'Publish all'. The data flow includes a source 'Geography_5', a transformation 'Product_5', and a sink 'Subcategory_5'. A new sink is being added to the data flow, highlighted by a red box. The 'Set properties' dialog box is open, showing the configuration for the new sink. The 'Name' is 'DwysalesGold', the 'Linked service' is 'AzureSqlDatabase1', and the 'Table name' is 'Select...'. The 'Import schema' is set to 'None'. The 'Sink type' is 'Dataset', and the 'Dataset' is 'Select...'. The 'Options' section shows 'Allow schema drift' is checked.

Set properties

Name: DwysalesGold

Linked service *: AzureSqlDatabase1

☒ Select from existing table ☐ New table

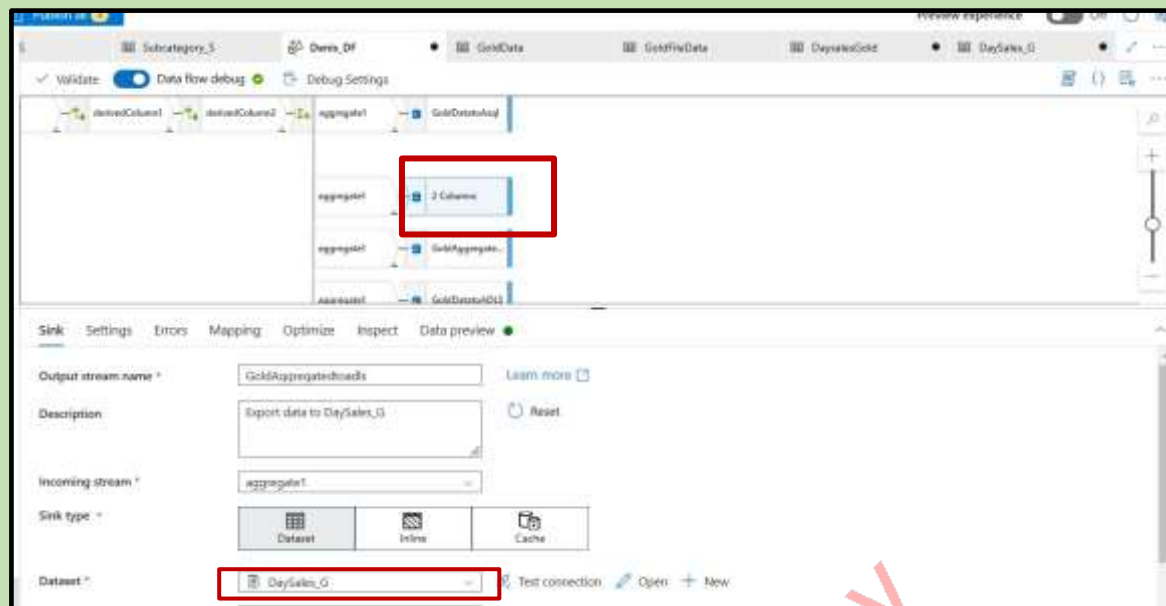
Table name: Select... ☐ Enter manually

Import schema: ☐ From connection/store ☒ None

> Advanced

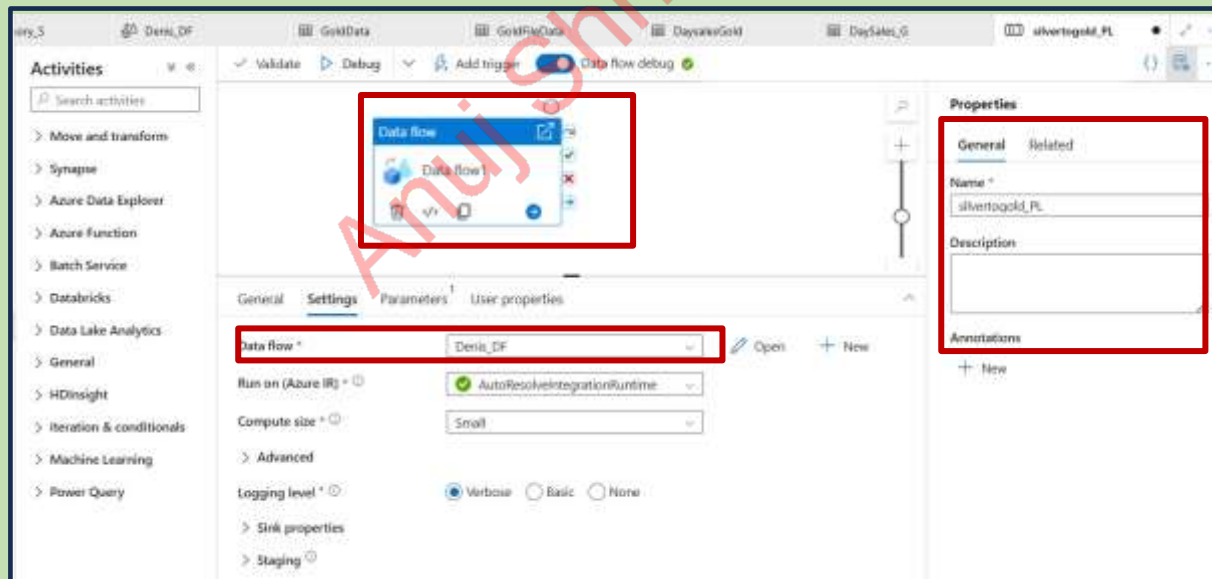
OK Back Cancel

Add one more sink for GoldAggregatedtoadls

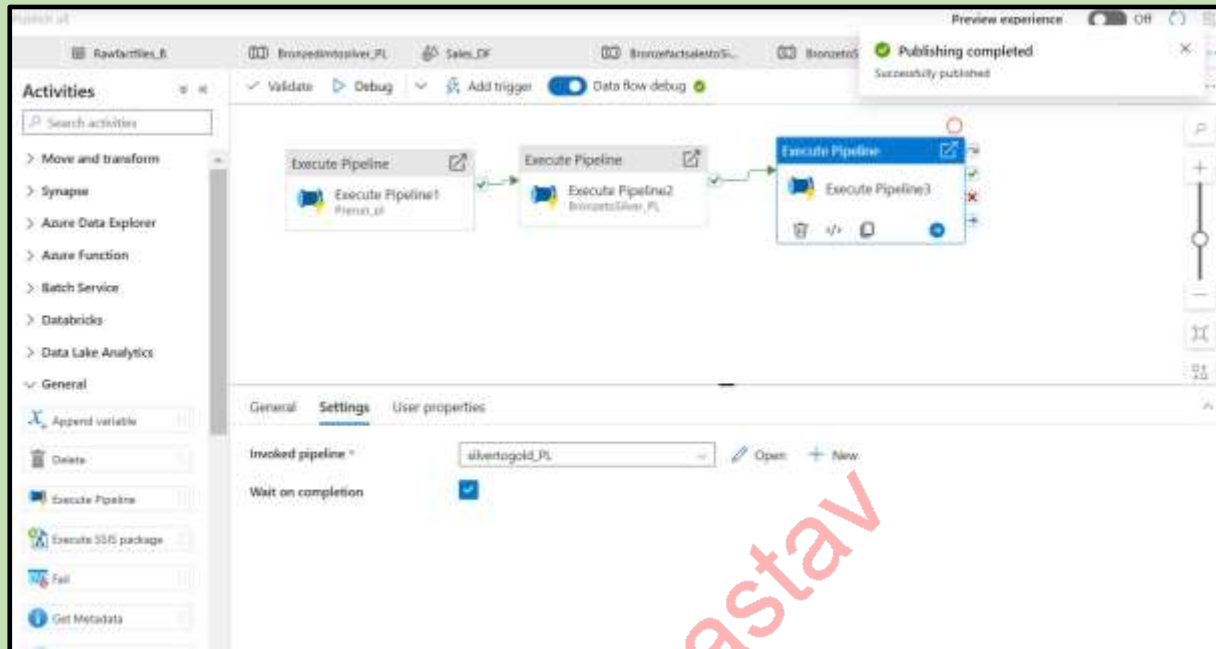


Pipeline:

Create new pipeline – silver to gold drag data flow and add the denies df



Now open the final pipeline denies pipeline and drag execute pipeline and invoke the silvertogold pipeline and run the pipeline



Pipelines got run successful

Showing 1 - 25 items										Last refreshed 11 minutes ago	
Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters	...			
silvertogold_PL	11/2/2024, 3:47:07 PM	11/2/2024, 4:00:20 PM	13m 14s	Manual trigger	✓ Succeeded	Original					
silvertogold_PL	11/2/2024, 2:13:08 PM	11/2/2024, 2:25:07 PM	12m 0s	45331802-e251-42b...	✓ Succeeded	Original					
BronzeSilver_PL	11/2/2024, 2:09:42 PM	11/2/2024, 2:13:07 PM	3m 25s	20b1b3de-af44-41e6...	✓ Succeeded	Original					
BronzeSilver_PL	11/2/2024, 2:05:43 PM	11/2/2024, 2:09:42 PM	3m 59s	17aa75ad-f026-4300...	✓ Succeeded	Original					
BronzeSilver_PL	11/2/2024, 2:05:43 PM	11/2/2024, 2:13:07 PM	7m 25s	ab6a1790-4a0f-4793...	✓ Succeeded	Original					
onpresto2cloudgen	11/2/2024, 2:05:04 PM	11/2/2024, 2:05:36 PM	32s	74ef9fc2-e6b6-4826...	✓ Succeeded	Original					
Onpresto2cloudgen	11/2/2024, 2:05:04 PM	11/2/2024, 2:05:29 PM	36s	bc8b7128-3d8a-4c7...	✓ Succeeded	Original					
Rawdata2Bronze_PL	11/2/2024, 2:05:04 PM	11/2/2024, 2:05:21 PM	17s	89b7d8e4-3370-43af...	✓ Succeeded	Original					
Rawdata2Bronze_PL	11/2/2024, 2:05:04 PM	11/2/2024, 2:05:21 PM	18s	3598970d-1249-42f1...	✓ Succeeded	Original					
Preproc_PL	11/2/2024, 2:05:01 PM	11/2/2024, 2:05:42 PM	29s	53425da-a634-488...	✓ Succeeded	Original					
Devic_PL	11/2/2024, 2:05:01 PM	11/2/2024, 2:25:08 PM	20m 8s	Manual trigger	✓ Succeeded	Original					
BronzeSilver_PL	11/2/2024, 9:39:10 AM	11/2/2024, 9:42:40 AM	3m 30s	06d1f172-d759-4ec7...	✓ Succeeded	Original					
BronzeSilver_PL	11/2/2024, 9:38:52 AM	11/2/2024, 9:39:08 AM	4m 17s	d1a0b0d7-132b-4d2...	✓ Succeeded	Original					
BronzeSilver_PL	11/2/2024, 9:38:51 AM	11/2/2024, 9:42:47 AM	7m 57s	Manual trigger	✓ Succeeded	Original					

We can see the output

Upload + Add Directory Refresh Remove Delete Change type Archive/Unarchive Feedback							
Authentication method: Access key (Switch to Microsoft Entra user account) Location: storageaccount / Gold							
Search blobs by prefix (case-sensitive) ☐ Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	...
[[							...
Aggregatetables.csv	11/2/2024, 3:55:38 PM	Hot (Inferred)		Block blob	33.01 KB	Available	...
Device_G.csv	11/2/2024, 3:56:32 PM	Hot (Inferred)		Block blob	8.36 MB	Available	...

Bronze

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Blob lease Give feedback							
Authentication method: Access key (Switch to Microsoft Entra user account)							
Location: development / Bronze							
Search blobs by prefix (case-sensitive) Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐  .							...
☐  sales	10/29/2024, 6:57:22 PM						...
☐  Categories.xlsx	11/2/2024, 2:05:18 PM	Hot (Inferred)		Block blob	17.03 KB	Available	...
☐  Geography.xlsx	11/2/2024, 2:05:32 PM	Hot (Inferred)		Block blob	472 B	Available	...
☐  Product.csv	11/2/2024, 2:05:34 PM	Hot (Inferred)		Block blob	778 B	Available	...
☐  SalesRep.xlsx	11/2/2024, 2:05:18 PM	Hot (Inferred)		Block blob	17.62 KB	Available	...
☐  SubCategories.xlsx	11/2/2024, 2:05:18 PM	Hot (Inferred)		Block blob	18.09 KB	Available	...

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Blob lease Give feedback							
Authentication method: Access key (Switch to Microsoft Entra user account)							
Location: development / Bronze / sales							
Search blobs by prefix (case-sensitive) Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐  .							...
☐  Sales 2014.csv	11/2/2024, 2:05:17 PM	Hot (Inferred)		Block blob	617.38 KB	Available	...
☐  Sales 2015.csv	11/2/2024, 2:05:17 PM	Hot (Inferred)		Block blob	640.35 KB	Available	...
☐  Sales 2016.csv	11/2/2024, 2:05:17 PM	Hot (Inferred)		Block blob	625.03 KB	Available	...
☐  Sales 2017.csv	11/2/2024, 2:05:17 PM	Hot (Inferred)		Block blob	915.34 KB	Available	...

SILVER

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Blob lease Give feedback							
Authentication method: Access key (Switch to Microsoft Entra user account)							
Location: development / Silver							
Search blobs by prefix (case-sensitive) Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐  .							...
☐  Sales	11/2/2024, 8:54:25 AM						...
☐  Category5.csv	11/2/2024, 2:12:26 PM	Hot (Inferred)		Block blob	41 B	Available	...
☐  Geography5.csv	11/2/2024, 2:12:44 PM	Hot (Inferred)		Block blob	166 B	Available	...
☐  Product5.csv	11/2/2024, 2:12:05 PM	Hot (Inferred)		Block blob	355 B	Available	...
☐  SalesRep5.csv	11/2/2024, 2:12:54 PM	Hot (Inferred)		Block blob	117 B	Available	...
☐  Subcategory5.csv	11/2/2024, 2:12:54 PM	Hot (Inferred)		Block blob	86 B	Available	...

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Blob lease Give feedback							
Authentication method: Access key (Switch to Microsoft Entra user account)							
Location: development / Silver / Sales							
Search blobs by prefix (case-sensitive) Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐  .							...
☐  Sales5.csv	11/2/2024, 2:09:27 PM	Hot (Inferred)		Block blob	2.74 MB	Available	...

GOLD

Upload + Add Directory Refresh Remove Delete Change tier Archive blob Blob lease Give feedback							
Authentication method: Access key (Switch to Microsoft Entra user account)							
Location: development / Gold							
Search blobs by prefix (case-sensitive) Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐  .							...
☐  AggregateSales.csv	11/2/2024, 3:55:38 PM	Hot (Inferred)		Block blob	33.01 KB	Available	...
☐  Denset_G.csv	11/2/2024, 2:58:52 PM	Hot (Inferred)		Block blob	8.36 MB	Available	...

Azure sql Database:

The screenshot shows the Azure SQL Database interface. On the left, the 'Servers' pane displays a tree view of the database structure, including 'dbo.DemoDaySales_TB'. The main query editor shows a SQL query to select the top 1000 rows from the 'Date' column, ordered by 'Total Aggregated Revenue'.

```

SELECT TOP (1000) [Date]
, (Total Aggregated Revenue)
FROM [dbo].[DemoDaySales_TB]
  
```

The 'Results' pane displays the following data:

	Date	Total Aggregated Revenue
1	2015-04-04	142108.55
2	2015-09-02	123051.09
3	2015-11-22	110754.50000000001
4	2016-06-03	101079.0
5	2016-07-26	99677.10000000001
6	2016-06-11	117189.00000000001
7	2016-10-01	100111.40000000001
8	2016-06-01	100001.75
9	2017-06-11	112801.50000000001
10	2017-06-11	90002.00000000001
11	2015-09-10	96732.05
12	2016-01-01	87706.00000000001
13	2016-06-26	113803.00000000001

The screenshot shows the Azure SQL Database interface. On the left, the 'Servers' pane displays a tree view of the database structure, including 'dbo.DemoDaySales_TB'. The main query editor shows a SQL query to select the top 1000 rows from the 'ProductID' column, ordered by 'RevenueDiscount'.

```

SELECT TOP (1000) [ProductID]
, [ProductID]
, [ProductID]
FROM [dbo].[DemoDaySales_TB]
  
```

The 'Results' pane displays the following data:

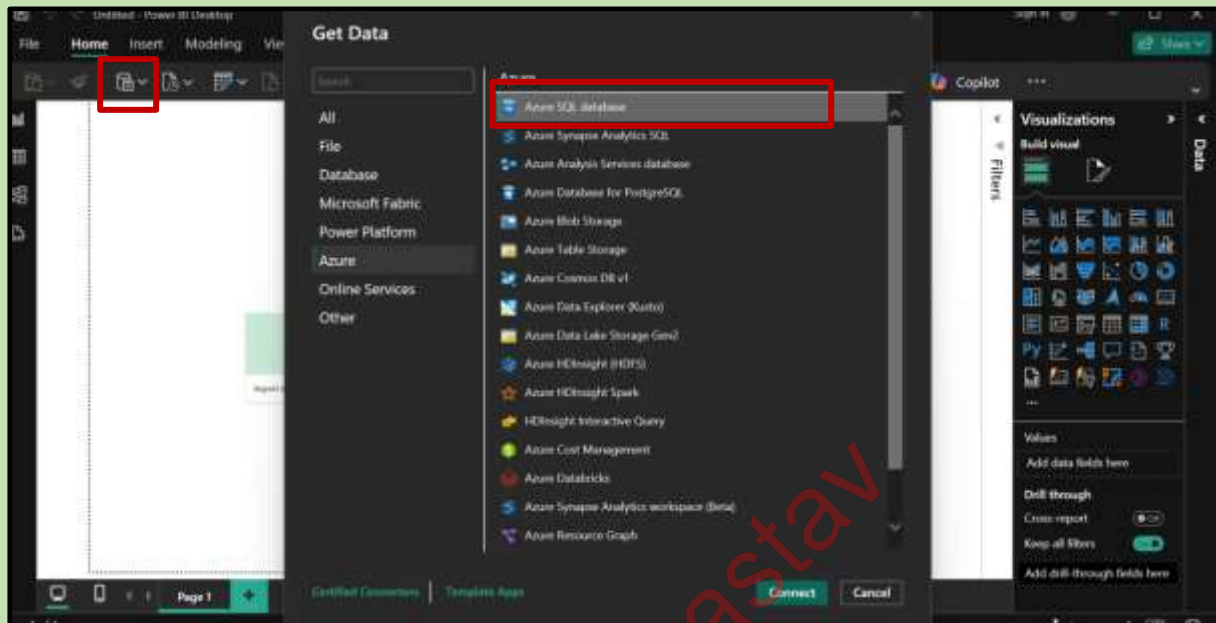
	ProductID	ProductID	ProductID	Units	PercentOfStandardCost	RevenueDiscount	RetailPrice
1	29974	4	Wend	130	0.901	0.55	47.99
2	29975	1	Wend	130	0.901	0.55	22.99
3	29976	13	Car Data	127	0.901	0.55	20.00
4	29977	1	Alider	124	0.901	0.55	22.99
5	29978	1	Quod	113	0.901	0.55	41.99
6	29979	1	linder	120	0.901	0.55	21.99
7	29980	6	Quod	107	0.901	0.55	41.99
8	29981	6	Varmelen	106	0.901	0.55	26.99
9	29982	1	Car Data	104	0.901	0.55	20.99
10	29983	2	linder	100	0.901	0.55	23.99
11	29984	5	Black mark	95	0.901	0.8	41.99
12	29985	5	Black mark	94	0.901	0.8	43.99
13	29986	7	Ring	88	0.901	0.8	26.99
14	29987	7	Ring	84	0.901	0.8	26.99
15	29988	7	Ring	87	0.901	0.8	26.99
16	29989	7	Ring	70	0.901	0.8	26.99
17	29990	7	Ring	70	0.901	0.15	26.99
18	29991	5	Black mark	64	0.901	0.15	41.99

The screenshot shows the Azure Data Factory interface. The 'Factory Resources' pane on the left lists various resources, including 'Power Query'. The 'Activities' pane in the center shows a list of activities, including 'Move and transform', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch', 'Data Lake Analytics', 'General', 'HDInsight', 'Integration & connect', 'Machine Learning', and 'Power Query'. The main canvas displays a pipeline with two 'Execute Pipeline' activities, 'Execute Pipeline1' and 'Execute Pipeline2', connected by a dependency arrow.

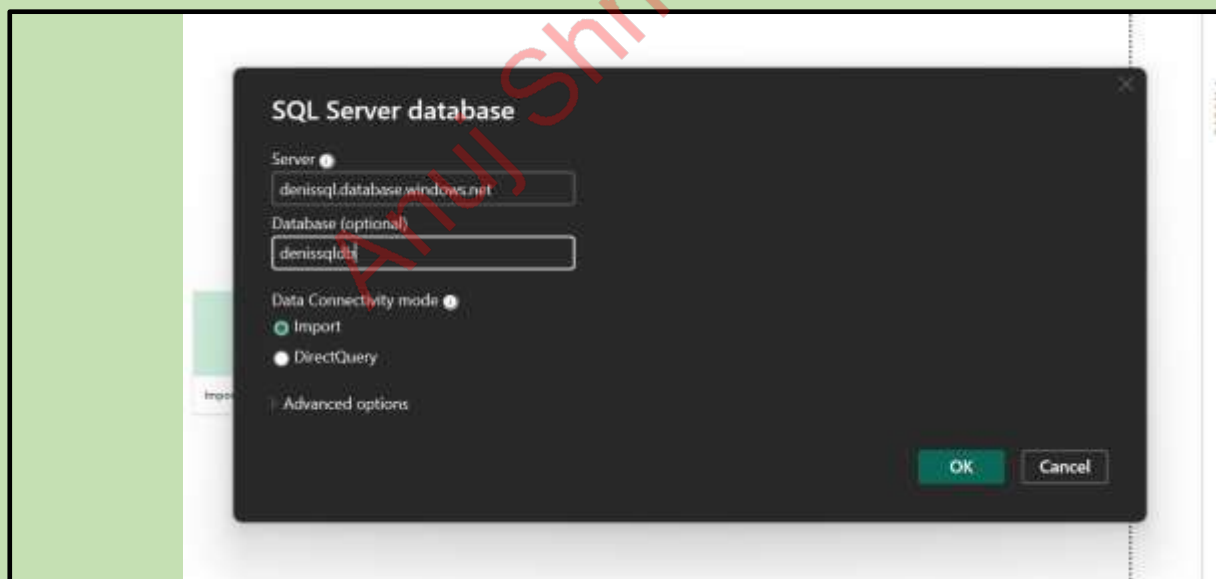
FINAL STEP:

Loading the data in power bi desktop

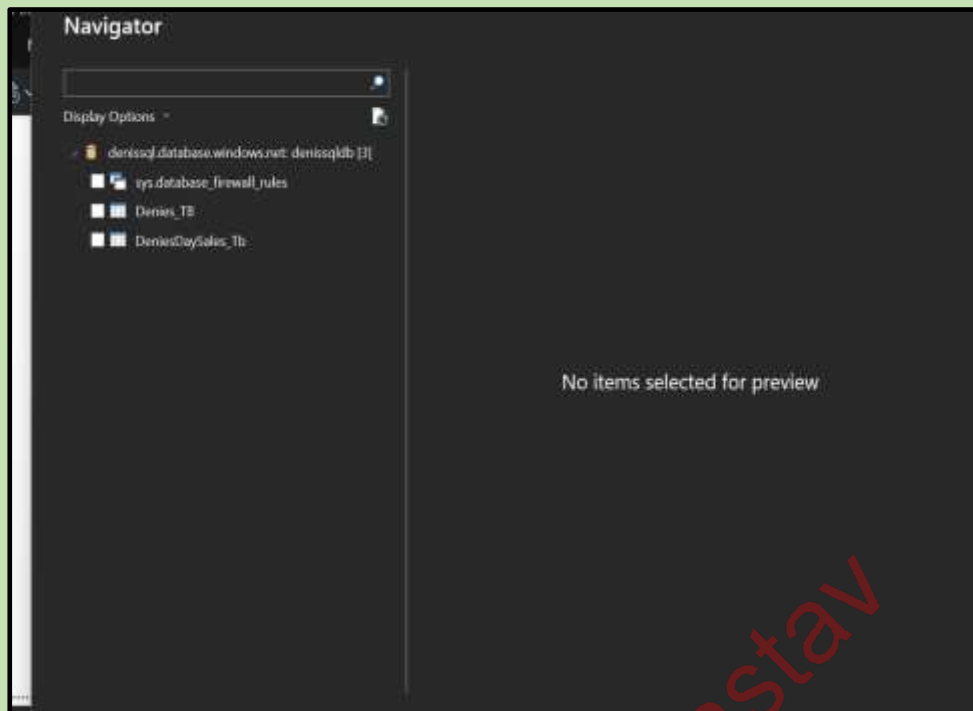
Open power Bi App → open azure sql database



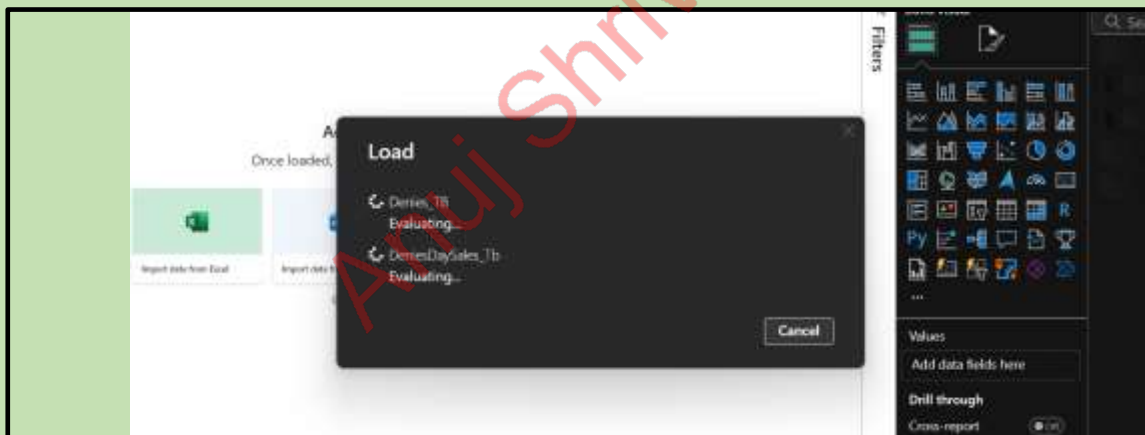
Give the server details from the azure sql data base and give the sql name and password



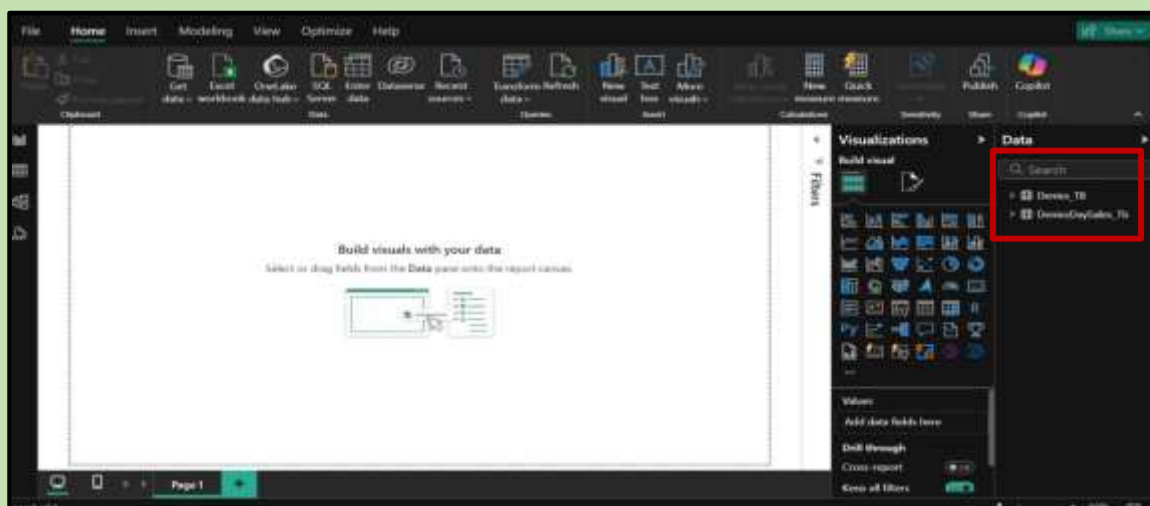
SQL TABLES:

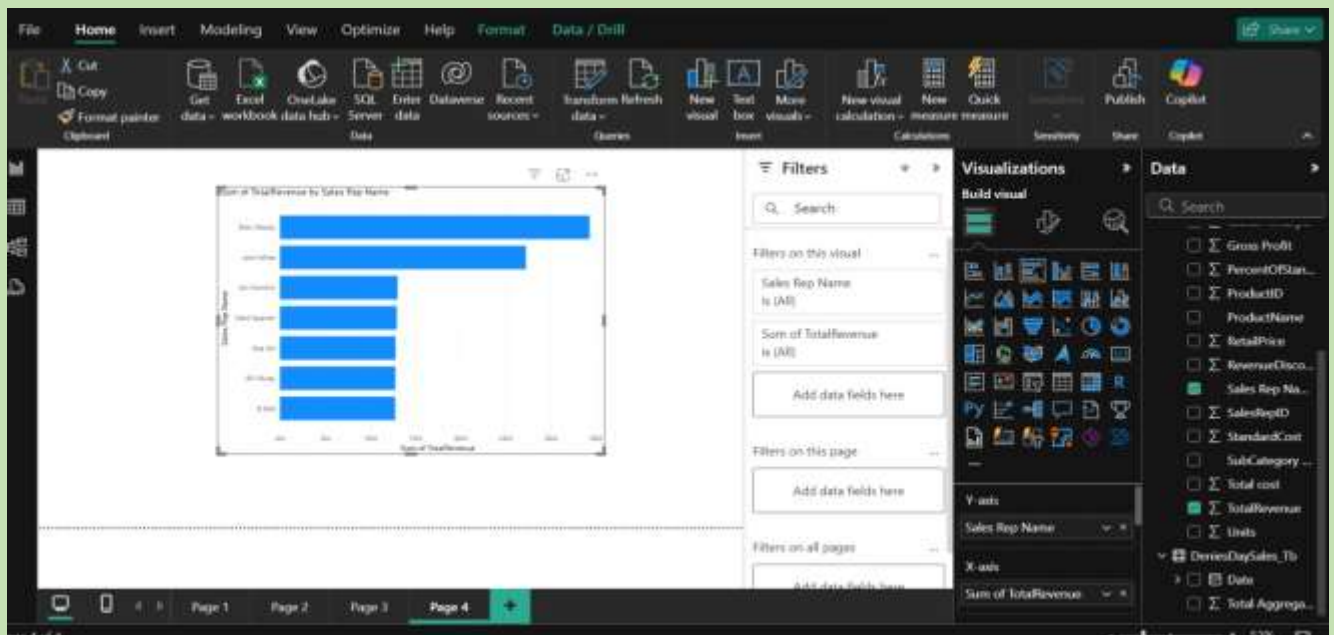
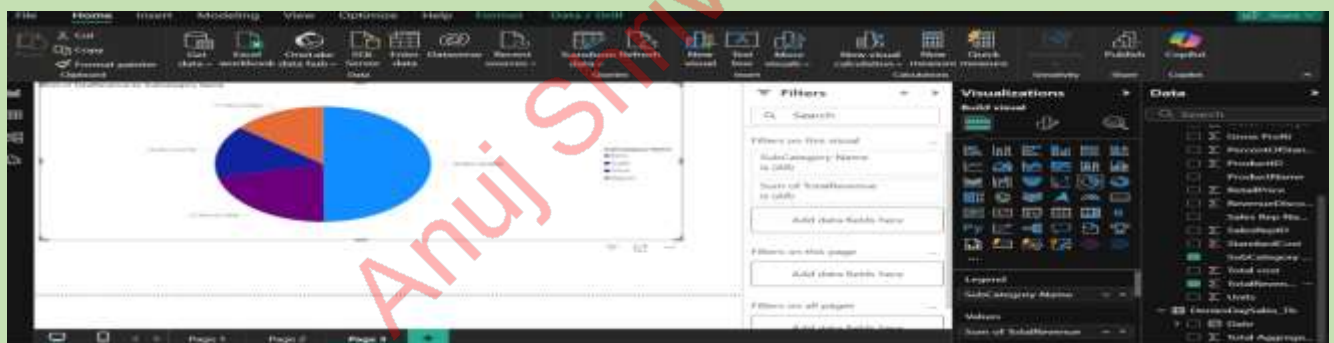
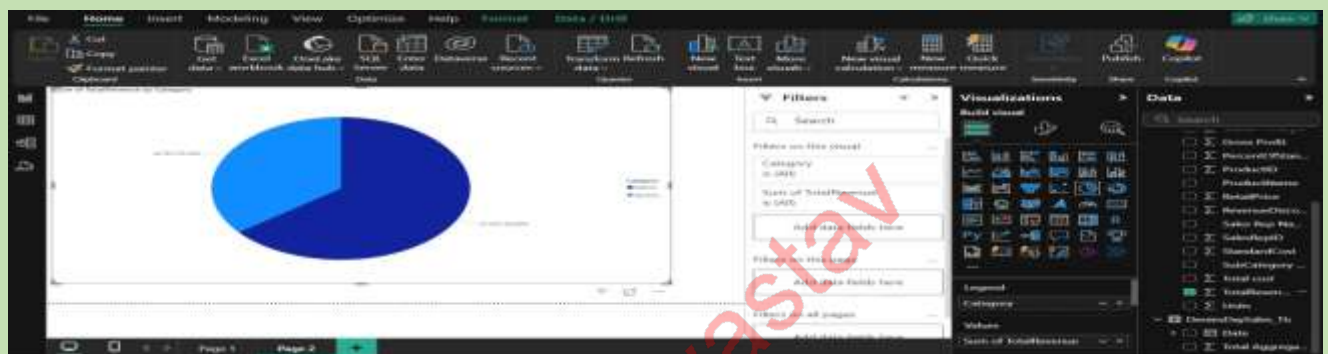


As we can see data is loading



As we can see data is loaded in power Bi





-----**END**-----

Anuj Shrivastav

Lead Data Engineer Pepsico

Anuj Shrivastav